
**STM32H742, STM32H743/753 and STM32H750 Value line
advanced Arm[®]-based 32-bit MCUs**

Introduction

This reference manual targets application developers. It provides complete information on how to use the STM32H742xx, STM32H743/53xx and STM32H750xB microcontroller memory and peripherals.

The STM32H742, STM32H743/753 and STM32H750 are lines of microcontrollers with different memory sizes, packages and peripherals.

For ordering information, mechanical, and electrical device characteristics refer to the corresponding datasheets.

For information on the Arm[®] Cortex[®]-M7 with FPU core, refer to the corresponding Arm *Technical Reference Manuals*.

Related documents

- Arm[®] Cortex[®]-M7 Technical Reference Manual, available from www.arm.com.
- Cortex[®]-M7 programming manual (PM0253).
- STM32H742xx, STM32H743xx and STM32H753xx datasheets
- STM32H750xB datasheet

Contents

1	Documentation conventions	100
1.1	General information	100
1.2	List of abbreviations for registers	100
1.3	Glossary	101
1.4	Availability of peripherals	101
1.5	Availability of security features	102
2	Memory and bus architecture	103
2.1	System architecture	103
2.1.1	Bus matrices	105
2.1.2	TCM buses	105
2.1.3	Bus-to-bus bridges	105
2.1.4	Inter-domain buses	106
2.1.5	CPU buses	106
2.1.6	Bus master peripherals	107
2.1.7	Clocks to functional blocks	108
2.2	AXI interconnect matrix (AXIM)	108
2.2.1	AXI introduction	108
2.2.2	AXI interconnect main features	108
2.2.3	AXI interconnect functional description	109
2.2.4	AXI interconnect registers	111
2.2.5	AXI interconnect register map	120
2.3	Memory organization	128
2.3.1	Introduction	128
2.3.2	Memory map and register boundary addresses	129
2.4	Embedded SRAM	135
2.5	Flash memory overview	136
2.6	Boot configuration	136
3	RAM ECC monitoring (RAMECC)	139
3.1	Introduction	139
3.2	RAMECC main features	139
3.3	RAMECC functional description	139

3.3.1	RAMECC block diagram	139
3.3.2	RAMECC internal signals	141
3.3.3	RAMECC monitor mapping	141
3.4	RAMECC registers	142
3.4.1	RAMECC interrupt enable register (RAMECC_IER)	142
3.4.2	RAMECC monitor x configuration register (RAMECC_MxCR)	143
3.4.3	RAMECC monitor x status register (RAMECC_MxSR)	143
3.4.4	RAMECC monitor x failing address register (RAMECC_MxFAR)	144
3.4.5	RAMECC monitor x failing data low register (RAMECC_MxFDRL)	144
3.4.6	RAMECC monitor x failing data high register (RAMECC_MxFDRH)	145
3.4.7	RAMECC monitor x failing ECC error code register RAMECC_MxFECR)	145
3.4.8	RAMECC register map	146
4	Embedded Flash memory (FLASH)	147
4.1	Introduction	147
4.2	FLASH main features	147
4.3	FLASH functional description	148
4.3.1	FLASH block diagram	148
4.3.2	FLASH internal signals	149
4.3.3	FLASH architecture and integration in the system	149
4.3.4	Flash memory architecture and usage	151
4.3.5	FLASH system performance enhancements	155
4.3.6	FLASH data protection schemes	155
4.3.7	Overview of FLASH operations	156
4.3.8	FLASH read operations	157
4.3.9	FLASH program operations	160
4.3.10	FLASH erase operations	164
4.3.11	FLASH parallel operations (STM32H742/743/753 devices only)	167
4.3.12	Flash memory error protections	167
4.3.13	Flash bank and register swapping (STM32H742/743/753 devices only)	169
4.3.14	FLASH reset and clocks	172
4.4	FLASH option bytes	172
4.4.1	About option bytes	172
4.4.2	Option byte loading	173
4.4.3	Option byte modification	173

4.4.4	Option bytes overview	176
4.4.5	Description of user and system option bytes	178
4.4.6	Description of data protection option bytes	179
4.4.7	Description of boot address option bytes	180
4.5	FLASH protection mechanisms	181
4.5.1	FLASH configuration protection	181
4.5.2	Write protection	182
4.5.3	Readout protection (RDP)	183
4.5.4	Proprietary code readout protection (PCROP)	188
4.5.5	Secure access mode	189
4.6	FLASH low-power modes	190
4.6.1	Introduction	190
4.6.2	Managing the FLASH domain switching to DStop or DStandby	191
4.7	FLASH error management	192
4.7.1	Introduction	192
4.7.2	Write protection error (WRPERR)	192
4.7.3	Programming sequence error (PGSERR)	193
4.7.4	Strobe error (STRBERR)	194
4.7.5	Inconsistency error (INCERR)	194
4.7.6	Operation error (OPERR)	195
4.7.7	Error correction code error (SNECCERR/DBECCERR)	195
4.7.8	Read protection error (RDPERR)	196
4.7.9	Read secure error (RDSERR)	196
4.7.10	CRC read error (CRCRDERR)	196
4.7.11	Option byte change error (OPTCHANGEERR)	197
4.7.12	Miscellaneous HardFault errors	197
4.8	FLASH interrupts	197
4.9	FLASH registers	200
4.9.1	FLASH access control register (FLASH_ACR)	200
4.9.2	FLASH key register for bank 1 (FLASH_KEYR1)	200
4.9.3	FLASH option key register (FLASH_OPTKEYR)	201
4.9.4	FLASH control register for bank 1 (FLASH_CR1)	201
4.9.5	FLASH status register for bank 1 (FLASH_SR1)	206
4.9.6	FLASH clear control register for bank 1 (FLASH_CCR1)	209
4.9.7	FLASH option control register (FLASH_OPTCR)	210
4.9.8	FLASH option status register (FLASH_OPTSR_CUR)	211

4.9.9	FLASH option status register (FLASH_OPTSR_PRG)	214
4.9.10	FLASH option clear control register (FLASH_OPTCCR)	216
4.9.11	FLASH protection address for bank 1 (FLASH_PRAR_CUR1)	216
4.9.12	FLASH protection address for bank 1 (FLASH_PRAR_PRG1)	217
4.9.13	FLASH secure address for bank 1 (FLASH_SCAR_CUR1)	218
4.9.14	FLASH secure address for bank 1 (FLASH_SCAR_PRG1)	218
4.9.15	FLASH write sector protection for bank 1 (FLASH_WPSN_CUR1R)	219
4.9.16	FLASH write sector protection for bank 1 (FLASH_WPSN_PRG1R)	220
4.9.17	FLASH register boot address FLASH_BOOT_CURR)	220
4.9.18	FLASH register boot address FLASH_BOOT_PRGR)	220
4.9.19	FLASH CRC control register for bank 1 (FLASH_CRCCR1)	221
4.9.20	FLASH CRC start address register for bank 1 (FLASH_CRCSADD1R)	222
4.9.21	FLASH CRC end address register for bank 1 (FLASH_CRCEADD1R)	223
4.9.22	FLASH CRC data register (FLASH_CRCDATAR)	223
4.9.23	FLASH ECC fail address for bank 1 (FLASH_ECC_FA1R)	224
4.9.24	FLASH key register for bank 2 (FLASH_KEYR2)	224
4.9.25	FLASH control register for bank 2 (FLASH_CR2)	225
4.9.26	FLASH status register for bank 2 (FLASH_SR2)	229
4.9.27	FLASH clear control register for bank 2 (FLASH_CCR2)	232
4.9.28	FLASH protection address for bank 2 (FLASH_PRAR_CUR2)	233
4.9.29	FLASH protection address for bank 2 (FLASH_PRAR_PRG2)	233
4.9.30	FLASH secure address for bank 2 (FLASH_SCAR_CUR2)	234
4.9.31	FLASH secure address for bank 2 (FLASH_SCAR_PRG2)	235
4.9.32	FLASH write sector protection for bank 2 (FLASH_WPSN_CUR2R)	236
4.9.33	FLASH write sector protection for bank 2 (FLASH_WPSN_PRG2R)	236
4.9.34	FLASH CRC control register for bank 2 (FLASH_CRCCR2)	237
4.9.35	FLASH CRC start address register for bank 2 (FLASH_CRCSADD2R)	238
4.9.36	FLASH CRC end address register for bank 2 (FLASH_CRCEADD2R)	239
4.9.37	FLASH ECC fail address for bank 2 (FLASH_ECC_FA2R)	239
4.9.38	FLASH register map and reset values	240

5	Secure memory management (SMM)	245
5.1	Introduction	245
5.2	Glossary	245
5.3	Secure access mode	246
5.3.1	Associated features	247
5.3.2	Boot state machine	247
5.3.3	Secure access mode configuration	248
5.4	Root secure services (RSS)	249
5.4.1	Secure area setting service	249
5.4.2	Secure area exiting service	249
5.5	Secure user software	250
5.5.1	Access rules	250
5.5.2	Setting secure user memory areas	250
5.6	Summary of Flash protection mechanisms	251
6	Power control (PWR)	252
6.1	Introduction	252
6.2	PWR main features	252
6.3	PWR block diagram	253
6.3.1	PWR pins and internal signals	254
6.4	Power supplies	256
6.4.1	System supply startup	259
6.4.2	Core domain	261
6.4.3	PWR external supply	262
6.4.4	Backup domain	263
6.4.5	VBAT battery charging	265
6.4.6	Analog supply	265
6.4.7	USB regulator	266
6.5	Power supply supervision	266
6.5.1	Power-on reset (POR)/power-down reset (PDR)	267
6.5.2	Brownout reset (BOR)	267
6.5.3	Programmable voltage detector (PVD)	268
6.5.4	Analog voltage detector (AVD)	270
6.5.5	Battery voltage thresholds	271
6.5.6	Temperature thresholds	272
6.6	Power management	273

6.6.1	Operating modes	274
6.6.2	Voltage scaling	277
6.6.3	Power control modes	279
6.6.4	Power management examples	283
6.7	Low-power modes	289
6.7.1	Slowing down system clocks	289
6.7.2	Controlling peripheral clocks	289
6.7.3	Entering low-power modes	289
6.7.4	Exiting from low-power modes	290
6.7.5	CSleep mode	291
6.7.6	CStop mode	292
6.7.7	DStop mode	293
6.7.8	Stop mode	294
6.7.9	DStandby mode	296
6.7.10	Standby mode	298
6.7.11	Monitoring low-power modes	300
6.8	PWR register description	301
6.8.1	PWR control register 1 (PWR_CR1)	301
6.8.2	PWR control status register 1 (PWR_CSR1)	302
6.8.3	PWR control register 2 (PWR_CR2)	303
6.8.4	PWR control register 3 (PWR_CR3)	304
6.8.5	PWR CPU control register (PWR_CPUCR)	306
6.8.6	PWR D3 domain control register (PWR_D3CR)	307
6.8.7	PWR wakeup clear register (PWR_WKUPCR)	308
6.8.8	PWR wakeup flag register (PWR_WKUPFR)	308
6.8.9	PWR wakeup enable and polarity register (PWR_WKUPEPR)	309
6.8.10	PWR register map	310
7	Low-power D3 domain	311
7.1	Introduction	311
7.2	EXTI, RCC and PWR interconnections	311
7.2.1	Interrupts and wakeup	313
7.2.2	Block interactions	313
7.2.3	Role of D3 domain DMAMUX2	314
7.3	Low-power application example based on LPUART1 transmission	315
7.3.1	Memory retention	315

7.3.2	Memory-to-peripheral transfer using LPUART1 interface	315
7.3.3	Overall description of the low-power application example based on LPUART1 transmission	320
7.3.4	Alternate implementations	321
7.4	Other low-power applications	322
8	Reset and Clock Control (RCC)	323
8.1	RCC main features	323
8.2	RCC block diagram	324
8.3	RCC pins and internal signals	324
8.4	RCC reset block functional description	326
8.4.1	Power-on/off reset	326
8.4.2	System reset	327
8.4.3	Local resets	328
8.4.4	Reset source identification	330
8.4.5	Low-power mode security reset (lpwr_rst)	331
8.4.6	Backup domain reset	331
8.4.7	Power-on and wakeup sequences	331
8.5	RCC clock block functional description	334
8.5.1	Clock naming convention	336
8.5.2	Oscillators description	336
8.5.3	Clock Security System (CSS)	341
8.5.4	Clock output generation (MCO1/MCO2)	342
8.5.5	PLL description	343
8.5.6	System clock (sys_ck)	347
8.5.7	Handling clock generators in Stop and Standby mode	349
8.5.8	Kernel clock selection	351
8.5.9	General clock concept overview	364
8.5.10	Peripheral allocation	368
8.5.11	Peripheral clock gating control	370
8.5.12	CPU and bus matrix clock gating control	375
8.6	RCC Interrupts	377
8.7	RCC register description	378
8.7.1	Register mapping overview	378
8.7.2	RCC Source Control Register (RCC_CR)	379
8.7.3	RCC Internal Clock Source Calibration Register (RCC_ICSCR)	383

8.7.4	RCC HSI configuration register (RCC_HSI CFGR)	384
8.7.5	RCC Clock Recovery RC Register (RCC_CRRCR)	385
8.7.6	RCC CSI configuration register (RCC_CSICFGR)	386
8.7.7	RCC Clock Configuration Register (RCC_CFGR)	387
8.7.8	RCC Domain 1 Clock Configuration Register (RCC_D1CFGR)	390
8.7.9	RCC Domain 2 Clock Configuration Register (RCC_D2CFGR)	392
8.7.10	RCC Domain 3 Clock Configuration Register (RCC_D3CFGR)	393
8.7.11	RCC PLLs Clock Source Selection Register (RCC_PLLCKSELR)	394
8.7.12	RCC PLLs Configuration Register (RCC_PLLCFGR)	396
8.7.13	RCC PLL1 Dividers Configuration Register (RCC_PLL1DIVR)	399
8.7.14	RCC PLL1 Fractional Divider Register (RCC_PLL1FRACR)	401
8.7.15	RCC PLL2 Dividers Configuration Register (RCC_PLL2DIVR)	402
8.7.16	RCC PLL2 Fractional Divider Register (RCC_PLL2FRACR)	404
8.7.17	RCC PLL3 Dividers Configuration Register (RCC_PLL3DIVR)	405
8.7.18	RCC PLL3 Fractional Divider Register (RCC_PLL3FRACR)	407
8.7.19	RCC Domain 1 Kernel Clock Configuration Register (RCC_D1CCIPR)	408
8.7.20	RCC Domain 2 Kernel Clock Configuration Register (RCC_D2CCIP1R)	409
8.7.21	RCC Domain 2 Kernel Clock Configuration Register (RCC_D2CCIP2R)	412
8.7.22	RCC Domain 3 Kernel Clock Configuration Register (RCC_D3CCIPR)	414
8.7.23	RCC Clock Source Interrupt Enable Register (RCC_CIER)	417
8.7.24	RCC Clock Source Interrupt Flag Register (RCC_CIFR)	419
8.7.25	RCC Clock Source Interrupt Clear Register (RCC_CICR)	421
8.7.26	RCC Backup Domain Control Register (RCC_BDCR)	423
8.7.27	RCC Clock Control and Status Register (RCC_CSR)	425
8.7.28	RCC AHB3 Reset Register (RCC_AHB3RSTR)	426
8.7.29	RCC AHB1 Peripheral Reset Register (RCC_AHB1RSTR)	428
8.7.30	RCC AHB2 Peripheral Reset Register (RCC_AHB2RSTR)	430
8.7.31	RCC AHB4 Peripheral Reset Register (RCC_AHB4RSTR)	431
8.7.32	RCC APB3 Peripheral Reset Register (RCC_APB3RSTR)	433
8.7.33	RCC APB1 Peripheral Reset Register (RCC_APB1LRSTR)	434
8.7.34	RCC APB1 Peripheral Reset Register (RCC_APB1HRSTR)	437
8.7.35	RCC APB2 Peripheral Reset Register (RCC_APB2RSTR)	438
8.7.36	RCC APB4 Peripheral Reset Register (RCC_APB4RSTR)	440
8.7.37	RCC Global Control Register (RCC_GCR)	442

8.7.38	RCC D3 Autonomous mode Register (RCC_D3AMR)	443
8.7.39	RCC Reset Status Register (RCC_RSR)	446
8.7.40	RCC AHB3 Clock Register (RCC_AHB3ENR)	448
8.7.41	RCC AHB1 Clock Register (RCC_AHB1ENR)	450
8.7.42	RCC AHB2 Clock Register (RCC_AHB2ENR)	452
8.7.43	RCC AHB4 Clock Register (RCC_AHB4ENR)	454
8.7.44	RCC APB3 Clock Register (RCC_APB3ENR)	457
8.7.45	RCC APB1 Clock Register (RCC_APB1LENR)	458
8.7.46	RCC APB1 Clock Register (RCC_APB1HENR)	462
8.7.47	RCC APB2 Clock Register (RCC_APB2ENR)	464
8.7.48	RCC APB4 Clock Register (RCC_APB4ENR)	467
8.7.49	RCC AHB3 Sleep Clock Register (RCC_AHB3LPENR)	470
8.7.50	RCC AHB1 Sleep Clock Register (RCC_AHB1LPENR)	472
8.7.51	RCC AHB2 Sleep Clock Register (RCC_AHB2LPENR)	474
8.7.52	RCC AHB4 Sleep Clock Register (RCC_AHB4LPENR)	476
8.7.53	RCC APB3 Sleep Clock Register (RCC_APB3LPENR)	479
8.7.54	RCC APB1 Low Sleep Clock Register (RCC_APB1LLPENR)	480
8.7.55	RCC APB1 High Sleep Clock Register (RCC_APB1HLPENR)	484
8.7.56	RCC APB2 Sleep Clock Register (RCC_APB2LPENR)	486
8.7.57	RCC APB4 Sleep Clock Register (RCC_APB4LPENR)	489
8.8	RCC register map	492
9	Clock recovery system (CRS)	502
9.1	Introduction	502
9.2	CRS main features	502
9.3	CRS implementation	502
9.4	CRS functional description	503
9.4.1	CRS block diagram	503
9.5	CRS internal signals	503
9.5.1	Synchronization input	504
9.5.2	Frequency error measurement	504
9.5.3	Frequency error evaluation and automatic trimming	506
9.5.4	CRS initialization and configuration	506
9.6	CRS low-power modes	507
9.7	CRS interrupts	507
9.8	CRS registers	508

9.8.1	CRS control register (CRS_CR)	508
9.8.2	CRS configuration register (CRS_CFGR)	509
9.8.3	CRS interrupt and status register (CRS_ISR)	510
9.8.4	CRS interrupt flag clear register (CRS_ICR)	512
9.8.5	CRS register map	513
10	Hardware semaphore (HSEM)	514
10.1	Hardware semaphore introduction	514
10.2	Hardware semaphore main features	514
10.3	HSEM functional description	515
10.3.1	HSEM block diagram	515
10.3.2	HSEM internal signals	515
10.3.3	HSEM lock procedures	516
10.3.4	HSEM write/read/readlock register address	517
10.3.5	HSEM clear procedures	517
10.3.6	HSEM MASTERID semaphore clear	518
10.3.7	HSEM interrupts	518
10.3.8	AHB bus master ID verification	520
10.4	HSEM registers	521
10.4.1	HSEM register semaphore x (HSEM_Rx)	521
10.4.2	HSEM read lock register semaphore x (HSEM_RLRx)	522
10.4.3	HSEM interrupt enable register (HSEM_IER)	523
10.4.4	HSEM interrupt clear register (HSEM_ICR)	523
10.4.5	HSEM interrupt status register (HSEM_ISR)	523
10.4.6	HSEM interrupt status register (HSEM_MISR)	524
10.4.7	HSEM clear register (HSEM_CR)	524
10.4.8	HSEM interrupt clear register (HSEM_KEYR)	525
10.4.9	HSEM register map	526
11	General-purpose I/Os (GPIO)	527
11.1	Introduction	527
11.2	GPIO main features	527
11.3	GPIO functional description	527
11.3.1	General-purpose I/O (GPIO)	530
11.3.2	I/O pin alternate function multiplexer and mapping	530
11.3.3	I/O port control registers	531

11.3.4	I/O port data registers	531
11.3.5	I/O data bitwise handling	531
11.3.6	GPIO locking mechanism	532
11.3.7	I/O alternate function input/output	532
11.3.8	External interrupt/wakeup lines	532
11.3.9	Input configuration	533
11.3.10	Output configuration	533
11.3.11	I/O compensation cell	534
11.3.12	Alternate function configuration	534
11.3.13	Analog configuration	535
11.3.14	Using the HSE or LSE oscillator pins as GPIOs	536
11.3.15	Using the GPIO pins in the backup supply domain	536
11.4	GPIO registers	537
11.4.1	GPIO port mode register (GPIOx_MODER) (x = A to K)	537
11.4.2	GPIO port output type register (GPIOx_OTYPER) (x = A to K)	537
11.4.3	GPIO port output speed register (GPIOx_OSPEEDR) (x = A to K)	538
11.4.4	GPIO port pull-up/pull-down register (GPIOx_PUPDR) (x = A to K)	538
11.4.5	GPIO port input data register (GPIOx_IDR) (x = A to K)	539
11.4.6	GPIO port output data register (GPIOx_ODR) (x = A to K)	539
11.4.7	GPIO port bit set/reset register (GPIOx_BSRR) (x = A to K)	539
11.4.8	GPIO port configuration lock register (GPIOx_LCKR) (x = A to K)	540
11.4.9	GPIO alternate function low register (GPIOx_AFRL) (x = A to K)	541
11.4.10	GPIO alternate function high register (GPIOx_AFRH) (x = A to J)	542
11.4.11	GPIO register map	544
12	System configuration controller (SYSCFG)	546
12.1	Introduction	546
12.2	SYSCFG main features	546
12.3	SYSCFG registers	546
12.3.1	SYSCFG peripheral mode configuration register (SYSCFG_PMCRR) ..	546

12.3.2	SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1)	548
12.3.3	SYSCFG external interrupt configuration register 2 (SYSCFG_EXTICR2)	549
12.3.4	SYSCFG external interrupt configuration register 3 (SYSCFG_EXTICR3)	550
12.3.5	SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4)	551
12.3.6	SYSCFG configuration register (SYSCFG_CFGR)	552
12.3.7	SYSCFG compensation cell control/status register (SYSCFG_CCCSR)	555
12.3.8	SYSCFG compensation cell value register (SYSCFG_CCVR)	556
12.3.9	SYSCFG compensation cell code register (SYSCFG_CCCR)	556
12.3.10	SYSCFG power control register (SYSCFG_PWRCR)	557
12.3.11	SYSCFG package register (SYSCFG_PKGR)	557
12.3.12	SYSCFG user register 0 (SYSCFG_UR0)	558
12.3.13	SYSCFG user register 2 (SYSCFG_UR2)	558
12.3.14	SYSCFG user register 3 (SYSCFG_UR3)	559
12.3.15	SYSCFG user register 4 (SYSCFG_UR4)	559
12.3.16	SYSCFG user register 5 (SYSCFG_UR5)	560
12.3.17	SYSCFG user register 6 (SYSCFG_UR6)	560
12.3.18	SYSCFG user register 7 (SYSCFG_UR7)	561
12.3.19	SYSCFG user register 8 (SYSCFG_UR8)	561
12.3.20	SYSCFG user register 9 (SYSCFG_UR9)	562
12.3.21	SYSCFG user register 10 (SYSCFG_UR10)	562
12.3.22	SYSCFG user register 11 (SYSCFG_UR11)	563
12.3.23	SYSCFG user register 12 (SYSCFG_UR12)	563
12.3.24	SYSCFG user register 13 (SYSCFG_UR13)	564
12.3.25	SYSCFG user register 14 (SYSCFG_UR14)	565
12.3.26	SYSCFG user register 15 (SYSCFG_UR15)	566
12.3.27	SYSCFG user register 16 (SYSCFG_UR16)	567
12.3.28	SYSCFG user register 17 (SYSCFG_UR17)	567
12.3.29	SYSCFG register maps	568

13 Block interconnect 571

13.1	Peripheral interconnect	571
13.1.1	Introduction	571
13.1.2	Connection overview	571

13.2	Wakeup from low power modes	590
13.3	DMA	595
13.3.1	MDMA (D1 domain)	596
13.3.2	DMAMUX1, DMA1 and DMA2 (D2 domain)	598
13.3.3	DMAMUX2, BDMA (D3 domain)	603
14	MDMA controller (MDMA)	606
14.1	MDMA introduction	606
14.2	MDMA main features	606
14.3	MDMA functional description	608
14.3.1	MDMA block diagram	608
14.3.2	MDMA internal signals	608
14.3.3	MDMA overview	609
14.3.4	MDMA channel	610
14.3.5	Source, destination and transfer modes	610
14.3.6	Pointer update	610
14.3.7	MDMA buffer transfer	611
14.3.8	Request arbitration	612
14.3.9	FIFO	612
14.3.10	Block transfer	613
14.3.11	Block repeat mode	613
14.3.12	Linked list mode	613
14.3.13	MDMA transfer completion	613
14.3.14	MDMA transfer suspension	614
14.3.15	Error management	614
14.4	MDMA interrupts	614
14.5	MDMA registers	615
14.5.1	MDMA global interrupt/status register (MDMA_GISR0)	615
14.5.2	MDMA channel x interrupt/status register (MDMA_CxISR)	616
14.5.3	MDMA channel x interrupt flag clear register (MDMA_CxIFCR)	617
14.5.4	MDMA channel x error status register (MDMA_CxESR)	618
14.5.5	MDMA channel x control register (MDMA_CxCR)	620
14.5.6	MDMA channel x transfer configuration register (MDMA_CxTCR)	622
14.5.7	MDMA channel x block number of data register (MDMA_CxBNDTR)	625
14.5.8	MDMA channel x source address register (MDMA_CxSAR)	627
14.5.9	MDMA channel x destination address register (MDMA_CxDAR)	627

14.5.10	MDMA channel x block repeat address update register (MDMA_CxBRUR)	629
14.5.11	MDMA channel x link address register (MDMA_CxLAR)	630
14.5.12	MDMA channel x trigger and bus selection register (MDMA_CxTBR)	631
14.5.13	MDMA channel x mask address register (MDMA_CxMAR)	632
14.5.14	MDMA channel x mask data register (MDMA_CxMDR)	632
14.5.15	MDMA register map	633
15	Direct memory access controller (DMA)	635
15.1	DMA introduction	635
15.2	DMA main features	635
15.3	DMA functional description	637
15.3.1	DMA block diagram	637
15.3.2	DMA internal signals	637
15.3.3	DMA overview	637
15.3.4	DMA transactions	638
15.3.5	DMA request mapping	638
15.3.6	Arbiter	639
15.3.7	DMA streams	639
15.3.8	Source, destination and transfer modes	639
15.3.9	Pointer incrementation	643
15.3.10	Circular mode	644
15.3.11	Double-buffer mode	644
15.3.12	Programmable data width, packing/unpacking, endianness	645
15.3.13	Single and burst transfers	647
15.3.14	FIFO	648
15.3.15	DMA transfer completion	650
15.3.16	DMA transfer suspension	651
15.3.17	Flow controller	652
15.3.18	Summary of the possible DMA configurations	653
15.3.19	Stream configuration procedure	653
15.3.20	Error management	654
15.4	DMA interrupts	655
15.5	DMA registers	656
15.5.1	DMA low interrupt status register (DMA_LISR)	656
15.5.2	DMA high interrupt status register (DMA_HISR)	657
15.5.3	DMA low interrupt flag clear register (DMA_LIFCR)	658

15.5.4	DMA high interrupt flag clear register (DMA_HIFCR)	658
15.5.5	DMA stream x configuration register (DMA_SxCR)	659
15.5.6	DMA stream x number of data register (DMA_SxNDTR)	662
15.5.7	DMA stream x peripheral address register (DMA_SxPAR)	663
15.5.8	DMA stream x memory 0 address register (DMA_SxM0AR)	663
15.5.9	DMA stream x memory 1 address register (DMA_SxM1AR)	663
15.5.10	DMA stream x FIFO control register (DMA_SxFCR)	664
15.5.11	DMA register map	666
16	Basic direct memory access controller (BDMA)	670
16.1	Introduction	670
16.2	BDMA main features	670
16.3	BDMA implementation	671
16.3.1	BDMA	671
16.3.2	BDMA request mapping	671
16.4	BDMA functional description	671
16.4.1	BDMA block diagram	671
16.4.2	BDMA pins and internal signals	672
16.4.3	BDMA transfers	672
16.4.4	BDMA arbitration	673
16.4.5	BDMA channels	673
16.4.6	BDMA data width, alignment and endianness	678
16.4.7	BDMA error management	679
16.5	BDMA interrupts	680
16.6	BDMA registers	680
16.6.1	BDMA interrupt status register (BDMA_ISR)	680
16.6.2	BDMA interrupt flag clear register (BDMA_IFCR)	683
16.6.3	BDMA channel x configuration register (BDMA_CCRx)	684
16.6.4	BDMA channel x number of data to transfer register (BDMA_CNDTRx)	688
16.6.5	BDMA channel x peripheral address register (BDMA_CPARx)	688
16.6.6	BDMA channel x memory 0 address register (BDMA_CM0ARx)	689
16.6.7	BDMA channel x memory 1 address register (BDMA_CM1ARx)	690
16.6.8	BDMA register map	690

17	DMA request multiplexer (DMAMUX)	693
17.1	Introduction	693
17.2	DMAMUX main features	694
17.3	DMAMUX implementation	694
17.3.1	DMAMUX1 and DMAMUX2 instantiation	694
17.3.2	DMAMUX1 mapping	694
17.3.3	DMAMUX2 mapping	696
17.4	DMAMUX functional description	699
17.4.1	DMAMUX block diagram	699
17.4.2	DMAMUX signals	700
17.4.3	DMAMUX channels	700
17.4.4	DMAMUX request line multiplexer	700
17.4.5	DMAMUX request generator	703
17.5	DMAMUX interrupts	704
17.6	DMAMUX registers	705
17.6.1	DMAMUX1 request line multiplexer channel x configuration register (DMAMUX1_CxCR)	705
17.6.2	DMAMUX2 request line multiplexer channel x configuration register (DMAMUX2_CxCR)	706
17.6.3	DMAMUX1 request line multiplexer interrupt channel status register (DMAMUX1_CSR)	707
17.6.4	DMAMUX2 request line multiplexer interrupt channel status register (DMAMUX2_CSR)	707
17.6.5	DMAMUX1 request line multiplexer interrupt clear flag register (DMAMUX1_CFR)	708
17.6.6	DMAMUX2 request line multiplexer interrupt clear flag register (DMAMUX2_CFR)	708
17.6.7	DMAMUX1 request generator channel x configuration register (DMAMUX1_RGxCR)	709
17.6.8	DMAMUX2 request generator channel x configuration register (DMAMUX2_RGxCR)	709
17.6.9	DMAMUX1 request generator interrupt status register (DMAMUX1_RGSR)	710
17.6.10	DMAMUX2 request generator interrupt status register (DMAMUX2_RGSR)	711
17.6.11	DMAMUX1 request generator interrupt clear flag register (DMAMUX1_RGCFR)	711
17.6.12	DMAMUX2 request generator interrupt clear flag register (DMAMUX2_RGCFR)	712
17.6.13	DMAMUX register map	713

18	Chrom-Art Accelerator™ controller (DMA2D)	715
18.1	DMA2D introduction	715
18.2	DMA2D main features	715
18.3	DMA2D functional description	716
18.3.1	General description	716
18.4	DMA2D pins and internal signals	717
18.4.1	DMA2D control	718
18.4.2	DMA2D foreground and background FIFOs	718
18.4.3	DMA2D foreground and background pixel format converter (PFC)	718
18.4.4	DMA2D foreground and background CLUT interface	721
18.4.5	DMA2D blender	722
18.4.6	DMA2D output PFC	722
18.4.7	DMA2D output FIFO	723
18.4.8	DMA2D output FIFO byte reordering	723
18.4.9	DMA2D AXI master port timer	725
18.4.10	DMA2D transactions	725
18.4.11	DMA2D configuration	726
18.4.12	YCbCr support	730
18.4.13	DMA2D transfer control (start, suspend, abort and completion)	730
18.4.14	Watermark	730
18.4.15	Error management	731
18.4.16	AXI dead time	731
18.5	DMA2D interrupts	731
18.6	DMA2D registers	732
18.6.1	DMA2D control register (DMA2D_CR)	732
18.6.2	DMA2D interrupt status register (DMA2D_ISR)	734
18.6.3	DMA2D interrupt flag clear register (DMA2D_IFCR)	735
18.6.4	DMA2D foreground memory address register (DMA2D_FGMAR)	736
18.6.5	DMA2D foreground offset register (DMA2D_FGOR)	736
18.6.6	DMA2D background memory address register (DMA2D_BGMAR)	737
18.6.7	DMA2D background offset register (DMA2D_BGOR)	737
18.6.8	DMA2D foreground PFC control register (DMA2D_FGPFCCR)	738
18.6.9	DMA2D foreground color register (DMA2D_FGCOLOR)	740
18.6.10	DMA2D background PFC control register (DMA2D_BGPFCCR)	741
18.6.11	DMA2D background color register (DMA2D_BGCOLOR)	743

18.6.12	DMA2D foreground CLUT memory address register (DMA2D_FGCMAR)	743
18.6.13	DMA2D background CLUT memory address register (DMA2D_BGCMAR)	744
18.6.14	DMA2D output PFC control register (DMA2D_OPFCCR)	744
18.6.15	DMA2D output color register (DMA2D_OCOLR)	745
18.6.16	DMA2D output memory address register (DMA2D_OMAR)	747
18.6.17	DMA2D output offset register (DMA2D_OOR)	747
18.6.18	DMA2D number of line register (DMA2D_NLR)	748
18.6.19	DMA2D line watermark register (DMA2D_LWR)	748
18.6.20	DMA2D AXI master timer configuration register (DMA2D_AMTCR) ..	749
18.6.21	DMA2D foreground CLUT (DMA2D_FGCLUT[y])	749
18.6.22	DMA2D background CLUT (DMA2D_BGCLUT[y])	750
18.6.23	DMA2D register map	751
19	Nested vectored interrupt controller (NVIC)	753
19.1	NVIC features	753
19.1.1	SysTick calibration value register	753
19.1.2	Interrupt and exception vectors	754
20	Extended interrupt and event controller (EXTI)	762
20.1	EXTI main features	762
20.2	EXTI block diagram	762
20.2.1	EXTI connections between peripherals, CPU, and D3 domain	763
20.3	EXTI functional description	764
20.3.1	EXTI Configurable event input CPU wakeup	765
20.3.2	EXTI configurable event input Any wakeup	766
20.3.3	EXTI direct event input CPU wakeup	768
20.3.4	EXTI direct event input Any wakeup	769
20.3.5	EXTI D3 pending request clear selection	770
20.4	EXTI event input mapping	770
20.5	EXTI functional behavior	773
20.5.1	EXTI CPU interrupt procedure	774
20.5.2	EXTI CPU event procedure	774
20.5.3	EXTI CPU wakeup procedure	775
20.5.4	EXTI D3 domain wakeup for autonomous Run mode procedure	775
20.5.5	EXTI software interrupt/event trigger procedure	775

20.6	EXTI registers	776
20.6.1	EXTI rising trigger selection register (EXTI_RTISR1)	776
20.6.2	EXTI falling trigger selection register (EXTI_FTSR1)	776
20.6.3	EXTI software interrupt event register (EXTI_SWIER1)	777
20.6.4	EXTI D3 pending mask register (EXTI_D3PMR1)	777
20.6.5	EXTI D3 pending clear selection register low (EXTI_D3PCR1L)	778
20.6.6	EXTI D3 pending clear selection register high (EXTI_D3PCR1H)	778
20.6.7	EXTI rising trigger selection register (EXTI_RTISR2)	779
20.6.8	EXTI falling trigger selection register (EXTI_FTSR2)	780
20.6.9	EXTI software interrupt event register (EXTI_SWIER2)	780
20.6.10	EXTI D3 pending mask register (EXTI_D3PMR2)	781
20.6.11	EXTI D3 pending clear selection register low (EXTI_D3PCR2L)	782
20.6.12	EXTI D3 pending clear selection register high (EXTI_D3PCR2H)	782
20.6.13	EXTI rising trigger selection register (EXTI_RTISR3)	783
20.6.14	EXTI falling trigger selection register (EXTI_FTSR3)	783
20.6.15	EXTI software interrupt event register (EXTI_SWIER3)	784
20.6.16	EXTI D3 pending mask register (EXTI_D3PMR3)	784
20.6.17	EXTI D3 pending clear selection register low (EXTI_D3PCR3L)	785
20.6.18	EXTI D3 pending clear selection register high (EXTI_D3PCR3H)	785
20.6.19	EXTI interrupt mask register (EXTI_CPUIMR1)	786
20.6.20	EXTI event mask register (EXTI_CPUEMR1)	786
20.6.21	EXTI pending register (EXTI_CPUPR1)	787
20.6.22	EXTI interrupt mask register (EXTI_CPUIMR2)	787
20.6.23	EXTI event mask register (EXTI_CPUEMR2)	788
20.6.24	EXTI pending register (EXTI_CPUPR2)	788
20.6.25	EXTI interrupt mask register (EXTI_CPUIMR3)	789
20.6.26	EXTI event mask register (EXTI_CPUEMR3)	790
20.6.27	EXTI pending register (EXTI_CPUPR3)	790
20.6.28	EXTI register map	791
21	Cyclic redundancy check calculation unit (CRC)	794
21.1	Introduction	794
21.2	CRC main features	794
21.3	CRC functional description	795
21.3.1	CRC block diagram	795
21.3.2	CRC internal signals	795
21.3.3	CRC operation	795

21.4	CRC registers	797
21.4.1	CRC data register (CRC_DR)	797
21.4.2	CRC independent data register (CRC_IDR)	797
21.4.3	CRC control register (CRC_CR)	798
21.4.4	CRC initial value (CRC_INIT)	798
21.4.5	CRC polynomial (CRC_POL)	799
21.4.6	CRC register map	799
22	Flexible memory controller (FMC)	800
22.1	FMC main features	800
22.2	FMC block diagram	801
22.3	FMC internal signals	803
22.4	AHB interface	803
22.5	AXI interface	803
22.5.1	Supported memories and transactions	804
22.6	External device address mapping	805
22.6.1	NOR/PSRAM address mapping	806
22.6.2	NAND Flash memory address mapping	807
22.6.3	SDRAM address mapping	807
22.7	NOR Flash/PSRAM controller	811
22.7.1	External memory interface signals	812
22.7.2	Supported memories and transactions	814
22.7.3	General timing rules	815
22.7.4	NOR Flash/PSRAM controller asynchronous transactions	816
22.7.5	Synchronous transactions	835
22.7.6	NOR/PSRAM controller registers	841
22.8	NAND Flash controller	850
22.8.1	External memory interface signals	850
22.8.2	NAND Flash supported memories and transactions	851
22.8.3	Timing diagrams for NAND Flash memories	852
22.8.4	NAND Flash operations	853
22.8.5	NAND Flash prewait feature	854
22.8.6	Computation of the error correction code (ECC) in NAND Flash memory	855
22.8.7	NAND Flash controller registers	856
22.9	SDRAM controller	862

22.9.1	SDRAM controller main features	862
22.9.2	SDRAM External memory interface signals	862
22.9.3	SDRAM controller functional description	863
22.9.4	Low-power modes	870
22.9.5	SDRAM controller registers	873
22.9.6	FMC register map	879
23	Quad-SPI interface (QUADSPI)	882
23.1	Introduction	882
23.2	QUADSPI main features	882
23.3	QUADSPI functional description	883
23.3.1	QUADSPI block diagram	883
23.3.2	QUADSPI pins and internal signals	884
23.3.3	QUADSPI command sequence	884
23.3.4	QUADSPI signal interface protocol modes	887
23.3.5	QUADSPI indirect mode	889
23.3.6	QUADSPI status flag polling mode	891
23.3.7	QUADSPI memory-mapped mode	891
23.3.8	QUADSPI Free running clock mode	892
23.3.9	QUADSPI Flash memory configuration	892
23.3.10	QUADSPI delayed data sampling	892
23.3.11	QUADSPI configuration	893
23.3.12	QUADSPI usage	893
23.3.13	Sending the instruction only once	895
23.3.14	QUADSPI error management	896
23.3.15	QUADSPI busy bit and abort functionality	896
23.3.16	nCS behavior	896
23.4	QUADSPI interrupts	898
23.5	QUADSPI registers	899
23.5.1	QUADSPI control register (QUADSPI_CR)	899
23.5.2	QUADSPI device configuration register (QUADSPI_DCR)	902
23.5.3	QUADSPI status register (QUADSPI_SR)	903
23.5.4	QUADSPI flag clear register (QUADSPI_FCR)	904
23.5.5	QUADSPI data length register (QUADSPI_DLR)	904
23.5.6	QUADSPI communication configuration register (QUADSPI_CCR)	905
23.5.7	QUADSPI address register (QUADSPI_AR)	907
23.5.8	QUADSPI alternate bytes registers (QUADSPI_ABR)	908

	23.5.9	QUADSPI data register (QUADSPI_DR)	908
	23.5.10	QUADSPI polling status mask register (QUADSPI_PSMKR)	909
	23.5.11	QUADSPI polling status match register (QUADSPI_PSMAR)	909
	23.5.12	QUADSPI polling interval register (QUADSPI_PIR)	910
	23.5.13	QUADSPI low-power timeout register (QUADSPI_LPTR)	910
	23.5.14	QUADSPI register map	912
24		Delay block (DLYB)	913
	24.1	Introduction	913
	24.2	DLYB main features	913
	24.3	DLYB functional description	913
	24.3.1	DLYB diagram	913
	24.3.2	DLYB pins and internal signals	914
	24.3.3	General description	914
	24.3.4	Delay line length configuration procedure	915
	24.3.5	Output clock phase configuration procedure	915
	24.4	DLYB registers	916
	24.4.1	DLYB control register (DLYB_CR)	916
	24.4.2	DLYB configuration register (DLYB_CFGR)	916
	24.4.3	DLYB register map	917
25		Analog-to-digital converters (ADC)	918
	25.1	Introduction	918
	25.2	ADC main features	919
	25.3	ADC implementation	920
	25.4	ADC functional description	921
	25.4.1	ADC block diagram	921
	25.4.2	ADC pins and internal signals	922
	25.4.3	ADC clocks	923
	25.4.4	ADC1/2/3 connectivity	927
	25.4.5	Slave AHB interface	930
	25.4.6	ADC deep-power-down mode (DEEPPWD) and ADC voltage regulator (ADVREGEN)	930
	25.4.7	Single-ended and differential input channels	931
	25.4.8	Calibration (ADCAL, ADCALDIF, ADCALLIN, ADC_CALFACT)	931
	25.4.9	ADC on-off control (ADEN, ADDIS, ADRDY)	937
	25.4.10	Constraints when writing the ADC control bits	938

25.4.11	Channel selection (SQRx, JSQRx)	938
25.4.12	Channel preselection register (ADC_PCSEL)	939
25.4.13	Channel-wise programmable sampling time (SMPR1, SMPR2)	940
25.4.14	Single conversion mode (CONT=0)	940
25.4.15	Continuous conversion mode (CONT=1)	941
25.4.16	Starting conversions (ADSTART, JADSTART)	942
25.4.17	Timing	943
25.4.18	Stopping an ongoing conversion (ADSTP, JADSTP)	943
25.4.19	Conversion on external trigger and trigger polarity (EXTSEL, EXTEN, JEXTSEL, JEXTEN)	945
25.4.20	Injected channel management	948
25.4.21	Discontinuous mode (DISCEN, DISCNUM, JDISCEN)	950
25.4.22	Queue of context for injected conversions	951
25.4.23	Programmable resolution (RES) - fast conversion mode	959
25.4.24	End of conversion, end of sampling phase (EOC, JEOC, EOSMP)	959
25.4.25	End of conversion sequence (EOS, JEOS)	959
25.4.26	Timing diagrams example (single/continuous modes, hardware/software triggers)	960
25.4.27	Data management	961
25.4.28	Managing conversions using the DFSDM	968
25.4.29	Dynamic low-power features	969
25.4.30	Analog window watchdog (AWD1EN, JAWD1EN, AWD1SGL, AWD1CH, AWD2CH, AWD3CH, AWD_HTRy, AWD_LTRy, AWDy)	974
25.4.31	Oversampler	977
25.4.32	Dual ADC modes	983
25.4.33	Temperature sensor	998
25.4.34	VBAT supply monitoring	999
25.4.35	Monitoring the internal voltage reference	1000
25.5	ADC interrupts	1002
25.6	ADC registers (for each ADC)	1003
25.6.1	ADC interrupt and status register (ADC_ISR)	1003
25.6.2	ADC interrupt enable register (ADC_IER)	1006
25.6.3	ADC control register (ADC_CR)	1008
25.6.4	ADC configuration register (ADC_CFGR)	1017
25.6.5	ADC configuration register 2 (ADC_CFGR2)	1021
25.6.6	ADC sample time register 1 (ADC_SMPR1)	1023
25.6.7	ADC sample time register 2 (ADC_SMPR2)	1024
25.6.8	ADC channel preselection register (ADC_PCSEL)	1025

25.6.9	ADC watchdog threshold register 1 (ADC_LTR1)	1025
25.6.10	ADC watchdog threshold register 1 (ADC_HTR1)	1026
25.6.11	ADC regular sequence register 1 (ADC_SQR1)	1027
25.6.12	ADC regular sequence register 2 (ADC_SQR2)	1028
25.6.13	ADC regular sequence register 3 (ADC_SQR3)	1029
25.6.14	ADC regular sequence register 4 (ADC_SQR4)	1030
25.6.15	ADC regular Data Register (ADC_DR)	1031
25.6.16	ADC injected sequence register (ADC_JSQR)	1032
25.6.17	ADC injected channel y offset register (ADC_OFRy)	1034
25.6.18	ADC injected channel y data register (ADC_JDRy)	1035
25.6.19	ADC analog watchdog 2 configuration register (ADC_AWD2CR)	1035
25.6.20	ADC analog watchdog 3 configuration register (ADC_AWD3CR)	1036
25.6.21	ADC watchdog lower threshold register 2 (ADC_LTR2)	1036
25.6.22	ADC watchdog higher threshold register 2 (ADC_HTR2)	1037
25.6.23	ADC watchdog lower threshold register 3 (ADC_LTR3)	1037
25.6.24	ADC watchdog higher threshold register 3 (ADC_HTR3)	1038
25.6.25	ADC differential mode selection register (ADC_DIFSEL)	1038
25.6.26	ADC calibration factors register (ADC_CALFACT)	1039
25.6.27	ADC calibration factor register 2 (ADC_CALFACT2)	1039
25.7	ADC common registers	1040
25.7.1	ADC x common status register (ADCx_CSR) (x=1/2 or 3)	1040
25.7.2	ADC x common control register (ADCx_CCR) (x=1/2 or 3)	1042
25.7.3	ADC x common regular data register for dual mode (ADCx_CDR) (x=1/2 or 3)	1046
25.7.4	ADC x common regular data register for 32-bit dual mode (ADCx_CDR2) (x=1/2 or 3)	1046
25.8	ADC register map	1047
26	Digital-to-analog converter (DAC)	1051
26.1	Introduction	1051
26.2	DAC main features	1051
26.3	DAC implementation	1052
26.4	DAC functional description	1053
26.4.1	DAC block diagram	1053
26.4.2	DAC pins and internal signals	1054
26.4.3	DAC channel enable	1055

26.4.4	DAC data format	1055
26.4.5	DAC conversion	1057
26.4.6	DAC output voltage	1057
26.4.7	DAC trigger selection	1057
26.4.8	DMA requests	1058
26.4.9	DAC noise generation	1058
26.4.10	DAC triangle-wave generation	1060
26.4.11	DAC channel modes	1061
26.4.12	DAC channel buffer calibration	1064
26.4.13	Dual DAC channel conversion modes (if dual channels are available)	1065
26.5	DAC low-power modes	1069
26.6	DAC interrupts	1070
26.7	DAC registers	1071
26.7.1	DAC control register (DAC_CR)	1071
26.7.2	DAC software trigger register (DAC_SWTRGR)	1074
26.7.3	DAC channel1 12-bit right-aligned data holding register (DAC_DHR12R1)	1075
26.7.4	DAC channel1 12-bit left aligned data holding register (DAC_DHR12L1)	1075
26.7.5	DAC channel1 8-bit right aligned data holding register (DAC_DHR8R1)	1076
26.7.6	DAC channel2 12-bit right aligned data holding register (DAC_DHR12R2)	1076
26.7.7	DAC channel2 12-bit left aligned data holding register (DAC_DHR12L2)	1077
26.7.8	DAC channel2 8-bit right-aligned data holding register (DAC_DHR8R2)	1077
26.7.9	Dual DAC 12-bit right-aligned data holding register (DAC_DHR12RD)	1078
26.7.10	Dual DAC 12-bit left aligned data holding register (DAC_DHR12LD)	1078
26.7.11	Dual DAC 8-bit right aligned data holding register (DAC_DHR8RD)	1079
26.7.12	DAC channel1 data output register (DAC_DOR1)	1079
26.7.13	DAC channel2 data output register (DAC_DOR2)	1080
26.7.14	DAC status register (DAC_SR)	1080
26.7.15	DAC calibration control register (DAC_CCR)	1082
26.7.16	DAC mode control register (DAC_MCR)	1082

	26.7.17	DAC channel1 sample and hold sample time register (DAC_SHSR1)	1084
	26.7.18	DAC channel2 sample and hold sample time register (DAC_SHSR2)	1084
	26.7.19	DAC sample and hold time register (DAC_SHHR)	1085
	26.7.20	DAC sample and hold refresh time register (DAC_SHRR)	1085
	26.7.21	DAC register map	1087
27		Voltage reference buffer (VREFBUF)	1089
	27.1	Introduction	1089
	27.2	VREFBUF functional description	1089
	27.3	VREFBUF registers	1090
	27.3.1	VREFBUF control and status register (VREFBUF_CSR)	1090
	27.3.2	VREFBUF calibration control register (VREFBUF_CCR)	1091
	27.3.3	VREFBUF register map	1091
28		Comparator (COMP)	1092
	28.1	Introduction	1092
	28.2	COMP main features	1092
	28.3	COMP functional description	1093
	28.3.1	COMP block diagram	1093
	28.3.2	COMP pins and internal signals	1093
	28.3.3	COMP reset and clocks	1095
	28.3.4	Comparator LOCK mechanism	1095
	28.3.5	Window comparator	1095
	28.3.6	Hysteresis	1095
	28.3.7	Comparator output blanking function	1096
	28.3.8	Comparator output on GPIOs	1097
	28.3.9	Comparator output redirection	1098
	28.3.10	COMP power and speed modes	1098
	28.4	COMP low-power modes	1099
	28.5	COMP interrupts	1099
	28.5.1	Interrupt through EXTI block	1099
	28.5.2	Interrupt through NVIC of the CPU	1100
	28.6	SCALER function	1100
	28.7	COMP registers	1101
	28.7.1	Comparator status register (COMP_SR)	1101

28.7.2	Comparator interrupt clear flag register (COMP_ICFR)	1101
28.7.3	Comparator option register (COMP_OR)	1102
28.7.4	Comparator configuration register 1 (COMP_CFGR1)	1102
28.7.5	Comparator configuration register 2 (COMP_CFGR2)	1104
28.7.6	COMP register map	1107
29	Operational amplifiers (OPAMP)	1108
29.1	Introduction	1108
29.2	OPAMP main features	1108
29.3	OPAMP functional description	1108
29.3.1	OPAMP reset and clocks	1108
29.3.2	Initial configuration	1109
29.3.3	Signal routing	1109
29.3.4	OPAMP modes	1110
29.3.5	Calibration	1116
29.4	OPAMP low-power modes	1118
29.5	OPAMP PGA gain	1118
29.6	OPAMP registers	1118
29.6.1	OPAMP1 control/status register (OPAMP1_CSR)	1118
29.6.2	OPAMP1 trimming register in normal mode (OPAMP1_OTR)	1120
29.6.3	OPAMP1 trimming register in high-speed mode (OPAMP1_HSOTR)	1121
29.6.4	OPAMP option register (OPAMP_OR)	1121
29.6.5	OPAMP2 control/status register (OPAMP2_CSR)	1121
29.6.6	OPAMP2 trimming register in normal mode (OPAMP2_OTR)	1123
29.6.7	OPAMP2 trimming register in high-speed mode (OPAMP2_HSOTR)	1124
29.6.8	OPAMP register map	1125
30	Digital filter for sigma delta modulators (DFSDM)	1126
30.1	Introduction	1126
30.2	DFSDM main features	1127
30.3	DFSDM implementation	1128
30.4	DFSDM functional description	1129
30.4.1	DFSDM block diagram	1129
30.4.2	DFSDM pins and internal signals	1130
30.4.3	DFSDM reset and clocks	1131
30.4.4	Serial channel transceivers	1132

30.4.5	Configuring the input serial interface	1142
30.4.6	Parallel data inputs	1142
30.4.7	Channel selection	1144
30.4.8	Digital filter configuration	1145
30.4.9	Integrator unit	1146
30.4.10	Analog watchdog	1147
30.4.11	Short-circuit detector	1149
30.4.12	Extreme detector	1150
30.4.13	Data unit block	1150
30.4.14	Signed data format	1151
30.4.15	Launching conversions	1152
30.4.16	Continuous and fast continuous modes	1152
30.4.17	Request precedence	1153
30.4.18	Power optimization in run mode	1154
30.5	DFSDM interrupts	1154
30.6	DFSDM DMA transfer	1156
30.7	DFSDM channel y registers (y=0..7)	1156
30.7.1	DFSDM channel y configuration register (DFSDM_CHyCFGR1)	1156
30.7.2	DFSDM channel y configuration register (DFSDM_CHyCFGR2)	1158
30.7.3	DFSDM channel y analog watchdog and short-circuit detector register (DFSDM_CHyAWSCDR)	1159
30.7.4	DFSDM channel y watchdog filter data register (DFSDM_CHyWDATR)	1160
30.7.5	DFSDM channel y data input register (DFSDM_CHyDATINR)	1160
30.8	DFSDM filter x module registers (x=0..3)	1161
30.8.1	DFSDM filter x control register 1 (DFSDM_FLTxCR1)	1161
30.8.2	DFSDM filter x control register 2 (DFSDM_FLTxCR2)	1164
30.8.3	DFSDM filter x interrupt and status register (DFSDM_FLTxISR)	1165
30.8.4	DFSDM filter x interrupt flag clear register (DFSDM_FLTxICR)	1167
30.8.5	DFSDM filter x injected channel group selection register (DFSDM_FLTxJCHGR)	1168
30.8.6	DFSDM filter x control register (DFSDM_FLTxFCR)	1168
30.8.7	DFSDM filter x data register for injected group (DFSDM_FLTxJDATAR)	1169
30.8.8	DFSDM filter x data register for the regular channel (DFSDM_FLTxRDATAR)	1170
30.8.9	DFSDM filter x analog watchdog high threshold register (DFSDM_FLTxAWHTR)	1171

30.8.10	DFSDM filter x analog watchdog low threshold register (DFSDM_FLTxAWLTR)	1171
30.8.11	DFSDM filter x analog watchdog status register (DFSDM_FLTxAWSR)	1172
30.8.12	DFSDM filter x analog watchdog clear flag register (DFSDM_FLTxAWCFR)	1173
30.8.13	DFSDM filter x extremes detector maximum register (DFSDM_FLTxEXMAX)	1173
30.8.14	DFSDM filter x extremes detector minimum register (DFSDM_FLTxEXMIN)	1174
30.8.15	DFSDM filter x conversion timer register (DFSDM_FLTxCNVTIMR) .	1174
30.8.16	DFSDM register map	1175
31	Digital camera interface (DCMI)	1185
31.1	Introduction	1185
31.2	DCMI main features	1185
31.3	DCMI functional description	1185
31.3.1	DCMI block diagram	1186
31.3.2	DCMI pins and internal signals	1186
31.3.3	DCMI clocks	1187
31.3.4	DCMI DMA interface	1187
31.3.5	DCMI physical interface	1187
31.3.6	DCMI synchronization	1189
31.3.7	DCMI capture modes	1191
31.3.8	DCMI crop feature	1192
31.3.9	DCMI JPEG format	1193
31.3.10	DCMI FIFO	1193
31.3.11	DCMI data format description	1194
31.4	DCMI interrupts	1196
31.5	DCMI registers	1197
31.5.1	DCMI control register (DCMI_CR)	1197
31.5.2	DCMI status register (DCMI_SR)	1199
31.5.3	DCMI raw interrupt status register (DCMI_RIS)	1200
31.5.4	DCMI interrupt enable register (DCMI_IER)	1201
31.5.5	DCMI masked interrupt status register (DCMI_MIS)	1202
31.5.6	DCMI interrupt clear register (DCMI_ICR)	1203
31.5.7	DCMI embedded synchronization code register (DCMI_ESCR)	1203
31.5.8	DCMI embedded synchronization unmask register (DCMI_ESUR) .	1205

31.5.9	DCMI crop window start (DCMI_CWSTRT)	1205
31.5.10	DCMI crop window size (DCMI_CWSIZE)	1206
31.5.11	DCMI data register (DCMI_DR)	1206
31.5.12	DCMI register map	1208
32	LCD-TFT display controller (LTDC)	1209
32.1	Introduction	1209
32.2	LTDC main features	1209
32.3	LTDC functional description	1210
32.3.1	LTDC block diagram	1210
32.3.2	LCD-TFT internal signals	1210
32.3.3	LTDC pins and external signal interface	1211
32.3.4	LTDC reset and clocks	1211
32.4	LTDC programmable parameters	1213
32.4.1	LTDC global configuration parameters	1213
32.4.2	Layer programmable parameters	1216
32.5	LTDC interrupts	1221
32.6	LTDC programming procedure	1222
32.7	LTDC registers	1223
32.7.1	LTDC synchronization size configuration register (LTDC_SSCR) ...	1223
32.7.2	LTDC back porch configuration register (LTDC_BPCR)	1223
32.7.3	LTDC active width configuration register (LTDC_AWCR)	1224
32.7.4	LTDC total width configuration register (LTDC_TWCR)	1225
32.7.5	LTDC global control register (LTDC_GCR)	1225
32.7.6	LTDC shadow reload configuration register (LTDC_SRCR)	1227
32.7.7	LTDC background color configuration register (LTDC_BCCR)	1227
32.7.8	LTDC interrupt enable register (LTDC_IER)	1228
32.7.9	LTDC interrupt status register (LTDC_ISR)	1229
32.7.10	LTDC Interrupt Clear Register (LTDC_ICR)	1229
32.7.11	LTDC line interrupt position configuration register (LTDC_LIPCR) ..	1230
32.7.12	LTDC current position status register (LTDC_CPSR)	1230
32.7.13	LTDC current display status register (LTDC_CDSR)	1231
32.7.14	LTDC layer x control register (LTDC_LxCR)	1231
32.7.15	LTDC layer x window horizontal position configuration register (LTDC_LxWHPCR)	1232
32.7.16	LTDC layer x window vertical position configuration register (LTDC_LxWVPCR)	1233

32.7.17	LTDC layer x color keying configuration register (LTDC_LxCKCR)	1234
32.7.18	LTDC layer x pixel format configuration register (LTDC_LxPFCR)	1234
32.7.19	LTDC layer x constant alpha configuration register (LTDC_LxCACR)	1235
32.7.20	LTDC layer x default color configuration register (LTDC_LxDCCR)	1236
32.7.21	LTDC layer x blending factors configuration register (LTDC_LxBFCR)	1237
32.7.22	LTDC layer x color frame buffer address register (LTDC_LxCFBAR)	1238
32.7.23	LTDC layer x color frame buffer length register (LTDC_LxCFBLR)	1238
32.7.24	LTDC layer x color frame buffer line number register (LTDC_LxCFBLNR)	1239
32.7.25	LTDC layer x CLUT write register (LTDC_LxCLUTWR)	1240
32.7.26	LTDC register map	1241
33	JPEG codec (JPEG)	1244
33.1	Introduction	1244
33.2	JPEG codec main features	1244
33.3	JPEG codec block functional description	1245
33.3.1	General description	1245
33.3.2	JPEG internal signals	1245
33.3.3	JPEG decoding procedure	1246
33.3.4	JPEG encoding procedure	1247
33.4	JPEG codec interrupts	1249
33.5	JPEG codec registers	1250
33.5.1	JPEG codec control register (JPEG_CONFR0)	1250
33.5.2	JPEG codec configuration register 1 (JPEG_CONFR1)	1250
33.5.3	JPEG codec configuration register 2 (JPEG_CONFR2)	1251
33.5.4	JPEG codec configuration register 3 (JPEG_CONFR3)	1252
33.5.5	JPEG codec configuration register 4-7 (JPEG_CONFR4-7)	1252
33.5.6	JPEG control register (JPEG_CR)	1253
33.5.7	JPEG status register (JPEG_SR)	1254
33.5.8	JPEG clear flag register (JPEG_CFR)	1255
33.5.9	JPEG data input register (JPEG_DIR)	1256
33.5.10	JPEG data output register (JPEG_DOR)	1256

	33.5.11 JPEG quantization memory x (JPEG_QMEMx[y])	1257
	33.5.12 JPEG Huffman min (JPEG_HUFFMIN[4*y+x])	1257
	33.5.13 JPEG Huffman base (JPEG_HUFFBASE[y])	1258
	33.5.14 JPEG Huffman symbol (JPEG_HUFFSYMB[y])	1259
	33.5.15 JPEG DHT memory (JPEG_DHTMEM[y])	1259
	33.5.16 JPEG Huffman encoder ACx (JPEG_HUFFENC_ACx[y])	1260
	33.5.17 JPEG Huffman encoder DCx (JPEG_HUFFENC_DCx[y])	1260
	33.5.18 JPEG codec register map	1262
34	True random number generator (RNG)	1264
34.1	Introduction	1264
34.2	RNG main features	1264
34.3	RNG functional description	1265
34.3.1	RNG block diagram	1265
34.3.2	RNG internal signals	1265
34.3.3	Random number generation	1266
34.3.4	RNG initialization	1268
34.3.5	RNG operation	1269
34.3.6	RNG clocking	1270
34.3.7	Error management	1270
34.3.8	RNG low-power usage	1271
34.4	RNG interrupts	1272
34.5	RNG processing time	1272
34.6	RNG entropy source validation	1272
34.6.1	Introduction	1272
34.6.2	Validation conditions	1272
34.6.3	Data collection	1273
34.7	RNG registers	1274
34.7.1	RNG control register (RNG_CR)	1274
34.7.2	RNG status register (RNG_SR)	1275
34.7.3	RNG data register (RNG_DR)	1276
34.7.4	RNG register map	1277
35	Cryptographic processor (CRYP)	1278
35.1	Introduction	1278
35.2	CRYP main features	1278

35.3	CRYP functional description	1280
35.3.1	CRYP block diagram	1280
35.3.2	CRYP internal signals	1281
35.3.3	CRYP DES/TDES cryptographic core	1281
35.3.4	CRYP AES cryptographic core	1282
35.3.5	CRYP procedure to perform a cipher operation	1288
35.3.6	CRYP busy state	1290
35.3.7	Preparing the CRYP AES key for decryption	1290
35.3.8	CRYP stealing and data padding	1291
35.3.9	CRYP suspend/resume operations	1293
35.3.10	CRYP DES/TDES basic chaining modes (ECB, CBC)	1294
35.3.11	CRYP AES basic chaining modes (ECB, CBC)	1299
35.3.12	CRYP AES counter mode (AES-CTR)	1304
35.3.13	CRYP AES Galois/counter mode (GCM)	1308
35.3.14	CRYP AES Galois message authentication code (GMAC)	1313
35.3.15	CRYP AES Counter with CBC-MAC (CCM)	1314
35.3.16	CRYP data registers and data swapping	1320
35.3.17	CRYP key registers	1324
35.3.18	CRYP initialization vector registers	1325
35.3.19	CRYP DMA interface	1326
35.3.20	CRYP error management	1328
35.4	CRYP interrupts	1328
35.5	CRYP processing time	1330
35.6	CRYP registers	1331
35.6.1	CRYP control register (CRYP_CR)	1331
35.6.2	CRYP status register (CRYP_SR)	1333
35.6.3	CRYP data input register (CRYP_DIN)	1333
35.6.4	CRYP data output register (CRYP_DOUT)	1334
35.6.5	CRYP DMA control register (CRYP_DMACR)	1335
35.6.6	CRYP interrupt mask set/clear register (CRYP_IMSCR)	1335
35.6.7	CRYP raw interrupt status register (CRYP_RISR)	1336
35.6.8	CRYP masked interrupt status register (CRYP_MISR)	1336
35.6.9	CRYP key register 0L (CRYP_K0LR)	1337
35.6.10	CRYP key register 0R (CRYP_K0RR)	1338
35.6.11	CRYP key register 1L (CRYP_K1LR)	1338
35.6.12	CRYP key register 1R (CRYP_K1RR)	1339
35.6.13	CRYP key register 2L (CRYP_K2LR)	1339

35.6.14	CRYP key register 2R (CRYP_K2RR)	1339
35.6.15	CRYP key register 3L (CRYP_K3LR)	1340
35.6.16	CRYP key register 3R (CRYP_K3RR)	1340
35.6.17	CRYP initialization vector register 0L (CRYP_IV0LR)	1341
35.6.18	CRYP initialization vector register 0R (CRYP_IV0RR)	1341
35.6.19	CRYP initialization vector register 1L (CRYP_IV1LR)	1342
35.6.20	CRYP initialization vector register 1R (CRYP_IV1RR)	1342
35.6.21	CRYP context swap GCM-CCM registers (CRYP_CSGCMCCMxR)	1342
35.6.22	CRYP context swap GCM registers (CRYP_CSGCMxR)	1343
35.6.23	CRYP register map	1344
36	Hash processor (HASH)	1347
36.1	Introduction	1347
36.2	HASH main features	1347
36.3	HASH implementation	1348
36.4	HASH functional description	1348
36.4.1	HASH block diagram	1348
36.4.2	HASH internal signals	1349
36.4.3	About secure hash algorithms	1349
36.4.4	Message data feeding	1349
36.4.5	Message digest computing	1351
36.4.6	Message padding	1352
36.4.7	HMAC operation	1354
36.4.8	HASH suspend/resume operations	1356
36.4.9	HASH DMA interface	1358
36.4.10	HASH error management	1358
36.5	HASH interrupts	1358
36.6	HASH processing time	1359
36.7	HASH registers	1360
36.7.1	HASH control register (HASH_CR)	1360
36.7.2	HASH data input register (HASH_DIN)	1363
36.7.3	HASH start register (HASH_STR)	1364
36.7.4	HASH digest registers	1365
36.7.5	HASH interrupt enable register (HASH_IMR)	1366
36.7.6	HASH status register (HASH_SR)	1367
36.7.7	HASH context swap registers	1367

	36.7.8	HASH register map	1369
37		High-Resolution Timer (HRTIM)	1371
	37.1	Introduction	1371
	37.2	Main features	1372
	37.3	Functional description	1373
	37.3.1	General description	1373
	37.3.2	HRTIM pins and internal signals	1375
	37.3.3	Clocks	1376
	37.3.4	Timer A..E timing units	1379
	37.3.5	Master timer	1396
	37.3.6	Set/reset events priorities and narrow pulses management	1397
	37.3.7	External events global conditioning	1398
	37.3.8	External event filtering in timing units	1403
	37.3.9	Delayed Protection	1408
	37.3.10	Register preload and update management	1414
	37.3.11	Events propagation within or across multiple timers	1417
	37.3.12	Output management	1421
	37.3.13	Burst mode controller	1423
	37.3.14	Chopper	1432
	37.3.15	Fault protection	1433
	37.3.16	Auxiliary outputs	1436
	37.3.17	Synchronizing the HRTIM with other timers or HRTIM instances	1439
	37.3.18	ADC triggers	1442
	37.3.19	DAC triggers	1443
	37.3.20	HRTIM Interrupts	1445
	37.3.21	DMA	1447
	37.3.22	HRTIM initialization	1450
	37.3.23	Debug	1451
	37.4	Application use cases	1452
	37.4.1	Buck converter	1452
	37.4.2	Buck converter with synchronous rectification	1453
	37.4.3	Multiphase converters	1454
	37.4.4	Transition mode Power Factor Correction	1456
	37.5	HRTIM registers	1458
	37.5.1	HRTIM Master Timer Control Register (HRTIM_MCR)	1458
	37.5.2	HRTIM Master Timer Interrupt Status Register (HRTIM_MISR)	1461

37.5.3	HRTIM Master Timer Interrupt Clear Register (HRTIM_MICR)	1462
37.5.4	HRTIM Master Timer DMA / Interrupt Enable Register (HRTIM_MDIER)	1463
37.5.5	HRTIM Master Timer Counter Register (HRTIM_MCNTR)	1465
37.5.6	HRTIM Master Timer Period Register (HRTIM_MPER)	1465
37.5.7	HRTIM Master Timer Repetition Register (HRTIM_MREP)	1466
37.5.8	HRTIM Master Timer Compare 1 Register (HRTIM_MCMP1R)	1466
37.5.9	HRTIM Master Timer Compare 2 Register (HRTIM_MCMP2R)	1467
37.5.10	HRTIM Master Timer Compare 3 Register (HRTIM_MCMP3R)	1467
37.5.11	HRTIM Master Timer Compare 4 Register (HRTIM_MCMP4R)	1468
37.5.12	HRTIM Timerx Control Register (HRTIM_TIMxCR)	1469
37.5.13	HRTIM Timerx Interrupt Status Register (HRTIM_TIMxISR)	1473
37.5.14	HRTIM Timerx Interrupt Clear Register (HRTIM_TIMxICR)	1475
37.5.15	HRTIM Timerx DMA / Interrupt Enable Register (HRTIM_TIMxDIER)	1476
37.5.16	HRTIM Timerx Counter Register (HRTIM_CNTxR)	1479
37.5.17	HRTIM Timerx Period Register (HRTIM_PERxR)	1479
37.5.18	HRTIM Timerx Repetition Register (HRTIM_REPxR)	1480
37.5.19	HRTIM Timerx Compare 1 Register (HRTIM_CMP1xR)	1480
37.5.20	HRTIM Timerx Compare 1 Compound Register (HRTIM_CMP1CxR)	1481
37.5.21	HRTIM Timerx Compare 2 Register (HRTIM_CMP2xR)	1481
37.5.22	HRTIM Timerx Compare 3 Register (HRTIM_CMP3xR)	1482
37.5.23	HRTIM Timerx Compare 4 Register (HRTIM_CMP4xR)	1482
37.5.24	HRTIM Timerx Capture 1 Register (HRTIM_CPT1xR)	1483
37.5.25	HRTIM Timerx Capture 2 Register (HRTIM_CPT2xR)	1483
37.5.26	HRTIM Timerx Deadtime Register (HRTIM_DTxR)	1484
37.5.27	HRTIM Timerx Output1 Set Register (HRTIM_SETx1R)	1486
37.5.28	HRTIM Timerx Output1 Reset Register (HRTIM_RSTx1R)	1488
37.5.29	HRTIM Timerx Output2 Set Register (HRTIM_SETx2R)	1488
37.5.30	HRTIM Timerx Output2 Reset Register (HRTIM_RSTx2R)	1489
37.5.31	HRTIM Timerx External Event Filtering Register 1 (HRTIM_EEFxR1)	1490
37.5.32	HRTIM Timerx External Event Filtering Register 2 (HRTIM_EEFxR2)	1492
37.5.33	HRTIM Timerx Reset Register (HRTIM_RSTxR)	1493
37.5.34	HRTIM Timerx Chopper Register (HRTIM_CHPxR)	1496
37.5.35	HRTIM Timerx Capture 1 Control Register (HRTIM_CPT1xCR)	1498

37.5.36	HRTIM Timerx Capture 2 Control Register (HRTIM_CPT2xCR)	1499
37.5.37	HRTIM Timerx Output Register (HRTIM_OUTxR)	1502
37.5.38	HRTIM Timerx Fault Register (HRTIM_FLTxR)	1505
37.5.39	HRTIM Control Register 1 (HRTIM_CR1)	1506
37.5.40	HRTIM Control Register 2 (HRTIM_CR2)	1508
37.5.41	HRTIM Interrupt Status Register (HRTIM_ISR)	1509
37.5.42	HRTIM Interrupt Clear Register (HRTIM_ICR)	1510
37.5.43	HRTIM Interrupt Enable Register (HRTIM_IER)	1511
37.5.44	HRTIM Output Enable Register (HRTIM_OENR)	1512
37.5.45	HRTIM Output Disable Register (HRTIM_ODISR)	1513
37.5.46	HRTIM Output Disable Status Register (HRTIM_ODSR)	1514
37.5.47	HRTIM Burst Mode Control Register (HRTIM_BMCR)	1515
37.5.48	HRTIM Burst Mode Trigger Register (HRTIM_BMTRGR)	1517
37.5.49	HRTIM Burst Mode Compare Register (HRTIM_BMCMPR)	1519
37.5.50	HRTIM Burst Mode Period Register (HRTIM_BMPER)	1519
37.5.51	HRTIM Timer External Event Control Register 1 (HRTIM_EECCR1) .	1520
37.5.52	HRTIM Timer External Event Control Register 2 (HRTIM_EECCR2) .	1522
37.5.53	HRTIM Timer External Event Control Register 3 (HRTIM_EECCR3) .	1523
37.5.54	HRTIM ADC Trigger 1 Register (HRTIM_ADC1R)	1524
37.5.55	HRTIM ADC Trigger 2 Register (HRTIM_ADC2R)	1525
37.5.56	HRTIM ADC Trigger 3 Register (HRTIM_ADC3R)	1526
37.5.57	HRTIM ADC Trigger 4 Register (HRTIM_ADC4R)	1528
37.5.58	HRTIM Fault Input Register 1 (HRTIM_FLTINR1)	1530
37.5.59	HRTIM Fault Input Register 2 (HRTIM_FLTINR2)	1532
37.5.60	HRTIM Burst DMA Master timer update Register (HRTIM_BDMUPR)	1534
37.5.61	HRTIM Burst DMA Timerx update Register (HRTIM_BDTxUPR) . . .	1535
37.5.62	HRTIM Burst DMA Data Register (HRTIM_BDMADR)	1536
37.5.63	HRTIM register map	1537
38	Advanced-control timers (TIM1/TIM8)	1546
38.1	TIM1/TIM8 introduction	1546
38.2	TIM1/TIM8 main features	1546
38.3	TIM1/TIM8 functional description	1548
38.3.1	Time-base unit	1548
38.3.2	Counter modes	1550
38.3.3	Repetition counter	1561

38.3.4	External trigger input	1563
38.3.5	Clock selection	1564
38.3.6	Capture/compare channels	1568
38.3.7	Input capture mode	1570
38.3.8	PWM input mode	1571
38.3.9	Forced output mode	1572
38.3.10	Output compare mode	1573
38.3.11	PWM mode	1574
38.3.12	Asymmetric PWM mode	1577
38.3.13	Combined PWM mode	1578
38.3.14	Combined 3-phase PWM mode	1579
38.3.15	Complementary outputs and dead-time insertion	1580
38.3.16	Using the break function	1582
38.3.17	Bidirectional break inputs	1588
38.3.18	Clearing the OCxREF signal on an external event	1588
38.3.19	6-step PWM generation	1590
38.3.20	One-pulse mode	1591
38.3.21	Retriggerable one pulse mode (OPM)	1592
38.3.22	Encoder interface mode	1593
38.3.23	UIF bit remapping	1595
38.3.24	Timer input XOR function	1596
38.3.25	Interfacing with Hall sensors	1596
38.3.26	Timer synchronization	1599
38.3.27	ADC synchronization	1603
38.3.28	DMA burst mode	1603
38.3.29	Debug mode	1604
38.4	TIM1/TIM8 registers	1605
38.4.1	TIMx control register 1 (TIMx_CR1)(x = 1, 8)	1605
38.4.2	TIMx control register 2 (TIMx_CR2)(x = 1, 8)	1606
38.4.3	TIMx slave mode control register (TIMx_SMCR)(x = 1, 8)	1609
38.4.4	TIMx DMA/interrupt enable register (TIMx_DIER)(x = 1, 8)	1611
38.4.5	TIMx status register (TIMx_SR)(x = 1, 8)	1613
38.4.6	TIMx event generation register (TIMx_EGR)(x = 1, 8)	1615
38.4.7	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 1, 8)	1616
38.4.8	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 1, 8)	1617

38.4.9	TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2)(x = 1, 8)	1620
38.4.10	TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2)(x = 1, 8)	1621
38.4.11	TIMx capture/compare enable register (TIMx_CCER)(x = 1, 8)	1623
38.4.12	TIMx counter (TIMx_CNT)(x = 1, 8)	1626
38.4.13	TIMx prescaler (TIMx_PSC)(x = 1, 8)	1626
38.4.14	TIMx auto-reload register (TIMx_ARR)(x = 1, 8)	1626
38.4.15	TIMx repetition counter register (TIMx_RCR)(x = 1, 8)	1627
38.4.16	TIMx capture/compare register 1 (TIMx_CCR1)(x = 1, 8)	1627
38.4.17	TIMx capture/compare register 2 (TIMx_CCR2)(x = 1, 8)	1628
38.4.18	TIMx capture/compare register 3 (TIMx_CCR3)(x = 1, 8)	1628
38.4.19	TIMx capture/compare register 4 (TIMx_CCR4)(x = 1, 8)	1629
38.4.20	TIMx break and dead-time register (TIMx_BDTR)(x = 1, 8)	1629
38.4.21	TIMx DMA control register (TIMx_DCR)(x = 1, 8)	1632
38.4.22	TIMx DMA address for full transfer (TIMx_DMAR)(x = 1, 8)	1633
38.4.23	TIMx capture/compare mode register 3 (TIMx_CCMR3)(x = 1, 8)	1634
38.4.24	TIMx capture/compare register 5 (TIMx_CCR5)(x = 1, 8)	1635
38.4.25	TIMx capture/compare register 6 (TIMx_CCR6)(x = 1, 8)	1636
38.4.26	TIM1 alternate function option register 1 (TIM1_AF1)	1636
38.4.27	TIM1 Alternate function register 2 (TIM1_AF2)	1638
38.4.28	TIM8 Alternate function option register 1 (TIM8_AF1)	1639
38.4.29	TIM8 Alternate function option register 2 (TIM8_AF2)	1641
38.4.30	TIM1 timer input selection register (TIM1_TISEL)	1643
38.4.31	TIM8 timer input selection register (TIM8_TISEL)	1643
38.4.32	TIM1 register map	1645
38.4.33	TIM8 register map	1647
39	General-purpose timers (TIM2/TIM3/TIM4/TIM5)	1650
39.1	TIM2/TIM3/TIM4/TIM5 introduction	1650
39.2	TIM2/TIM3/TIM4/TIM5 main features	1650
39.3	TIM2/TIM3/TIM4/TIM5 functional description	1652
39.3.1	Time-base unit	1652
39.3.2	Counter modes	1654
39.3.3	Clock selection	1664

39.3.4	Capture/Compare channels	1668
39.3.5	Input capture mode	1670
39.3.6	PWM input mode	1671
39.3.7	Forced output mode	1672
39.3.8	Output compare mode	1672
39.3.9	PWM mode	1673
39.3.10	Asymmetric PWM mode	1677
39.3.11	Combined PWM mode	1677
39.3.12	Clearing the OCxREF signal on an external event	1678
39.3.13	One-pulse mode	1680
39.3.14	Retriggerable one pulse mode (OPM)	1681
39.3.15	Encoder interface mode	1682
39.3.16	UIF bit remapping	1684
39.3.17	Timer input XOR function	1684
39.3.18	Timers and external trigger synchronization	1685
39.3.19	Timer synchronization	1688
39.3.20	DMA burst mode	1692
39.3.21	Debug mode	1693
39.4	TIM2/TIM3/TIM4/TIM5 registers	1694
39.4.1	TIMx control register 1 (TIMx_CR1)(x = 2 to 5)	1694
39.4.2	TIMx control register 2 (TIMx_CR2)(x = 2 to 5)	1695
39.4.3	TIMx slave mode control register (TIMx_SMCR)(x = 2 to 5)	1697
39.4.4	TIMx DMA/Interrupt enable register (TIMx_DIER)(x = 2 to 5)	1700
39.4.5	TIMx status register (TIMx_SR)(x = 2 to 5)	1701
39.4.6	TIMx event generation register (TIMx_EGR)(x = 2 to 5)	1702
39.4.7	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 2 to 5)	1704
39.4.8	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 2 to 5)	1706
39.4.9	TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2) (x = 2 to 5)	1708
39.4.10	TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2) (x = 2 to 5)	1709
39.4.11	TIMx capture/compare enable register (TIMx_CCER)(x = 2 to 5)	1710
39.4.12	TIMx counter [alternate] (TIMx_CNT)(x = 2 to 5)	1711
39.4.13	TIMx counter [alternate] (TIMx_CNT)(x = 2 to 5)	1712
39.4.14	TIMx prescaler (TIMx_PSC)(x = 2 to 5)	1712

39.4.15	TIMx auto-reload register (TIMx_ARR)(x = 2 to 5)	1713
39.4.16	TIMx capture/compare register 1 (TIMx_CCR1)(x = 2 to 5)	1713
39.4.17	TIMx capture/compare register 2 (TIMx_CCR2)(x = 2 to 5)	1714
39.4.18	TIMx capture/compare register 3 (TIMx_CCR3)(x = 2 to 5)	1714
39.4.19	TIMx capture/compare register 4 (TIMx_CCR4)(x = 2 to 5)	1715
39.4.20	TIMx DMA control register (TIMx_DCR)(x = 2 to 5)	1716
39.4.21	TIMx DMA address for full transfer (TIMx_DMAR)(x = 2 to 5)	1716
39.4.22	TIM2 alternate function option register 1 (TIM2_AF1)	1717
39.4.23	TIM3 alternate function option register 1 (TIM3_AF1)	1717
39.4.24	TIM4 alternate function option register 1 (TIM4_AF1)	1718
39.4.25	TIM5 alternate function option register 1 (TIM5_AF1)	1718
39.4.26	TIM2 timer input selection register (TIM2_TISEL)	1719
39.4.27	TIM3 timer input selection register (TIM3_TISEL)	1719
39.4.28	TIM4 timer input selection register (TIM4_TISEL)	1720
39.4.29	TIM5 timer input selection register (TIM5_TISEL)	1721
39.4.30	TIMx register map	1723
40	General-purpose timers (TIM12/TIM13/TIM14)	1726
40.1	TIM12/TIM13/TIM14 introduction	1726
40.2	TIM12/TIM13/TIM14 main features	1726
40.2.1	TIM12 main features	1726
40.2.2	TIM13/TIM14 main features	1727
40.3	TIM12/TIM13/TIM14 functional description	1729
40.3.1	Time-base unit	1729
40.3.2	Counter modes	1731
40.3.3	Clock selection	1734
40.3.4	Capture/compare channels	1736
40.3.5	Input capture mode	1738
40.3.6	PWM input mode (only for TIM12)	1739
40.3.7	Forced output mode	1740
40.3.8	Output compare mode	1741
40.3.9	PWM mode	1742
40.3.10	Combined PWM mode (TIM12 only)	1743
40.3.11	One-pulse mode	1744
40.3.12	Retriggerable one pulse mode (OPM) (TIM12 only)	1746
40.3.13	UIF bit remapping	1747
40.3.14	Timer input XOR function	1747

40.3.15	TIM12 external trigger synchronization	1747
40.3.16	Slave mode – combined reset + trigger mode	1750
40.3.17	Timer synchronization (TIM12)	1751
40.3.18	Debug mode	1751
40.4	TIM12 registers	1751
40.4.1	TIM12 control register 1 (TIM12_CR1)	1751
40.4.2	TIM12 control register 2 (TIM12_CR2)	1752
40.4.3	TIM12 slave mode control register (TIM12_SMCR)	1753
40.4.4	TIM12 Interrupt enable register (TIM12_DIER)	1755
40.4.5	TIM12 status register (TIM12_SR)	1755
40.4.6	TIM12 event generation register (TIM12_EGR)	1756
40.4.7	TIM12 capture/compare mode register 1 [alternate] (TIM12_CCMR1)	1757
40.4.8	TIM12 capture/compare mode register 1 [alternate] (TIM12_CCMR1)	1758
40.4.9	TIM12 capture/compare enable register (TIM12_CCER)	1761
40.4.10	TIM12 counter (TIM12_CNT)	1762
40.4.11	TIM12 prescaler (TIM12_PSC)	1763
40.4.12	TIM12 auto-reload register (TIM12_ARR)	1763
40.4.13	TIM12 capture/compare register 1 (TIM12_CCR1)	1763
40.4.14	TIM12 capture/compare register 2 (TIM12_CCR2)	1764
40.4.15	TIM12 timer input selection register (TIM12_TISEL)	1764
40.4.16	TIM12 register map	1765
40.5	TIM13/TIM14 registers	1767
40.5.1	TIMx control register 1 (TIMx_CR1)(x = 13 to 14)	1767
40.5.2	TIMx Interrupt enable register (TIMx_DIER)(x = 13 to 14)	1768
40.5.3	TIMx status register (TIMx_SR)(x = 13 to 14)	1768
40.5.4	TIMx event generation register (TIMx_EGR)(x = 13 to 14)	1769
40.5.5	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 13 to 14)	1770
40.5.6	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 13 to 14)	1771
40.5.7	TIMx capture/compare enable register (TIMx_CCER)(x = 13 to 14)	1773
40.5.8	TIMx counter (TIMx_CNT)(x = 13 to 14)	1774
40.5.9	TIMx prescaler (TIMx_PSC)(x = 13 to 14)	1775
40.5.10	TIMx auto-reload register (TIMx_ARR)(x = 13 to 14)	1775
40.5.11	TIMx capture/compare register 1 (TIMx_CCR1)(x = 13 to 14)	1775

	40.5.12	TIM13 timer input selection register (TIM13_TISEL)	1776
	40.5.13	TIM14 timer input selection register (TIM14_TISEL)	1776
	40.5.14	TIM13/TIM14 register map	1777
41		General-purpose timers (TIM15/TIM16/TIM17)	1779
	41.1	TIM15/TIM16/TIM17 introduction	1779
	41.2	TIM15 main features	1779
	41.3	TIM16/TIM17 main features	1780
	41.4	TIM15/TIM16/TIM17 functional description	1783
	41.4.1	Time-base unit	1783
	41.4.2	Counter modes	1785
	41.4.3	Repetition counter	1789
	41.4.4	Clock selection	1790
	41.4.5	Capture/compare channels	1792
	41.4.6	Input capture mode	1794
	41.4.7	PWM input mode (only for TIM15)	1795
	41.4.8	Forced output mode	1796
	41.4.9	Output compare mode	1797
	41.4.10	PWM mode	1798
	41.4.11	Combined PWM mode (TIM15 only)	1799
	41.4.12	Complementary outputs and dead-time insertion	1800
	41.4.13	Using the break function	1802
	41.4.14	One-pulse mode	1807
	41.4.15	Retriggerable one pulse mode (OPM) (TIM15 only)	1809
	41.4.16	UIF bit remapping	1809
	41.4.17	Timer input XOR function (TIM15 only)	1811
	41.4.18	External trigger synchronization (TIM15 only)	1812
	41.4.19	Slave mode – combined reset + trigger mode	1814
	41.4.20	DMA burst mode	1814
	41.4.21	Timer synchronization (TIM15)	1816
	41.4.22	Debug mode	1816
	41.5	TIM15 registers	1817
	41.5.1	TIM15 control register 1 (TIM15_CR1)	1817
	41.5.2	TIM15 control register 2 (TIM15_CR2)	1818
	41.5.3	TIM15 slave mode control register (TIM15_SMCR)	1820
	41.5.4	TIM15 DMA/interrupt enable register (TIM15_DIER)	1821
	41.5.5	TIM15 status register (TIM15_SR)	1822

41.5.6	TIM15 event generation register (TIM15_EGR)	1824
41.5.7	TIM15 capture/compare mode register 1 [alternate] (TIM15_CCMR1)	1825
41.5.8	TIM15 capture/compare mode register 1 [alternate] (TIM15_CCMR1)	1826
41.5.9	TIM15 capture/compare enable register (TIM15_CCER)	1829
41.5.10	TIM15 counter (TIM15_CNT)	1832
41.5.11	TIM15 prescaler (TIM15_PSC)	1832
41.5.12	TIM15 auto-reload register (TIM15_ARR)	1832
41.5.13	TIM15 repetition counter register (TIM15_RCR)	1833
41.5.14	TIM15 capture/compare register 1 (TIM15_CCR1)	1833
41.5.15	TIM15 capture/compare register 2 (TIM15_CCR2)	1834
41.5.16	TIM15 break and dead-time register (TIM15_BDTR)	1834
41.5.17	TIM15 DMA control register (TIM15_DCR)	1837
41.5.18	TIM15 DMA address for full transfer (TIM15_DMAR)	1837
41.5.19	TIM15 alternate register 1 (TIM15_AF1)	1838
41.5.20	TIM15 input selection register (TIM15_TISEL)	1839
41.5.21	TIM15 register map	1840
41.6	TIM16/TIM17 registers	1842
41.6.1	TIMx control register 1 (TIMx_CR1)(x = 16 to 17)	1842
41.6.2	TIMx control register 2 (TIMx_CR2)(x = 16 to 17)	1843
41.6.3	TIMx DMA/interrupt enable register (TIMx_DIER)(x = 16 to 17)	1844
41.6.4	TIMx status register (TIMx_SR)(x = 16 to 17)	1845
41.6.5	TIMx event generation register (TIMx_EGR)(x = 16 to 17)	1846
41.6.6	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 16 to 17)	1847
41.6.7	TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 16 to 17)	1848
41.6.8	TIMx capture/compare enable register (TIMx_CCER)(x = 16 to 17) .	1850
41.6.9	TIMx counter (TIMx_CNT)(x = 16 to 17)	1852
41.6.10	TIMx prescaler (TIMx_PSC)(x = 16 to 17)	1853
41.6.11	TIMx auto-reload register (TIMx_ARR)(x = 16 to 17)	1853
41.6.12	TIMx repetition counter register (TIMx_RCR)(x = 16 to 17)	1854
41.6.13	TIMx capture/compare register 1 (TIMx_CCR1)(x = 16 to 17)	1854
41.6.14	TIMx break and dead-time register (TIMx_BDTR)(x = 16 to 17)	1855
41.6.15	TIMx DMA control register (TIMx_DCR)(x = 16 to 17)	1857
41.6.16	TIMx DMA address for full transfer (TIMx_DMAR)(x = 16 to 17)	1858
41.6.17	TIM16 alternate function register 1 (TIM16_AF1)	1859

	41.6.18	TIM16 input selection register (TIM16_TISEL)	1860
	41.6.19	TIM17 alternate function register 1 (TIM17_AF1)	1861
	41.6.20	TIM17 input selection register (TIM17_TISEL)	1862
	41.6.21	TIM16/TIM17 register map	1863
42	Basic timers (TIM6/TIM7)	1865	
	42.1	TIM6/TIM7 introduction	1865
	42.2	TIM6/TIM7 main features	1865
	42.3	TIM6/TIM7 functional description	1866
	42.3.1	Time-base unit	1866
	42.3.2	Counting mode	1868
	42.3.3	UIF bit remapping	1871
	42.3.4	Clock source	1871
	42.3.5	Debug mode	1872
	42.4	TIM6/TIM7 registers	1872
	42.4.1	TIMx control register 1 (TIMx_CR1)(x = 6 to 7)	1872
	42.4.2	TIMx control register 2 (TIMx_CR2)(x = 6 to 7)	1874
	42.4.3	TIMx DMA/Interrupt enable register (TIMx_DIER)(x = 6 to 7)	1874
	42.4.4	TIMx status register (TIMx_SR)(x = 6 to 7)	1875
	42.4.5	TIMx event generation register (TIMx_EGR)(x = 6 to 7)	1875
	42.4.6	TIMx counter (TIMx_CNT)(x = 6 to 7)	1875
	42.4.7	TIMx prescaler (TIMx_PSC)(x = 6 to 7)	1876
	42.4.8	TIMx auto-reload register (TIMx_ARR)(x = 6 to 7)	1876
	42.4.9	TIMx register map	1877
43	Low-power timer (LPTIM)	1878	
	43.1	Introduction	1878
	43.2	LPTIM main features	1878
	43.3	LPTIM implementation	1879
	43.4	LPTIM functional description	1879
	43.4.1	LPTIM block diagram	1879
	43.4.2	LPTIM pins and internal signals	1881
	43.4.3	LPTIM input and trigger mapping	1881
	43.4.4	LPTIM reset and clocks	1884
	43.4.5	Glitch filter	1884
	43.4.6	Prescaler	1885

43.4.7	Trigger multiplexer	1886
43.4.8	Operating mode	1886
43.4.9	Timeout function	1888
43.4.10	Waveform generation	1888
43.4.11	Register update	1889
43.4.12	Counter mode	1890
43.4.13	Timer enable	1890
43.4.14	Timer counter reset	1891
43.4.15	Encoder mode	1891
43.4.16	Debug mode	1893
43.5	LPTIM low-power modes	1893
43.6	LPTIM interrupts	1894
43.7	LPTIM registers	1894
43.7.1	LPTIM interrupt and status register (LPTIM_ISR)	1895
43.7.2	LPTIM interrupt clear register (LPTIM_ICR)	1896
43.7.3	LPTIM interrupt enable register (LPTIM_IER)	1896
43.7.4	LPTIM configuration register (LPTIM_CFGR)	1897
43.7.5	LPTIM control register (LPTIM_CR)	1900
43.7.6	LPTIM compare register (LPTIM_CMP)	1902
43.7.7	LPTIM autoreload register (LPTIM_ARR)	1902
43.7.8	LPTIM counter register (LPTIM_CNT)	1903
43.7.9	LPTIM configuration register 2 (LPTIM_CFGR2)	1903
43.7.10	LPTIM3 configuration register 2 (LPTIM3_CFGR2)	1904
43.7.11	LPTIM register map	1905
44	System window watchdog (WWDG)	1907
44.1	Introduction	1907
44.2	WWDG main features	1907
44.3	WWDG functional description	1907
44.3.1	WWDG block diagram	1908
44.3.2	WWDG internal signals	1908
44.3.3	Enabling the watchdog	1908
44.3.4	Controlling the down-counter	1908
44.3.5	How to program the watchdog timeout	1909
44.3.6	Debug mode	1910
44.4	WWDG interrupts	1910

44.5	WWDG registers	1910
44.5.1	WWDG control register (WWDG_CR)	1910
44.5.2	WWDG configuration register (WWDG_CFR)	1911
44.5.3	WWDG status register (WWDG_SR)	1912
44.5.4	WWDG register map	1912
45	Independent watchdog (IWDG)	1913
45.1	Introduction	1913
45.2	IWDG main features	1913
45.3	IWDG functional description	1913
45.3.1	IWDG block diagram	1913
45.3.2	IWDG internal signals	1914
45.3.3	Window option	1914
45.3.4	Hardware watchdog	1915
45.3.5	Low-power freeze	1915
45.3.6	Register access protection	1915
45.3.7	Debug mode	1916
45.4	IWDG registers	1917
45.4.1	IWDG key register (IWDG_KR)	1917
45.4.2	IWDG prescaler register (IWDG_PR)	1918
45.4.3	IWDG reload register (IWDG_RLR)	1919
45.4.4	IWDG status register (IWDG_SR)	1920
45.4.5	IWDG window register (IWDG_WINR)	1921
45.4.6	IWDG register map	1922
46	Real-time clock (RTC)	1923
46.1	Introduction	1923
46.2	RTC main features	1924
46.3	RTC functional description	1924
46.3.1	RTC block diagram	1924
46.3.2	RTC pins and internal signals	1927
46.3.3	GPIOs controlled by the RTC	1927
46.3.4	Clock and prescalers	1929
46.3.5	Real-time clock and calendar	1930
46.3.6	Programmable alarms	1930
46.3.7	Periodic auto-wakeup	1930

46.3.8	RTC initialization and configuration	1931
46.3.9	Reading the calendar	1933
46.3.10	Resetting the RTC	1934
46.3.11	RTC synchronization	1934
46.3.12	RTC reference clock detection	1935
46.3.13	RTC smooth digital calibration	1935
46.3.14	Time-stamp function	1937
46.3.15	Tamper detection	1938
46.3.16	Calibration clock output	1940
46.3.17	Alarm output	1941
46.4	RTC low-power modes	1941
46.5	RTC interrupts	1941
46.6	RTC registers	1942
46.6.1	RTC time register (RTC_TR)	1942
46.6.2	RTC date register (RTC_DR)	1943
46.6.3	RTC control register (RTC_CR)	1945
46.6.4	RTC initialization and status register (RTC_ISR)	1948
46.6.5	RTC prescaler register (RTC_PRER)	1951
46.6.6	RTC wakeup timer register (RTC_WUTR)	1952
46.6.7	RTC alarm A register (RTC_ALRMAR)	1953
46.6.8	RTC alarm B register (RTC_ALRMBR)	1954
46.6.9	RTC write protection register (RTC_WPR)	1955
46.6.10	RTC sub second register (RTC_SSR)	1955
46.6.11	RTC shift control register (RTC_SHIFTR)	1956
46.6.12	RTC timestamp time register (RTC_TSTR)	1957
46.6.13	RTC timestamp date register (RTC_TSDR)	1958
46.6.14	RTC time-stamp sub second register (RTC_TSSSR)	1959
46.6.15	RTC calibration register (RTC_CALR)	1960
46.6.16	RTC tamper configuration register (RTC_TAMPCR)	1961
46.6.17	RTC alarm A sub second register (RTC_ALRMASR)	1964
46.6.18	RTC alarm B sub second register (RTC_ALRMBSSR)	1965
46.6.19	RTC option register (RTC_OR)	1966
46.6.20	RTC backup registers (RTC_BKPxR)	1966
46.6.21	RTC register map	1967
47	Inter-integrated circuit (I2C) interface	1969
47.1	Introduction	1969

47.2	I2C main features	1969
47.3	I2C implementation	1970
47.4	I2C functional description	1970
47.4.1	I2C block diagram	1971
47.4.2	I2C pins and internal signals	1972
47.4.3	I2C clock requirements	1972
47.4.4	Mode selection	1972
47.4.5	I2C initialization	1973
47.4.6	Software reset	1978
47.4.7	Data transfer	1979
47.4.8	I2C slave mode	1981
47.4.9	I2C master mode	1990
47.4.10	I2C_TIMINGR register configuration examples	2002
47.4.11	SMBus specific features	2003
47.4.12	SMBus initialization	2006
47.4.13	SMBus: I2C_TIMEOCTR register configuration examples	2008
47.4.14	SMBus slave mode	2009
47.4.15	Wakeup from Stop mode on address match	2017
47.4.16	Error conditions	2017
47.4.17	DMA requests	2019
47.4.18	Debug mode	2020
47.5	I2C low-power modes	2020
47.6	I2C interrupts	2021
47.7	I2C registers	2022
47.7.1	I2C control register 1 (I2C_CR1)	2022
47.7.2	I2C control register 2 (I2C_CR2)	2025
47.7.3	I2C own address 1 register (I2C_OAR1)	2028
47.7.4	I2C own address 2 register (I2C_OAR2)	2029
47.7.5	I2C timing register (I2C_TIMINGR)	2030
47.7.6	I2C timeout register (I2C_TIMEOCTR)	2031
47.7.7	I2C interrupt and status register (I2C_ISR)	2032
47.7.8	I2C interrupt clear register (I2C_ICR)	2034
47.7.9	I2C PEC register (I2C_PECR)	2035
47.7.10	I2C receive data register (I2C_RXDR)	2036
47.7.11	I2C transmit data register (I2C_TXDR)	2036
47.7.12	I2C register map	2037

48	Universal synchronous/asynchronous receiver transmitter (USART/UART)	2039
48.1	USART introduction	2039
48.2	USART main features	2040
48.3	USART extended features	2041
48.4	USART implementation	2041
48.5	USART functional description	2042
48.5.1	USART block diagram	2042
48.5.2	USART signals	2043
48.5.3	USART character description	2044
48.5.4	USART FIFOs and thresholds	2046
48.5.5	USART transmitter	2046
48.5.6	USART receiver	2050
48.5.7	USART baud rate generation	2057
48.5.8	Tolerance of the USART receiver to clock deviation	2058
48.5.9	USART Auto baud rate detection	2060
48.5.10	USART multiprocessor communication	2062
48.5.11	USART Modbus communication	2064
48.5.12	USART parity control	2065
48.5.13	USART LIN (local interconnection network) mode	2066
48.5.14	USART synchronous mode	2068
48.5.15	USART single-wire Half-duplex communication	2072
48.5.16	USART receiver timeout	2072
48.5.17	USART Smartcard mode	2073
48.5.18	USART IrDA SIR ENDEC block	2077
48.5.19	Continuous communication using USART and DMA	2080
48.5.20	RS232 Hardware flow control and RS485 Driver Enable	2082
48.5.21	USART low-power management	2085
48.6	USART interrupts	2088
48.7	USART registers	2091
48.7.1	USART control register 1 [alternate] (USART_CR1)	2091
48.7.2	USART control register 1 [alternate] (USART_CR1)	2095
48.7.3	USART control register 2 (USART_CR2)	2098
48.7.4	USART control register 3 (USART_CR3)	2102
48.7.5	USART baud rate register (USART_BRR)	2107
48.7.6	USART guard time and prescaler register (USART_GTPR)	2107

48.7.7	USART receiver timeout register (USART_RTOR)	2108
48.7.8	USART request register (USART_RQR)	2109
48.7.9	USART interrupt and status register [alternate] (USART_ISR)	2110
48.7.10	USART interrupt and status register [alternate] (USART_ISR)	2116
48.7.11	USART interrupt flag clear register (USART_ICR)	2121
48.7.12	USART receive data register (USART_RDR)	2123
48.7.13	USART transmit data register (USART_TDR)	2123
48.7.14	USART prescaler register (USART_PRESC)	2124
48.7.15	USART register map	2125
49	Low-power universal asynchronous receiver transmitter (LPUART)	2127
49.1	LPUART introduction	2127
49.2	LPUART main features	2128
49.3	LPUART implementation	2129
49.4	LPUART functional description	2130
49.4.1	LPUART block diagram	2130
49.4.2	LPUART signals	2131
49.4.3	LPUART character description	2131
49.4.4	LPUART FIFOs and thresholds	2132
49.4.5	LPUART transmitter	2133
49.4.6	LPUART receiver	2136
49.4.7	LPUART baud rate generation	2140
49.4.8	Tolerance of the LPUART receiver to clock deviation	2141
49.4.9	LPUART multiprocessor communication	2142
49.4.10	LPUART parity control	2144
49.4.11	LPUART single-wire Half-duplex communication	2145
49.4.12	Continuous communication using DMA and LPUART	2145
49.4.13	RS232 Hardware flow control and RS485 Driver Enable	2148
49.4.14	LPUART low-power management	2150
49.5	LPUART interrupts	2153
49.6	LPUART registers	2155
49.6.1	LPUART control register 1 [alternate] (LPUART_CR1)	2155
49.6.2	LPUART control register 1 [alternate] (LPUART_CR1)	2158
49.6.3	LPUART control register 2 (LPUART_CR2)	2161
49.6.4	LPUART control register 3 (LPUART_CR3)	2163
49.6.5	LPUART baud rate register (LPUART_BRR)	2166

49.6.6	LPUART request register (LPUART_RQR)	2167
49.6.7	LPUART interrupt and status register [alternate] (LPUART_ISR) . . .	2167
49.6.8	LPUART interrupt and status register [alternate] (LPUART_ISR) . . .	2172
49.6.9	LPUART interrupt flag clear register (LPUART_ICR)	2175
49.6.10	LPUART receive data register (LPUART_RDR)	2176
49.6.11	LPUART transmit data register (LPUART_TDR)	2176
49.6.12	LPUART prescaler register (LPUART_PRESC)	2177
49.6.13	LPUART register map	2178
50	Serial peripheral interface (SPI)	2180
50.1	Introduction	2180
50.2	SPI main features	2181
50.3	SPI implementation	2181
50.4	SPI functional description	2182
50.4.1	SPI block diagram	2182
50.4.2	SPI signals	2183
50.4.3	SPI communication general aspects	2183
50.4.4	Communications between one master and one slave	2183
50.4.5	Standard multi-slave communication	2186
50.4.6	Multi-master communication	2189
50.4.7	Slave select (SS) pin management	2189
50.4.8	Communication formats	2193
50.4.9	Configuration of SPI	2195
50.4.10	Procedure for enabling SPI	2196
50.4.11	SPI data transmission and reception procedures	2196
50.4.12	Procedure for disabling the SPI	2200
50.4.13	Data packing	2201
50.4.14	Communication using DMA (direct memory addressing)	2202
50.5	SPI specific modes and control	2204
50.5.1	TI mode	2204
50.5.2	SPI error flags	2204
50.5.3	CRC computation	2206
50.6	Low-power mode management	2208
50.7	SPI wakeup and interrupts	2210
50.8	I2S main features	2211
50.9	I2S functional description	2212

50.9.1	I2S general description	2212
50.9.2	Pin sharing with SPI function	2212
50.9.3	Bits and fields usable in I2S/PCM mode	2213
50.9.4	Slave and master modes	2214
50.9.5	Supported audio protocols	2214
50.9.6	Additional Serial Interface Flexibility	2220
50.9.7	Start-up sequence	2222
50.9.8	Stop sequence	2223
50.9.9	Clock generator	2224
50.9.10	Internal FIFOs	2226
50.9.11	FIFOs status flags	2227
50.9.12	Handling of underrun situation	2228
50.9.13	Handling of overrun situation	2229
50.9.14	Frame error detection	2229
50.9.15	DMA Interface	2231
50.9.16	Programing examples	2232
50.9.17	Slave I2S Philips standard, receive	2234
50.10	I2S wakeup and interrupts	2235
50.11	SPI/I2S registers	2236
50.11.1	SPI/I2S control register 1 (SPI2S_CR1)	2236
50.11.2	SPI control register 2 (SPI_CR2)	2237
50.11.3	SPI configuration register 1 (SPI_CFG1)	2238
50.11.4	SPI configuration register 2 (SPI_CFG2)	2241
50.11.5	SPI/I2S interrupt enable register (SPI2S_IER)	2243
50.11.6	SPI/I2S status register (SPI2S_SR)	2244
50.11.7	SPI/I2S interrupt/status flags clear register (SPI2S_IFCR)	2246
50.11.8	SPI/I2S transmit data register (SPI2S_TXDR)	2247
50.11.9	SPI/I2S receive data register (SPI2S_RXDR)	2248
50.11.10	SPI polynomial register (SPI_CRCPOLY)	2248
50.11.11	SPI transmitter CRC register (SPI_TXCRC)	2249
50.11.12	SPI receiver CRC register (SPI_RXCRC)	2249
50.11.13	SPI underrun data register (SPI_UDRDR)	2250
50.11.14	SPI/I2S configuration register (SPI_I2SCFGR)	2250
50.12	SPI register map and reset values	2253
51	Serial audio interface (SAI)	2255
51.1	Introduction	2255

51.2	SAI main features	2256
51.3	SAI implementation	2256
51.4	SAI functional description	2257
51.4.1	SAI block diagram	2257
51.4.2	SAI pins and internal signals	2258
51.4.3	Main SAI modes	2259
51.4.4	SAI synchronization mode	2260
51.4.5	Audio data size	2261
51.4.6	Frame synchronization	2261
51.4.7	Slot configuration	2264
51.4.8	SAI clock generator	2266
51.4.9	Internal FIFOs	2268
51.4.10	PDM Interface	2270
51.4.11	AC'97 link controller	2278
51.4.12	SPDIF output	2280
51.4.13	Specific features	2283
51.4.14	Error flags	2287
51.4.15	Disabling the SAI	2290
51.4.16	SAI DMA interface	2290
51.5	SAI interrupts	2291
51.6	SAI registers	2293
51.6.1	SAI global configuration register (SAI_GCR)	2293
51.6.2	SAI configuration register 1 (SAI_ACR1)	2293
51.6.3	SAI configuration register 1 (SAI_BCR1)	2296
51.6.4	SAI configuration register 2 (SAI_ACR2)	2299
51.6.5	SAI configuration register 2 (SAI_BCR2)	2301
51.6.6	SAI frame configuration register (SAI_AFRCR)	2303
51.6.7	SAI frame configuration register (SAI_BFRCR)	2304
51.6.8	SAI slot register (SAI_ASLOTR)	2305
51.6.9	SAI slot register (SAI_BSLOTR)	2306
51.6.10	SAI interrupt mask register (SAI_AIM)	2307
51.6.11	SAI interrupt mask register (SAI_BIM)	2309
51.6.12	SAI status register (SAI_ASR)	2310
51.6.13	SAI status register (SAI_BSR)	2312
51.6.14	SAI clear flag register (SAI_ACLRFR)	2314
51.6.15	SAI clear flag register (SAI_BCLRFR)	2315

51.6.16	SAI data register (SAI_ADR)	2316
51.6.17	SAI data register (SAI_BDR)	2317
51.6.18	SAI PDM control register (SAI_PDMCR)	2317
51.6.19	SAI PDM delay register (SAI_PDMDLY)	2318
51.6.20	SAI register map	2321
52	SPDIF receiver interface (SPDIFRX)	2323
52.1	SPDIFRX interface introduction	2323
52.2	SPDIFRX main features	2323
52.3	SPDIFRX functional description	2323
52.3.1	SPDIFRX pins and internal signals	2324
52.3.2	S/PDIF protocol (IEC-60958)	2325
52.3.3	SPDIFRX decoder (SPDIFRX_DC)	2327
52.3.4	SPDIFRX tolerance to clock deviation	2331
52.3.5	SPDIFRX synchronization	2331
52.3.6	SPDIFRX handling	2333
52.3.7	Data reception management	2335
52.3.8	Dedicated control flow	2337
52.3.9	Reception errors	2338
52.3.10	Clocking strategy	2340
52.3.11	DMA Interface	2340
52.3.12	Interrupt Generation	2341
52.3.13	Register protection	2342
52.4	Programming procedures	2343
52.4.1	Initialization phase	2343
52.4.2	Handling of interrupts coming from SPDIFRX	2344
52.4.3	Handling of interrupts coming from DMA	2345
52.5	SPDIFRX interface registers	2345
52.5.1	Control register (SPDIFRX_CR)	2345
52.5.2	Interrupt mask register (SPDIFRX_IMR)	2348
52.5.3	Status register (SPDIFRX_SR)	2349
52.5.4	Interrupt flag clear register (SPDIFRX_IFCR)	2351
52.5.5	Data input register (SPDIFRX_FMT0_DR)	2352
52.5.6	Data input register (SPDIFRX_FMT1_DR)	2353
52.5.7	Data input register (SPDIFRX_FMT2_DR)	2354
52.5.8	Channel status register (SPDIFRX_CSR)	2355
52.5.9	Debug information register (SPDIFRX_DIR)	2355

	52.5.10 SPDIFRX interface register map	2357
53	Single Wire Protocol Master Interface (SWPMI)	2358
53.1	Introduction	2358
53.2	SWPMI main features	2359
53.3	SWPMI functional description	2360
53.3.1	SWPMI block diagram	2360
53.3.2	SWPMI pins and internal signals	2360
53.3.3	SWP initialization and activation	2361
53.3.4	SWP bus states	2362
53.3.5	SWPMI_IO (internal transceiver) bypass	2363
53.3.6	SWPMI Bit rate	2363
53.3.7	SWPMI frame handling	2364
53.3.8	Transmission procedure	2364
53.3.9	Reception procedure	2369
53.3.10	Error management	2373
53.3.11	Loopback mode	2375
53.4	SWPMI low-power modes	2375
53.5	SWPMI interrupts	2376
53.6	SWPMI registers	2377
53.6.1	SWPMI Configuration/Control register (SWPMI_CR)	2377
53.6.2	SWPMI Bitrate register (SWPMI_BRR)	2378
53.6.3	SWPMI Interrupt and Status register (SWPMI_ISR)	2379
53.6.4	SWPMI Interrupt Flag Clear register (SWPMI_ICR)	2380
53.6.5	SWPMI Interrupt Enable register (SWPMI_IER)	2381
53.6.6	SWPMI Receive Frame Length register (SWPMI_RFL)	2383
53.6.7	SWPMI Transmit data register (SWPMI_TDR)	2383
53.6.8	SWPMI Receive data register (SWPMI_RDR)	2383
53.6.9	SWPMI Option register (SWPMI_OR)	2384
53.6.10	SWPMI register map and reset value table	2385
54	Management data input/output (MDIOS)	2386
54.1	MDIOS introduction	2386
54.2	MDIOS main features	2386
54.3	MDIOS functional description	2387
54.3.1	MDIOS block diagram	2387

54.3.2	MDIOS pins and internal signals	2387
54.3.3	MDIOS protocol	2388
54.3.4	MDIOS enabling and disabling	2389
54.3.5	MDIOS data	2389
54.3.6	MDIOS APB frequency	2391
54.3.7	Write/read flags and interrupts	2391
54.3.8	MDIOS error management	2391
54.3.9	MDIOS in Stop mode	2392
54.3.10	MDIOS interrupts	2392
54.4	MDIOS registers	2393
54.4.1	MDIOS configuration register (MDIOS_CR)	2393
54.4.2	MDIOS write flag register (MDIOS_WRFR)	2394
54.4.3	MDIOS clear write flag register (MDIOS_CWRFR)	2394
54.4.4	MDIOS read flag register (MDIOS_RDFR)	2395
54.4.5	MDIOS clear read flag register (MDIOS_CRDFR)	2395
54.4.6	MDIOS status register (MDIOS_SR)	2396
54.4.7	MDIOS clear flag register (MDIOS_CLRFR)	2397
54.4.8	MDIOS input data register x (MDIOS_DINRx)	2398
54.4.9	MDIOS output data register x (MDIOS_DOUTRx)	2398
54.4.10	MDIOS register map	2399
55	Secure digital input/output MultiMediaCard interface (SDMMC) . .	2400
55.1	SDMMC main features	2400
55.2	SDMMC implementation	2400
55.3	SDMMC bus topology	2401
55.4	SDMMC operation modes	2403
55.5	SDMMC functional description	2404
55.5.1	SDMMC block diagram	2404
55.5.2	SDMMC pins and internal signals	2404
55.5.3	General description	2405
55.5.4	SDMMC adapter	2407
55.5.5	SDMMC AHB slave interface	2428
55.5.6	SDMMC AHB master interface	2429
55.5.7	MDMA request generation	2431
55.5.8	AHB and SDMMC_CK clock relation	2431
55.5.9	Hardware flow control	2432

55.6	Card functional description	2433
55.6.1	SD I/O mode	2433
55.6.2	CMD12 send timing	2441
55.6.3	Sleep (CMD5)	2444
55.6.4	Interrupt mode (Wait-IRQ)	2445
55.6.5	Boot operation	2446
55.6.6	Response R1b handling	2449
55.6.7	Reset and card cycle power	2450
55.7	Ultra-high-speed phase I (UHS-I) voltage switch	2451
55.8	SDMMC interrupts	2455
55.9	SDMMC registers	2457
55.9.1	SDMMC power control register (SDMMC_POWER)	2457
55.9.2	SDMMC clock control register (SDMMC_CLKCR)	2458
55.9.3	SDMMC argument register (SDMMC_ARGR)	2460
55.9.4	SDMMC command register (SDMMC_CMDR)	2460
55.9.5	SDMMC command response register (SDMMC_RESPCMDR)	2462
55.9.6	SDMMC response x register (SDMMC_RESPxR)	2463
55.9.7	SDMMC data timer register (SDMMC_DTIMER)	2463
55.9.8	SDMMC data length register (SDMMC_DLENR)	2464
55.9.9	SDMMC data control register (SDMMC_DCTRL)	2465
55.9.10	SDMMC data counter register (SDMMC_DCNTR)	2466
55.9.11	SDMMC status register (SDMMC_STAR)	2467
55.9.12	SDMMC interrupt clear register (SDMMC_ICR)	2470
55.9.13	SDMMC mask register (SDMMC_MASKR)	2472
55.9.14	SDMMC acknowledgment timer register (SDMMC_ACKTIMER)	2475
55.9.15	SDMMC data FIFO registers x (SDMMC_FIFORx)	2475
55.9.16	SDMMC DMA control register (SDMMC_IDMACTRLR)	2476
55.9.17	SDMMC IDMA buffer size register (SDMMC_IDMABSIZE)	2477
55.9.18	SDMMC IDMA buffer 0 base address register (SDMMC_IDMABASE0R)	2477
55.9.19	SDMMC IDMA buffer 1 base address register (SDMMC_IDMABASE1R)	2478
55.9.20	SDMMC register map	2479
56	Controller area network with flexible data rate (FDCAN)	2482
56.1	Introduction	2482
56.2	FDCAN main features	2485

56.3	FDCAN implementation	2485
56.4	FDCAN functional description	2486
56.4.1	Operating modes	2487
56.4.2	Message RAM	2496
56.4.3	FIFO acknowledge handling	2507
56.4.4	Bit timing	2508
56.4.5	Clock calibration on CAN	2509
56.4.6	Application	2513
56.4.7	TTCAN operations (FDCAN1 only)	2514
56.4.8	TTCAN configuration	2515
56.4.9	Message scheduling	2517
56.4.10	TTCAN gap control	2524
56.4.11	Stop watch	2525
56.4.12	Local time, cycle time, global time, and external clock synchronization	2525
56.4.13	TTCAN error level	2528
56.4.14	TTCAN message handling	2529
56.4.15	TTCAN interrupt and error handling	2532
56.4.16	Level 0	2533
56.4.17	Synchronization to external time schedule	2535
56.4.18	FDCAN Rx buffer and FIFO element	2536
56.4.19	FDCAN Tx buffer element	2538
56.4.20	FDCAN Tx event FIFO element	2540
56.4.21	FDCAN standard message ID filter element	2541
56.4.22	FDCAN extended message ID filter element	2543
56.4.23	FDCAN trigger memory element	2544
56.5	FDCAN registers	2546
56.5.1	FDCAN core release register (FDCAN_CREL)	2546
56.5.2	FDCAN Endian register (FDCAN_ENDN)	2546
56.5.3	FDCAN data bit timing and prescaler register (FDCAN_DBTP)	2546
56.5.4	FDCAN test register (FDCAN_TEST)	2547
56.5.5	FDCAN RAM watchdog register (FDCAN_RWD)	2548
56.5.6	FDCAN CC control register (FDCAN_CCCR)	2549
56.5.7	FDCAN nominal bit timing and prescaler register (FDCAN_NBTP)	2551
56.5.8	FDCAN timestamp counter configuration register (FDCAN_TSCC)	2552
56.5.9	FDCAN timestamp counter value register (FDCAN_TSCV)	2552
56.5.10	FDCAN timeout counter configuration register (FDCAN_TOCC)	2553

56.5.11	FDCAN timeout counter value register (FDCAN_TOCV)	2554
56.5.12	FDCAN error counter register (FDCAN_ECR)	2554
56.5.13	FDCAN protocol status register (FDCAN_PSR)	2555
56.5.14	FDCAN transmitter delay compensation register (FDCAN_TDCR) . .	2557
56.5.15	FDCAN interrupt register (FDCAN_IR)	2558
56.5.16	FDCAN interrupt enable register (FDCAN_IE)	2561
56.5.17	FDCAN interrupt line select register (FDCAN_ILS)	2563
56.5.18	FDCAN interrupt line enable register (FDCAN_ILE)	2564
56.5.19	FDCAN global filter configuration register (FDCAN_GFC)	2565
56.5.20	FDCAN standard ID filter configuration register (FDCAN_SIDFC) . .	2566
56.5.21	FDCAN extended ID filter configuration register (FDCAN_XIDFC) . .	2566
56.5.22	FDCAN extended ID and mask register (FDCAN_XIDAM)	2567
56.5.23	FDCAN high priority message status register (FDCAN_HPMS) . . .	2568
56.5.24	FDCAN new data 1 register (FDCAN_NDAT1)	2568
56.5.25	FDCAN new data 2 register (FDCAN_NDAT2)	2569
56.5.26	FDCAN Rx FIFO 0 configuration register (FDCAN_RXF0C)	2569
56.5.27	FDCAN Rx FIFO 0 status register (FDCAN_RXF0S)	2570
56.5.28	FDCAN Rx FIFO 0 acknowledge register (FDCAN_RXF0A)	2571
56.5.29	FDCAN Rx buffer configuration register (FDCAN_RXBC)	2571
56.5.30	FDCAN Rx FIFO 1 configuration register (FDCAN_RXF1C)	2572
56.5.31	FDCAN Rx FIFO 1 status register (FDCAN_RXF1S)	2573
56.5.32	FDCAN Rx FIFO 1 acknowledge register (FDCAN_RXF1A)	2574
56.5.33	FDCAN Rx buffer element size configuration register (FDCAN_RXESC)	2574
56.5.34	FDCAN Tx buffer configuration register (FDCAN_TXBC)	2575
56.5.35	FDCAN Tx FIFO/queue status register (FDCAN_TXFQS)	2576
56.5.36	FDCAN Tx buffer element size configuration register (FDCAN_TXESC)	2577
56.5.37	FDCAN Tx buffer request pending register (FDCAN_TXBRP)	2578
56.5.38	FDCAN Tx buffer add request register (FDCAN_TXBAR)	2579
56.5.39	FDCAN Tx buffer cancellation request register (FDCAN_TXBCR) . .	2579
56.5.40	FDCAN Tx buffer transmission occurred register (FDCAN_TXBTO) .	2580
56.5.41	FDCAN Tx buffer cancellation finished register (FDCAN_TXBCF) . .	2580
56.5.42	FDCAN Tx buffer transmission interrupt enable register (FDCAN_TXBTIE)	2580
56.5.43	FDCAN Tx buffer cancellation finished interrupt enable register (FDCAN_TXBCIE)	2581
56.5.44	FDCAN Tx event FIFO configuration register (FDCAN_TXEFC) . . .	2581

56.5.45	FDCAN Tx event FIFO status register (FDCAN_TXEFS)	2582
56.5.46	FDCAN Tx event FIFO acknowledge register (FDCAN_TXEFA) . . .	2583
56.5.47	FDCAN TT trigger memory configuration register (FDCAN_TTTMC) .	2583
56.5.48	FDCAN TT reference message configuration register (FDCAN_TTRMC)	2584
56.5.49	FDCAN TT operation configuration register (FDCAN_TTOCF)	2585
56.5.50	FDCAN TT matrix limits register (FDCAN_TTMLM)	2586
56.5.51	FDCAN TUR configuration register (FDCAN_TURCF)	2587
56.5.52	FDCAN TT operation control register (FDCAN_TTOCN)	2589
56.5.53	FDCAN TT global time preset register (FDCAN_TTGP)	2591
56.5.54	FDCAN TT time mark register (FDCAN_TTTMK)	2591
56.5.55	FDCAN TT interrupt register (FDCAN_TTIR)	2592
56.5.56	FDCAN TT interrupt enable register (FDCAN_TTIE)	2594
56.5.57	FDCAN TT interrupt line select register (FDCAN_TTILS)	2596
56.5.58	FDCAN TT operation status register (FDCAN_TTOST)	2598
56.5.59	FDCAN TUR numerator actual register (FDCAN_TURNA)	2599
56.5.60	FDCAN TT local and global time register (FDCAN_TTLGT)	2600
56.5.61	FDCAN TT cycle time and count register (FDCAN_TTCTC)	2600
56.5.62	FDCAN TT capture time register (FDCAN_TTCPT)	2601
56.5.63	FDCAN TT cycle sync mark register (FDCAN_TTCSM)	2601
56.5.64	FDCAN TT trigger select register (FDCAN_TTTS)	2602
56.5.65	FDCAN register map and reset value table	2603
56.6	CCU registers	2608
56.6.1	Clock calibration unit core release register (FDCAN_CCU_CREL) . .	2608
56.6.2	Calibration configuration register (FDCAN_CCU_CCFG)	2608
56.6.3	Calibration status register (FDCAN_CCU_CSTAT)	2610
56.6.4	Calibration watchdog register (FDCAN_CCU_CWD)	2610
56.6.5	Clock calibration unit interrupt register (FDCAN_CCU_IR)	2611
56.6.6	Clock calibration unit interrupt enable register (FDCAN_CCU_IE) . .	2612
56.6.7	CCU register map and reset value table	2613
57	USB on-the-go high-speed (OTG_HS)	2614
57.1	Introduction	2614
57.2	OTG_HS main features	2615
57.2.1	General features	2615
57.2.2	Host-mode features	2616
57.2.3	Peripheral-mode features	2616

57.3	OTG_HS implementation	2617
57.4	OTG_HS functional description	2617
57.4.1	OTG_HS block diagram	2617
57.4.2	OTG_HS pin and internal signals	2619
57.4.3	OTG_HS core	2619
57.4.4	Embedded full speed OTG PHY	2620
57.4.5	High-speed OTG PHY	2620
57.5	OTG_HS dual role device (DRD)	2620
57.5.1	ID line detection	2620
57.5.2	HNP dual role device	2621
57.5.3	SRP dual role device	2621
57.6	OTG_HS as a USB peripheral	2621
57.6.1	SRP-capable peripheral	2622
57.6.2	Peripheral states	2622
57.6.3	Peripheral endpoints	2623
57.7	OTG_HS as a USB host	2625
57.7.1	SRP-capable host	2625
57.7.2	USB host states	2626
57.7.3	Host channels	2627
57.7.4	Host scheduler	2628
57.8	OTG_HS SOF trigger	2629
57.8.1	Host SOFs	2630
57.8.2	Peripheral SOFs	2630
57.9	OTG_HS low-power modes	2630
57.10	OTG_HS Dynamic update of the OTG_HFIR register	2631
57.11	OTG_HS data FIFOs	2632
57.11.1	Peripheral FIFO architecture	2633
57.11.2	Host FIFO architecture	2634
57.11.3	FIFO RAM allocation	2635
57.12	OTG_HS interrupts	2637
57.13	OTG_HS control and status registers	2639
57.13.1	CSR memory map	2639
57.14	OTG_HS registers	2644
57.14.1	OTG control and status register (OTG_GOTGCTL)	2644
57.14.2	OTG interrupt register (OTG_GOTGINT)	2647
57.14.3	OTG AHB configuration register (OTG_GAHBCFG)	2649

57.14.4	OTG USB configuration register (OTG_GUSBCFG)	2650
57.14.5	OTG reset register (OTG_GRSTCTL)	2654
57.14.6	OTG core interrupt register (OTG_GINTSTS)	2656
57.14.7	OTG interrupt mask register (OTG_GINTMSK)	2660
57.14.8	OTG receive status debug read register (OTG_GRXSTSR)	2664
57.14.9	OTG receive status debug read [alternate] (OTG_GRXSTSR)	2665
57.14.10	OTG status read and pop registers (OTG_GRXSTSP)	2666
57.14.11	OTG status read and pop registers [alternate] (OTG_GRXSTSP)	2667
57.14.12	OTG receive FIFO size register (OTG_GRXFSIZ)	2668
57.14.13	OTG host non-periodic transmit FIFO size register (OTG_HNPTXFSIZ)/Endpoint 0 Transmit FIFO size (OTG_DIEPTXF0)	2668
57.14.14	OTG non-periodic transmit FIFO/queue status register (OTG_HNPTXSTS)	2669
57.14.15	OTG general core configuration register (OTG_GCCFG)	2670
57.14.16	OTG core ID register (OTG_CID)	2672
57.14.17	OTG core LPM configuration register (OTG_GLPMCFG)	2672
57.14.18	OTG host periodic transmit FIFO size register (OTG_HPTXFSIZ)	2676
57.14.19	OTG device IN endpoint transmit FIFO x size register (OTG_DIEPTFX)	2676
57.14.20	Host-mode registers	2677
57.14.21	OTG host configuration register (OTG_HCFG)	2677
57.14.22	OTG host frame interval register (OTG_HFIR)	2678
57.14.23	OTG host frame number/frame time remaining register (OTG_HFNUM)	2679
57.14.24	OTG_Host periodic transmit FIFO/queue status register (OTG_HPTXSTS)	2680
57.14.25	OTG host all channels interrupt register (OTG_HAINT)	2681
57.14.26	OTG host all channels interrupt mask register (OTG_HAINTMSK)	2681
57.14.27	OTG host frame list base address register (OTG_HFLBADDR)	2682
57.14.28	OTG host port control and status register (OTG_HPRT)	2682
57.14.29	OTG host channel x characteristics register (OTG_HCCHARx)	2685
57.14.30	OTG host channel x split control register (OTG_HCSPLTx)	2686
57.14.31	OTG host channel x interrupt register (OTG_HCINTx)	2687
57.14.32	OTG host channel x interrupt mask register (OTG_HCINTMSKx)	2688
57.14.33	OTG host channel x transfer size register (OTG_HCTSIZx)	2690
57.14.34	OTG host channel x transfer size register (OTG_HCTSIZSGx)	2691

57.14.35 OTG host channel x DMA address register in buffer DMA [alternate] (OTG_HCDMAx)	2693
57.14.36 OTG host channel x DMA address register in scatter/gather DMA [alternate] (OTG_HCDMASGx)	2693
57.14.37 OTG host channel-n DMA address buffer register (OTG_HCDMABx)	2694
57.14.38 Device-mode registers	2695
57.14.39 OTG device configuration register (OTG_DCFG)	2695
57.14.40 OTG device control register (OTG_DCTL)	2697
57.14.41 OTG device status register (OTG_DSTS)	2699
57.14.42 OTG device IN endpoint common interrupt mask register (OTG_DIEPMSK)	2700
57.14.43 OTG device OUT endpoint common interrupt mask register (OTG_DOEPMSK)	2701
57.14.44 OTG device all endpoints interrupt register (OTG_DAININT)	2702
57.14.45 OTG all endpoints interrupt mask register (OTG_DAININTMSK)	2703
57.14.46 OTG device V _{BUS} discharge time register (OTG_DVBUSDIS)	2704
57.14.47 OTG device V _{BUS} pulsing time register (OTG_DVBUSPULSE)	2704
57.14.48 OTG device threshold control register (OTG_DTHRCTL)	2705
57.14.49 OTG device IN endpoint FIFO empty interrupt mask register (OTG_DIEPEMPMSK)	2706
57.14.50 OTG device each endpoint interrupt register (OTG_DEACHINT) ...	2706
57.14.51 OTG device each endpoint interrupt mask register (OTG_DEACHINTMSK)	2707
57.14.52 OTG device each IN endpoint-1 interrupt mask register (OTG_HS_DIEPEACHMSK1)	2707
57.14.53 OTG device each OUT endpoint-1 interrupt mask register (OTG_HS_DOEPEACHMSK1)	2708
57.14.54 OTG device IN endpoint x control register (OTG_DIEPCTLx)	2710
57.14.55 OTG device IN endpoint x interrupt register (OTG_DIEPINTx)	2712
57.14.56 OTG device IN endpoint 0 transfer size register (OTG_DIEPTSIZ0)	2714
57.14.57 OTG device IN endpoint x DMA address register (OTG_DIEPDMAx)	2714
57.14.58 OTG device IN endpoint transmit FIFO status register (OTG_DTXFSTSx)	2715
57.14.59 OTG device IN endpoint x transfer size register (OTG_DIEPTSIZx) .	2715
57.14.60 OTG device control OUT endpoint 0 control register (OTG_DOEPCTL0)	2716

57.14.61	OTG device OUT endpoint x interrupt register (OTG_DOEPINTx) . .	2718
57.14.62	OTG device OUT endpoint 0 transfer size register (OTG_DOEPTSIZ0)	2720
57.14.63	OTG device OUT endpoint x DMA address register . . (OTG_DOEPDMAx) 2721	
57.14.64	OTG device OUT endpoint x control register OTG_DOEPCTLx)	2721
57.14.65	OTG device OUT endpoint x transfer size register (OTG_DOEPTSIZx)	2724
57.14.66	OTG power and clock gating control register (OTG_PCGCCTL) . . .	2725
57.14.67	OTG_HS register map	2726
57.15	OTG_HS programming model	2738
57.15.1	Core initialization	2738
57.15.2	Host initialization	2738
57.15.3	Device initialization	2739
57.15.4	DMA mode	2740
57.15.5	Host programming model	2740
57.15.6	Device programming model	2773
57.15.7	Worst case response time	2793
57.15.8	OTG programming model	2795
58	Ethernet (ETH): media access control (MAC) with DMA controller	2801
58.1	Ethernet introduction	2801
58.2	Ethernet main features	2801
58.2.1	MAC core features	2801
58.2.2	DMA features	2804
58.2.3	Bus interface features	2804
58.3	Ethernet pins and internal signals	2805
58.4	Ethernet architecture	2807
58.4.1	DMA controller	2808
58.4.2	MTL	2814
58.4.3	MAC	2815
58.5	Ethernet functional description: MAC	2820
58.5.1	Double VLAN processing	2820
58.5.2	Source Address and VLAN insertion, replacement, or deletion	2821
58.5.3	Packet filtering	2823
58.5.4	IEEE 1588 timestamps	2830

58.5.5	IPv4 ARP offload	2843
58.5.6	TCP segmentation offload	2843
58.5.7	Loopback	2847
58.5.8	Flow control	2847
58.5.9	Checksum offload engine	2849
58.5.10	MAC management counters	2854
58.5.11	Interrupts generated by the MAC	2854
58.5.12	MAC and MMC register descriptions	2854
58.6	Ethernet functional description: PHY interfaces	2855
58.6.1	Station management agent (SMA)	2855
58.6.2	Media Independent Interface (MII)	2858
58.6.3	Reduced media independent interface (RMII)	2860
58.7	Ethernet low-power modes	2863
58.7.1	Energy Efficient Ethernet	2863
58.7.2	Power management	2864
58.7.3	Power-down and wakeup sequence	2866
58.8	Ethernet interrupts	2867
58.8.1	DMA interrupts	2867
58.8.2	MTL interrupts	2869
58.8.3	MAC Interrupts	2869
58.9	Ethernet programming model	2870
58.9.1	DMA initialization	2870
58.9.2	MTL initialization	2871
58.9.3	MAC initialization	2871
58.9.4	Performing normal receive and transmit operation	2872
58.9.5	Stopping and starting transmission	2873
58.9.6	Programming guidelines for MII link state transitions	2873
58.9.7	Programming guidelines for IEEE 1588 timestamping	2874
58.9.8	Programming guidelines for Energy Efficient Ethernet (EEE)	2875
58.9.9	Programming guidelines for flexible pulse-per-second (PPS) output	2876
58.9.10	Programming guidelines for TSO	2878
58.9.11	Programming guidelines to perform VLAN filtering on the receive	2878
58.10	Descriptors	2879
58.10.1	Descriptor overview	2879
58.10.2	Descriptor structure	2879
58.10.3	Transmit descriptor	2881

58.10.4	Receive descriptor	2893
58.11	Ethernet registers	2905
58.11.1	Ethernet registers maps	2905
58.11.2	Ethernet DMA registers	2905
58.11.3	Ethernet MTL registers	2928
58.11.4	Ethernet MAC and MMC registers	2940
59	HDMI-CEC controller (CEC)	3037
59.1	Introduction	3037
59.2	HDMI-CEC controller main features	3037
59.3	HDMI-CEC functional description	3038
59.3.1	HDMI-CEC pin and internal signals	3038
59.3.2	HDMI-CEC block diagram	3039
59.3.3	Message description	3039
59.3.4	Bit timing	3040
59.4	Arbitration	3040
59.4.1	SFT option bit	3042
59.5	Error handling	3042
59.5.1	Bit error	3042
59.5.2	Message error	3043
59.5.3	Bit rising error (BRE)	3043
59.5.4	Short bit period error (SBPE)	3043
59.5.5	Long bit period error (LBPE)	3043
59.5.6	Transmission error detection (TXERR)	3045
59.6	HDMI-CEC interrupts	3046
59.7	HDMI-CEC registers	3047
59.7.1	CEC control register (CEC_CR)	3047
59.7.2	CEC configuration register (CEC_CFGR)	3048
59.7.3	CEC Tx data register (CEC_TXDR)	3050
59.7.4	CEC Rx data register (CEC_RXDR)	3050
59.7.5	CEC interrupt and status register (CEC_ISR)	3050
59.7.6	CEC interrupt enable register (CEC_IER)	3052
59.7.7	HDMI-CEC register map	3054
60	Debug infrastructure	3055
60.1	Introduction	3055

60.2	Debug infrastructure features	3056
60.3	Debug infrastructure functional description	3056
60.3.1	Debug infrastructure block diagram	3056
60.3.2	Debug infrastructure pins and internal signals	3057
60.3.3	Debug infrastructure powering, clocking and reset	3058
60.4	Debug access port functional description	3060
60.4.1	Serial-wire and JTAG debug port (SWJ-DP)	3060
60.4.2	Access ports	3074
60.5	Trace and debug subsystem functional description	3080
60.5.1	System ROM tables	3080
60.5.2	Global timestamp generator (TSG)	3089
60.5.3	Cross trigger interfaces (CTI) and matrix (CTM)	3096
60.5.4	Trace funnel (CSTF)	3116
60.5.5	Embedded trace FIFO (ETF)	3126
60.5.6	Trace port interface unit (TPIU)	3148
60.5.7	Serial wire output (SWO) and SWO trace funnel (SWTF)	3166
60.5.8	Microcontroller debug unit (DBGMCU)	3189
60.6	Cortex-M7 debug functional description	3195
60.6.1	Cortex-M7 ROM tables	3196
60.6.2	Cortex-M7 data watchpoint and trace unit (DWT)	3208
60.6.3	Cortex-M7 instrumentation trace macrocell (ITM)	3221
60.6.4	Cortex-M7 breakpoint unit (FPB)	3230
60.6.5	Cortex-M7 embedded trace macrocell (ETM)	3238
60.6.6	Cortex-M7 cross trigger interface (CTI)	3270
60.7	References for debug infrastructure	3270
61	Device electronic signature	3271
61.1	Unique device ID register (96 bits)	3271
61.2	Flash size	3272
61.3	Package data register	3272
62	Revision history	3273

List of tables

Table 1.	Peripherals versus products	101
Table 2.	Availability of security features	102
Table 3.	Bus-master-to-bus-slave interconnect	103
Table 4.	ASIB configuration	109
Table 5.	AMIB configuration	110
Table 6.	AXI interconnect register map and reset values	120
Table 7.	Memory map and default device memory area attributes	129
Table 8.	Register boundary addresses	131
Table 9.	Boot modes	137
Table 10.	RAMECC internal input/output signals	141
Table 11.	ECC controller mapping	141
Table 12.	RAMECC register map and reset values	146
Table 13.	FLASH internal input/output signals	149
Table 14.	Flash memory organization on STM32H750xB devices	152
Table 15.	Flash memory organization on STM32H742xI/743xI/753xI devices	152
Table 16.	Flash memory organization on STM32H742xG/743xG devices	153
Table 17.	FLASH recommended number of wait states and programming delay	159
Table 18.	FLASH parallelism parameter	163
Table 19.	FLASH AXI interface memory map vs swapping option	169
Table 20.	Flash register map vs swapping option	171
Table 21.	Option byte organization	176
Table 22.	Flash interface register protection summary	182
Table 23.	RDP value vs readout protection level	183
Table 24.	Protection vs RDP Level	185
Table 25.	RDP transition and its effects	186
Table 26.	Effect of low-power modes on the embedded Flash memory	190
Table 27.	Flash interrupt request	198
Table 28.	Register map and reset value table	240
Table 29.	List of preferred terms	245
Table 30.	Summary of Flash protected areas access rights	251
Table 31.	PWR input/output signals connected to package pins or balls	254
Table 32.	PWR internal input/output signals	254
Table 33.	Supply configuration control	258
Table 34.	Low-power mode summary	276
Table 35.	PDDS_Dn low-power mode control	280
Table 36.	Low-power exit mode flags	282
Table 37.	CSleep mode	291
Table 38.	CStop mode	292
Table 39.	DStop mode overview	293
Table 40.	DStop mode	294
Table 41.	Stop mode operation	295
Table 42.	Stop mode	296
Table 43.	DStandby mode	297
Table 44.	Standby and Stop flags	299
Table 45.	Standby mode	299
Table 46.	Low-power modes monitoring pin overview	300
Table 47.	GPIO state according to CPU and domain state	300
Table 48.	Power control register map and reset values	310

Table 49.	BDMA and DMAMUX2 initialization sequence (DMAMUX2_INIT)	318
Table 50.	LPUART1 Initial programming (LPUART1_INIT)	320
Table 51.	LPUART1 start programming (LPUART1_Start)	320
Table 52.	RCC input/output signals connected to package pins or balls	324
Table 53.	RCC internal input/output signals	325
Table 54.	Reset distribution summary	328
Table 55.	Reset source identification (RCC_RSR)	330
Table 56.	Ratio between clock timer and pclk	348
Table 57.	STOPWUCK and STOPKERWUCK description	349
Table 58.	HSIKERON and CSIKERON behavior	350
Table 59.	Kernel clock distribution overview	352
Table 60.	System states overview	367
Table 61.	Peripheral clock enabling for D1 and D2 peripherals	372
Table 62.	Peripheral clock enabling for D3 peripherals	373
Table 63.	Interrupt sources and control	377
Table 64.	RCC_RSR address offset and reset value	446
Table 65.	RCC_AHB3ENR address offset and reset value	448
Table 66.	RCC_AHB1ENR address offset and reset value	450
Table 67.	RCC_AHB2ENR address offset and reset value	452
Table 68.	RCC_AHB4ENR address offset and reset value	454
Table 69.	RCC_APB3ENR address offset and reset value	457
Table 70.	RCC_APB1ENR address offset and reset value	458
Table 71.	RCC_APB1ENR address offset and reset value	462
Table 72.	RCC_APB2ENR address offset and reset value	464
Table 73.	RCC_APB4ENR address offset and reset value	467
Table 74.	RCC_AHB3LPENR address offset and reset value	470
Table 75.	RCC_AHB1LPENR address offset and reset value	472
Table 76.	RCC_AHB2LPENR address offset and reset value	474
Table 77.	RCC_AHB4LPENR address offset and reset value	476
Table 78.	RCC_APB3LPENR address offset and reset value	479
Table 79.	RCC_APB1LLPENR address offset and reset value	480
Table 80.	RCC_APB1HLPENR address offset and reset value	484
Table 81.	RCC_APB2LPENR address offset and reset value	486
Table 82.	RCC_APB4LPENR address offset and reset value	489
Table 83.	RCC register map and reset values	492
Table 84.	CRS features	502
Table 85.	CRS internal input/output signals	503
Table 86.	Effect of low-power modes on CRS	507
Table 87.	Interrupt control bits	507
Table 88.	CRS register map and reset values	513
Table 89.	HSEM internal input/output signals	515
Table 90.	Authorized AHB bus master ID	520
Table 91.	HSEM register map and reset values	526
Table 92.	Port bit configuration table	529
Table 93.	GPIO register map and reset values	544
Table 94.	SYSCFG register map and reset values	568
Table 95.	Peripherals interconnect matrix (D2 domain)	572
Table 96.	Peripherals interconnect matrix (D3 domain)	573
Table 97.	Peripherals interconnect matrix details	574
Table 98.	EXTI wakeup inputs	591
Table 99.	EXTI pending requests clear inputs	594
Table 100.	MDMA	596

Table 101.	DMAMUX1, DMA1 and DMA2 connections	598
Table 102.	DMAMUX2 and BDMA connections	603
Table 103.	MDMA internal input/output signals	608
Table 104.	MDMA interrupt requests	615
Table 105.	MDMA register map and reset values	633
Table 106.	DMA internal input/output signals	637
Table 107.	Source and destination address	639
Table 108.	Source and destination address registers in double-buffer mode (DBM = 1)	645
Table 109.	Packing/unpacking and endian behavior (bit PINC = MINC = 1)	646
Table 110.	Restriction on NDT versus PSIZE and MSIZE	647
Table 111.	FIFO threshold configurations	649
Table 112.	Possible DMA configurations	653
Table 113.	DMA interrupt requests	655
Table 114.	DMA register map and reset values	666
Table 115.	BDMA implementation	671
Table 116.	BDMA internal input/output signals	672
Table 117.	Programmable data width and endian behavior (when PINC = MINC = 1)	678
Table 118.	BDMA interrupt requests	680
Table 119.	BDMA register map and reset values	690
Table 120.	DMAMUX1 and DMAMUX2 instantiation	694
Table 121.	DMAMUX1: assignment of multiplexer inputs to resources	695
Table 122.	DMAMUX1: assignment of trigger inputs to resources	696
Table 123.	DMAMUX1: assignment of synchronization inputs to resources	696
Table 124.	DMAMUX2: assignment of multiplexer inputs to resources	696
Table 125.	DMAMUX2: assignment of trigger inputs to resources	697
Table 126.	DMAMUX2: assignment of synchronization inputs to resources	697
Table 127.	DMAMUX signals	700
Table 128.	DMAMUX interrupts	704
Table 129.	DMAMUX register map and reset values	713
Table 130.	DMA2D internal input/output signals	717
Table 131.	Supported color mode in input	718
Table 132.	Data order in memory	720
Table 133.	Alpha mode configuration	720
Table 134.	Supported CLUT color mode	721
Table 135.	CLUT data order in memory	721
Table 136.	Supported color mode in output	722
Table 137.	Data order in memory	723
Table 138.	Standard data order in memory	724
Table 139.	Output FIFO byte reordering steps	725
Table 140.	MCU order in memory	730
Table 141.	DMA2D interrupt requests	731
Table 142.	DMA2D register map and reset values	751
Table 143.	NVIC	754
Table 144.	EXTI Event input configurations and register control	764
Table 145.	Configurable Event input Asynchronous Edge detector reset	766
Table 146.	EXTI Event input mapping	771
Table 147.	Masking functionality	773
Table 148.	Asynchronous interrupt/event controller register map and reset values	791
Table 149.	CRC internal input/output signals	795
Table 150.	CRC register map and reset values	799
Table 151.	FMC pins	803
Table 152.	FMC bank mapping options	806

Table 153.	NOR/PSRAM bank selection	806
Table 154.	NOR/PSRAM External memory address	806
Table 155.	NAND memory mapping and timing registers.	807
Table 156.	NAND bank selection	807
Table 157.	SDRAM bank selection	807
Table 158.	SDRAM address mapping	808
Table 159.	SDRAM address mapping with 8-bit data bus width.	808
Table 160.	SDRAM address mapping with 16-bit data bus width.	809
Table 161.	SDRAM address mapping with 32-bit data bus width.	810
Table 162.	Programmable NOR/PSRAM access parameters	812
Table 163.	Non-multiplexed I/O NOR Flash memory	812
Table 164.	16-bit multiplexed I/O NOR Flash memory	813
Table 165.	Non-multiplexed I/Os PSRAM/SRAM	813
Table 166.	16-Bit multiplexed I/O PSRAM	813
Table 167.	NOR Flash/PSRAM: Example of supported memories and transactions	814
Table 168.	FMC_BCRx bitfields (mode 1)	817
Table 169.	FMC_BTRx bitfields (mode 1)	818
Table 170.	FMC_BCRx bitfields (mode A)	820
Table 171.	FMC_BTRx bitfields (mode A)	821
Table 172.	FMC_BWTRx bitfields (mode A)	821
Table 173.	FMC_BCRx bitfields (mode 2/B)	823
Table 174.	FMC_BTRx bitfields (mode 2/B)	824
Table 175.	FMC_BWTRx bitfields (mode 2/B)	824
Table 176.	FMC_BCRx bitfields (mode C)	826
Table 177.	FMC_BTRx bitfields (mode C)	827
Table 178.	FMC_BWTRx bitfields (mode C)	827
Table 179.	FMC_BCRx bitfields (mode D)	829
Table 180.	FMC_BTRx bitfields (mode D)	829
Table 181.	FMC_BWTRx bitfields (mode D)	830
Table 182.	FMC_BCRx bitfields (Muxed mode)	832
Table 183.	FMC_BTRx bitfields (Muxed mode)	832
Table 184.	FMC_BCRx bitfields (Synchronous multiplexed read mode)	838
Table 185.	FMC_BTRx bitfields (Synchronous multiplexed read mode)	838
Table 186.	FMC_BCRx bitfields (Synchronous multiplexed write mode)	839
Table 187.	FMC_BTRx bitfields (Synchronous multiplexed write mode)	840
Table 188.	Programmable NAND Flash access parameters	850
Table 189.	8-bit NAND Flash memory	850
Table 190.	16-bit NAND Flash memory	851
Table 191.	Supported memories and transactions	851
Table 192.	ECC result relevant bits	861
Table 193.	SDRAM signals	862
Table 194.	FMC register map	879
Table 195.	QUADSPI internal signals	884
Table 196.	QUADSPI pins	884
Table 197.	QUADSPI interrupt requests	898
Table 198.	QUADSPI register map and reset values	912
Table 199.	DLYB internal input/output signals	914
Table 200.	Delay block control	914
Table 201.	DLYB register map and reset values	917
Table 202.	ADC features	920
Table 203.	ADC input/output pins	922

Table 204.	ADC internal input/output signals	922
Table 205.	ADC interconnection	923
Table 206.	Configuring the trigger polarity for regular external triggers	945
Table 207.	Configuring the trigger polarity for injected external triggers	945
Table 208.	ADC1, ADC2 and ADC3 - External triggers for regular channels	946
Table 209.	ADC1, ADC2 and ADC3 - External triggers for injected channels	947
Table 210.	TSAR timings depending on resolution	959
Table 211.	Offset computation versus data resolution	962
Table 212.	16-bit data formats	965
Table 213.	Numerical examples for 16-bit format (bold indicates saturation).	965
Table 214.	Analog watchdog channel selection	974
Table 215.	Analog watchdog 1,2,3 comparison	975
Table 216.	Oversampler operating modes summary	983
Table 217.	ADC interrupts per each ADC.	1002
Table 218.	DELAY bits versus ADC resolution.	1044
Table 219.	ADC global register map.	1047
Table 220.	ADC register map and reset values for each ADC (offset=0x000 for master ADC, 0x100 for slave ADC).	1047
Table 221.	ADC register map and reset values (master and slave ADC common registers) offset =0x300)	1050
Table 222.	DAC implementation	1052
Table 223.	DAC input/output pins.	1054
Table 224.	DAC internal input/output signals	1054
Table 225.	DAC interconnection	1055
Table 226.	Sample and refresh timings	1062
Table 227.	Channel output modes summary	1063
Table 228.	Effect of low-power modes on DAC	1069
Table 229.	DAC interrupts	1070
Table 230.	DAC register map and reset values	1087
Table 231.	VREF buffer modes	1089
Table 232.	VREFBUF register map and reset values.	1091
Table 233.	COMP input/output internal signals	1094
Table 234.	COMP input/output pins	1094
Table 235.	COMP1_OUT assignment to GPIOs	1097
Table 236.	COMP2_OUT assignment to GPIOs	1097
Table 237.	Comparator behavior in the low-power modes	1099
Table 238.	Interrupt control bits	1099
Table 239.	Interrupt control bits	1100
Table 240.	COMP register map and reset values.	1107
Table 241.	Operational amplifier possible connections	1109
Table 242.	Operating modes and calibration	1116
Table 243.	Effect of low-power modes on the OPAMP	1118
Table 244.	OPAMP register map and reset values	1125
Table 245.	DFSDM1 implementation	1128
Table 246.	DFSDM external pins	1130
Table 247.	DFSDM internal signals	1130
Table 248.	DFSDM triggers connection	1130
Table 249.	DFSDM break connection.	1131
Table 250.	Filter maximum output resolution (peak data values from filter output) for some FOSR values	1146
Table 251.	Integrator maximum output resolution (peak data values from integrator output) for some IOSR values and FOSR = 256 and Sinc3 filter type (largest data) . . .	1147

Table 252.	DFSDM interrupt requests	1155
Table 253.	DFSDM register map and reset values	1175
Table 254.	DCMI input/output pins	1186
Table 255.	DCMI internal input/output signals	1186
Table 256.	Positioning of captured data bytes in 32-bit words (8-bit width)	1188
Table 257.	Positioning of captured data bytes in 32-bit words (10-bit width)	1188
Table 258.	Positioning of captured data bytes in 32-bit words (12-bit width)	1188
Table 259.	Positioning of captured data bytes in 32-bit words (14-bit width)	1189
Table 260.	Data storage in monochrome progressive video format	1194
Table 261.	Data storage in RGB progressive video format	1195
Table 262.	Data storage in YCbCr progressive video format	1195
Table 263.	Data storage in YCbCr progressive video format - Y extraction mode	1196
Table 264.	DCMI interrupts	1196
Table 265.	DCMI register map and reset values	1208
Table 266.	LCD-TFT internal signals	1210
Table 267.	LTDC pins and signal interface	1211
Table 268.	Clock domain for each register	1212
Table 269.	LTDC register access and update durations	1213
Table 270.	Pixel data mapping versus color format	1217
Table 271.	LTDC interrupt requests	1221
Table 272.	LTDC register map and reset values	1241
Table 273.	JPEG internal signals	1245
Table 274.	JPEG codec interrupt requests	1249
Table 275.	JPEG codec register map and reset values	1262
Table 276.	RNG internal input/output signals	1265
Table 277.	RNG interrupt requests	1272
Table 278.	RNG register map and reset map	1277
Table 279.	CRYP internal input/output signals	1281
Table 280.	Counter mode initialization vector	1306
Table 281.	GCM last block definition	1309
Table 282.	GCM mode IV registers initialization	1309
Table 283.	CCM mode IV registers initialization	1316
Table 284.	DES/TDES data swapping feature	1322
Table 285.	AES data swapping feature	1324
Table 286.	Key registers CRYPT_KxR/LR endianness (TDES K1/2/3 and AES 128/192/256-bit keys)	1324
Table 287.	Initialization vector registers CRYPT_IVxR endianness	1325
Table 288.	Cryptographic processor configuration for memory-to-peripheral DMA transfers	1326
Table 289.	Cryptographic processor configuration for peripheral-to-memory DMA transfers	1327
Table 290.	CRYP interrupt requests	1328
Table 291.	Processing latency for ECB, CBC and CTR	1330
Table 292.	Processing time (in clock cycle) for GCM and CCM per 128-bit block	1330
Table 293.	CRYP register map and reset values	1344
Table 294.	HASH internal input/output signals	1349
Table 295.	Hash processor outputs	1352
Table 296.	HASH interrupt requests	1359
Table 297.	Processing time (in clock cycle)	1359
Table 298.	HASH1 register map and reset values	1369
Table 299.	HRTIM Input/output summary	1375
Table 300.	Timer resolution and min. PWM frequency for $f_{HRTIM} = 400$ MHz	1377

Table 301.	Period and Compare registers min and max values	1379
Table 302.	Timer operating modes	1380
Table 303.	Events mapping across Timer A to E	1385
Table 304.	Deadtime resolution and max absolute values	1393
Table 305.	External events mapping and associated features	1400
Table 306.	Output set/reset latency and jitter vs external event operating mode	1401
Table 307.	Filtering signals mapping per time	1404
Table 308.	Windowing signals mapping per timer (EEFLTR[3:0] = 1111)	1406
Table 309.	HRTIM preloadable control registers and associated update sources	1415
Table 310.	Update enable inputs and sources	1416
Table 311.	Master timer update event propagation	1418
Table 312.	TIMx update event propagation	1418
Table 313.	Reset events able to generate an update	1419
Table 314.	Update event propagation for a timer reset	1420
Table 315.	Output state programming, x= A..E, y = 1 or 2	1421
Table 316.	Timer output programming for burst mode	1424
Table 317.	Burst mode clock sources from general purpose timer	1426
Table 318.	Fault inputs	1434
Table 319.	Sampling rate and filter length vs FLTxFx[3:0] and clock setting	1435
Table 320.	Effect of sync event vs timer operating modes	1440
Table 321.	HRTIM interrupt summary	1446
Table 322.	HRTIM DMA request summary	1447
Table 323.	RTIM global register map	1537
Table 324.	HRTIM Register map and reset values: Master timer	1537
Table 325.	HRTIM Register map and reset values: TIMx (x= A..E)	1539
Table 326.	HRTIM Register map and reset values: Common functions	1543
Table 327.	Behavior of timer outputs versus BRK/BRK2 inputs	1587
Table 328.	Counting direction versus encoder signals	1594
Table 329.	TIMx internal trigger connection	1611
Table 330.	Output control bits for complementary OCx and OCxN channels with break feature	1625
Table 331.	TIM1 register map and reset values	1645
Table 332.	TIM8 register map and reset values	1647
Table 333.	Counting direction versus encoder signals	1683
Table 334.	TIMx internal trigger connection	1700
Table 335.	Output control bit for standard OCx channels	1711
Table 336.	TIM2/TIM3/TIM4/TIM5 register map and reset values	1723
Table 337.	TIMx internal trigger connection	1754
Table 338.	Output control bit for standard OCx channels	1762
Table 339.	TIM12 register map and reset values	1765
Table 340.	Output control bit for standard OCx channels	1774
Table 341.	TIM13/TIM14 register map and reset values	1777
Table 342.	TIMx Internal trigger connection	1821
Table 343.	Output control bits for complementary OCx and OCxN channels with break feature (TIM15)	1831
Table 344.	TIM15 register map and reset values	1840
Table 345.	Output control bits for complementary OCx and OCxN channels with break feature (TIM16/17)	1852
Table 346.	TIM16/TIM17 register map and reset values	1863
Table 347.	TIMx register map and reset values	1877
Table 348.	STM32H742, STM32H743/753 and STM32H750 LPTIM features	1879
Table 349.	LPTIM input/output pins	1881
Table 350.	LPTIM internal signals	1881

Table 351.	LPTIM1 external trigger connection	1881
Table 352.	LPTIM2 external trigger connection	1882
Table 353.	LPTIM3 external trigger connection	1882
Table 354.	LPTIM4 external trigger connection	1882
Table 355.	LPTIM5 external trigger connection	1883
Table 356.	LPTIM1 input 1 connection	1883
Table 357.	LPTIM1 input 2 connection	1883
Table 358.	LPTIM2 input 1 connection	1883
Table 359.	LPTIM2 input 2 connection	1884
Table 360.	LPTIM3 input 1 connection	1884
Table 361.	Prescaler division ratios	1885
Table 362.	Encoder counting scenarios	1892
Table 363.	Effect of low-power modes on the LPTIM	1893
Table 364.	Interrupt events	1894
Table 365.	LPTIM register map and reset values	1905
Table 366.	WWDG internal input/output signals	1908
Table 367.	WWDG register map and reset values	1912
Table 368.	IWDG internal input/output signals	1914
Table 369.	IWDG register map and reset values	1922
Table 370.	RTC pins and internal signals	1927
Table 371.	RTC pin PC13 configuration	1927
Table 372.	RTC_OUT mapping	1928
Table 373.	RTC functions over modes	1929
Table 374.	Effect of low-power modes on RTC	1941
Table 375.	Interrupt control bits	1942
Table 376.	RTC register map and reset values	1967
Table 377.	STM32H742, STM32H743/753 and STM32H750 I2C implementation	1970
Table 378.	I2C input/output pins	1972
Table 379.	I2C internal input/output signals	1972
Table 380.	Comparison of analog vs. digital filters	1974
Table 381.	I2C-SMBus specification data setup and hold times	1977
Table 382.	I2C configuration	1981
Table 383.	I2C-SMBus specification clock timings	1992
Table 384.	Examples of timing settings for fI2CCLK = 8 MHz	2002
Table 385.	Examples of timings settings for fI2CCLK = 16 MHz	2002
Table 386.	Examples of timings settings for fI2CCLK = 48 MHz	2003
Table 387.	SMBus timeout specifications	2005
Table 388.	SMBus with PEC configuration	2007
Table 389.	Examples of TIMEOUTA settings for various i2c_ker_ck frequencies (max t _{TIMEOUT} = 25 ms)	2008
Table 390.	Examples of TIMEOUTB settings for various i2c_ker_ck frequencies	2008
Table 391.	Examples of TIMEOUTA settings for various i2c_ker_ck frequencies (max t _{IDLE} = 50 μs)	2009
Table 392.	Effect of low-power modes on the I2C	2020
Table 393.	I2C Interrupt requests	2021
Table 394.	I2C register map and reset values	2037
Table 395.	USART/LPUART features	2041
Table 396.	Noise detection from sampled data	2056
Table 397.	Tolerance of the USART receiver when BRR[3:0] = 0000	2059
Table 398.	Tolerance of the USART receiver when BRR[3:0] is different from 0000	2060
Table 399.	USART frame formats	2065
Table 400.	USART interrupt requests	2088

Table 401.	USART register map and reset values	2125
Table 402.	USART/LPUART features	2129
Table 403.	Error calculation for programmed baud rates at lpuart_ker_ck_pres= 32,768 KHz	2140
Table 404.	Error calculation for programmed baud rates at fCK = 100 MHz	2141
Table 405.	Tolerance of the LPUART receiver	2142
Table 407.	LPUART interrupt requests	2153
Table 408.	LPUART register map and reset values	2178
Table 409.	STM32H7xx SPI features	2181
Table 410.	SPI wakeup and interrupt requests	2210
Table 411.	Bit fields usable in PCM/I2S mode	2213
Table 412.	WS and CK level before SPI/I2S is enabled when AFCNTR = 1	2221
Table 413.	Serial data line swapping	2221
Table 414.	CLKGEN programming examples for usual I2S frequencies	2226
Table 415.	I2S interrupt requests	2235
Table 416.	SPI register map and reset values	2253
Table 417.	STM32H743/753/745/755/747/757 SAI interfaces	2256
Table 418.	SAI internal input/output signals	2258
Table 419.	SAI input/output pins	2258
Table 420.	External synchronization selection	2261
Table 421.	Clock generator programming examples	2268
Table 422.	TDM settings	2275
Table 423.	Allowed TDM frame configuration	2277
Table 424.	SOPD pattern	2281
Table 425.	Parity bit calculation	2281
Table 426.	Audio sampling frequency versus symbol rates	2282
Table 427.	SAI interrupt sources	2291
Table 428.	SAI register map and reset values	2321
Table 429.	SPDIFRX internal input/output signals	2324
Table 430.	SPDIFRX pins	2324
Table 431.	Transition sequence for preamble	2330
Table 432.	Minimum spdifrx_ker_ck frequency versus audio sampling rate	2340
Table 433.	Bit field property versus SPDIFRX state	2342
Table 434.	SPDIFRX interface register map and reset values	2357
Table 435.	SWPMI input/output signals connected to package pins or balls	2360
Table 436.	SWPMI internal input/output signals	2361
Table 437.	Effect of low-power modes on SWPMI	2375
Table 438.	Interrupt control bits	2376
Table 439.	Buffer modes selection for transmission/reception	2378
Table 440.	SWPMI register map and reset values	2385
Table 441.	MDIOS input/output signals connected to package pins or balls	2387
Table 442.	MDIOS internal input/output signals	2387
Table 443.	Interrupt control bits	2392
Table 444.	MDIOS register map and reset values	2399
Table 445.	SDMMC features	2400
Table 446.	SDMMC operation modes SD & SDIO	2403
Table 447.	SDMMC operation modes eMMC	2403
Table 448.	SDMMC internal input/output signals	2404
Table 449.	SDMMC pins	2405
Table 450.	SDMMC Command and data phase selection	2406
Table 451.	Command token format	2412
Table 452.	Short response with CRC token format	2413
Table 453.	Short response without CRC token format	2413

Table 454.	Long response with CRC token format	2413
Table 455.	Specific Commands overview	2414
Table 456.	Command path status flags	2415
Table 457.	Command path error handling	2415
Table 458.	Data token format	2423
Table 459.	Data path status flags and clear bits	2423
Table 460.	Data path error handling	2425
Table 461.	Data FIFO access	2426
Table 462.	Transmit FIFO status flags	2427
Table 463.	Receive FIFO status flags	2428
Table 464.	SDMMC connections to MDMA	2431
Table 465.	AHB and SDMMC_CK clock frequency relation	2432
Table 466.	SDIO special operation control	2433
Table 467.	4-bit mode Start, interrupt, and CRC-status Signaling detection	2437
Table 468.	CMD12 use cases	2441
Table 469.	SDMMC interrupts	2455
Table 470.	Response type and SDMMC_RESPxR registers	2463
Table 471.	SDMMC register map	2479
Table 472.	CAN subsystem I/O signals	2482
Table 473.	CAN subsystem I/O pins	2483
Table 474.	Main features	2485
Table 475.	DLC coding in FDCAN	2490
Table 476.	Example of filter configuration for Rx buffers	2502
Table 477.	Example of filter configuration for Debug messages	2503
Table 478.	Possible configurations for frame transmission	2503
Table 479.	Tx buffer/FIFO - queue element size	2504
Table 480.	First byte of level 1 reference message	2514
Table 481.	First four bytes of level 2 reference message	2515
Table 482.	First four bytes of level 0 reference message	2515
Table 483.	TUR configuration example	2516
Table 484.	System matrix, Node A	2521
Table 485.	Trigger list, Node A	2522
Table 486.	Number of data bytes transmitted with a reference message	2529
Table 487.	Rx buffer and FIFO element	2536
Table 488.	Rx buffer and FIFO element description	2536
Table 489.	Tx buffer and FIFO element	2538
Table 490.	Tx buffer element description	2538
Table 491.	Tx Event FIFO element	2540
Table 492.	Tx Event FIFO element description	2540
Table 493.	Standard message ID filter element	2541
Table 494.	Standard message ID filter element field description	2542
Table 495.	Extended message ID filter element	2543
Table 496.	Extended message ID filter element field description	2543
Table 497.	Trigger memory element	2544
Table 498.	Trigger memory element description	2544
Table 499.	FDCAN register map and reset values	2603
Table 500.	CCU register map and reset values	2613
Table 501.	OTG_HS speeds supported	2615
Table 502.	OTG_HS implementation	2617
Table 503.	OTG_FS input/output pins	2619
Table 504.	OTG_HS input/output pins	2619
Table 505.	OTG_HS input/output signals	2619

Table 506.	Compatibility of STM32 low power modes with the OTG	2630
Table 507.	Core global control and status registers (CSRs).	2639
Table 508.	Host-mode control and status registers (CSRs)	2640
Table 509.	Device-mode control and status registers	2642
Table 510.	Data FIFO (DFIFO) access register map	2644
Table 511.	Power and clock gating control and status registers	2644
Table 512.	TRDT values	2653
Table 513.	TRDT values (HS)	2653
Table 514.	Minimum duration for soft disconnect	2698
Table 515.	OTG_HS register map and reset values	2726
Table 516.	Ethernet peripheral pins	2805
Table 517.	Ethernet internal input/output signals	2805
Table 518.	Double VLAN processing features in Tx path	2820
Table 519.	Double VLAN processing in Rx path	2821
Table 520.	VLAN insertion or replacement based on VLTi bit	2822
Table 521.	Destination address filtering	2826
Table 522.	Source address filtering	2827
Table 523.	VLAN match status	2828
Table 524.	Ordinary clock: PTP messages for snapshot	2833
Table 525.	End-to-end transparent clock: PTP messages for snapshot	2834
Table 526.	Peer-to-peer transparent clock: PTP messages for snapshot	2835
Table 527.	PTP message generation criteria	2842
Table 528.	TSO: TCP and IP header fields	2845
Table 529.	Pause packet fields	2847
Table 530.	Tx MAC flow control	2848
Table 531.	Rx MAC Flow Control	2849
Table 532.	Transmit checksum offload engine functions for different packet types	2851
Table 533.	Receive checksum offload engine functions for different packet types	2852
Table 534.	MCD clock selection	2856
Table 535.	MDIO packet field description	2856
Table 536.	RX interface signal encoding	2859
Table 537.	Transfer complete interrupt behavior	2868
Table 538.	TDES0 normal descriptor (read format)	2881
Table 539.	TDES1 normal descriptor (read format)	2882
Table 540.	TDES2 normal descriptor (read format)	2882
Table 541.	TDES3 normal descriptor (Read format)	2883
Table 542.	DES0 normal descriptor (write-back format)	2886
Table 543.	TDES1 normal descriptor (write-back format)	2886
Table 544.	TDES2 normal descriptor (write-back format)	2886
Table 545.	TDES3 normal descriptor (write-back format)	2887
Table 546.	TDES0 context descriptor	2890
Table 547.	TDES1 context descriptor	2890
Table 548.	TDES2 context descriptor	2891
Table 549.	TDES3 context descriptor	2891
Table 550.	RDES0 normal descriptor (read format)	2894
Table 551.	RDES1 normal descriptor (read format)	2894
Table 552.	RDES2 normal descriptor (read format)	2894
Table 553.	RDES3 normal descriptor (read format)	2895
Table 554.	RDES0 normal descriptor (write-back format)	2896
Table 555.	RDES1 normal descriptor (write-back format)	2897
Table 556.	RDES2 normal descriptor (write-back format)	2899
Table 557.	RDES3 normal descriptor (write-back format)	2900

Table 558.	RDES0 context descriptor	2903
Table 559.	RDES1 context descriptor	2904
Table 560.	RDES2 context descriptor	2904
Table 561.	RDES3 context descriptor	2904
Table 562.	ETH_DMA common register map and reset values	2926
Table 563.	ETH_DMA_CH register map and reset values	2926
Table 564.	ETH_MTL register map and reset values	2938
Table 565.	Giant Packet Status based on S2KP and JE Bits	2945
Table 566.	Packet Length based on the CST and ACS bits	2946
Table 567.	Remote wakeup packet filter register	2968
Table 568.	ETH_MACRWKPFRR	2969
Table 569.	Remote Wakeup Packet and PMT Interrupt Generation	2970
Table 570.	Timestamp Snapshot Dependency on Register Bits	3008
Table 571.	Ethernet MAC register map and reset values	3029
Table 572.	HDMI pin	3038
Table 573.	HDMI-CEC internal input/output signals	3038
Table 574.	Error handling timing parameters	3044
Table 575.	TXERR timing parameters	3045
Table 576.	HDMI-CEC interrupts	3046
Table 577.	HDMI-CEC register map and reset values	3054
Table 578.	JTAG/Serial-wire debug port pins	3057
Table 579.	Trace port pins	3057
Table 580.	Serial-wire trace port pins	3057
Table 581.	Trigger pins	3057
Table 582.	Packet request	3061
Table 583.	ACK response	3061
Table 584.	Data transfer	3061
Table 585.	JTAG-DP data registers	3064
Table 586.	Debug port registers	3066
Table 587.	MEM-AP registers	3076
Table 588.	System ROM table 1	3080
Table 589.	System ROM table 2	3081
Table 590.	System ROM table 1 register map and reset values	3087
Table 591.	System ROM table 2 register map and reset values	3088
Table 592.	TSG register map and reset values	3096
Table 593.	System CTI inputs	3097
Table 594.	System CTI outputs	3098
Table 595.	Cortex-M7 CTI inputs	3098
Table 596.	Cortex-M7 CTI outputs	3099
Table 597.	CTI register map and reset values	3114
Table 598.	CSTF register map and reset values	3125
Table 599.	ETF register map and reset values	3146
Table 600.	TPIU register map and reset values	3164
Table 601.	SWO register map and reset values	3176
Table 602.	SWTF register map and reset values	3187
Table 603.	DBGMCU register map and reset values	3195
Table 604.	Cortex-M7 processor ROM table	3196
Table 605.	Cortex-M7 PPB ROM table	3196
Table 606.	Cortex-M7 processor ROM table register map and reset values	3202
Table 607.	Cortex-M7 PPB ROM table register map and reset values	3207
Table 608.	Cortex-M7 DWT register map and reset values	3219
Table 609.	Cortex-M7 ITM register map and reset values	3229

Table 610.	Cortex-M7 FPB register map and reset values.	3236
Table 611.	Cortex-M7 ETM register map and reset values	3266
Table 612.	Document revision history	3273



List of figures

Figure 1.	System architecture for STM32H742xx, STM32H743/53xx and STM32H750xB devices	104
Figure 2.	AXI interconnect	109
Figure 3.	RAM ECC controller implementation schematic	140
Figure 4.	Connection between RAM ECC controller and RAMECC monitoring unit	140
Figure 5.	FLASH block diagram	148
Figure 6.	Detailed FLASH architecture	150
Figure 7.	Embedded Flash memory organization	151
Figure 8.	Embedded Flash memory usage	154
Figure 9.	FLASH protection mechanisms	155
Figure 10.	FLASH read pipeline architecture	158
Figure 11.	FLASH write pipeline architecture	161
Figure 12.	Flash bank swapping sequence	170
Figure 13.	RDP protection transition scheme	186
Figure 14.	Example of protected region overlapping	187
Figure 15.	Flash memory areas and services in Standard and Secure access modes	247
Figure 16.	Bootloader state machine in Secure access mode	248
Figure 17.	Core access to Flash memory areas	251
Figure 18.	Power control block diagram	253
Figure 19.	Power supply overview	257
Figure 20.	System supply configurations	258
Figure 21.	Device startup with V_{CORE} supplied from voltage regulator	260
Figure 22.	Device startup with V_{CORE} supplied in Bypass mode from external regulator	261
Figure 23.	Backup domain	265
Figure 24.	USB supply configurations	266
Figure 25.	Power-on reset/power-down reset waveform	267
Figure 26.	BOR thresholds	268
Figure 27.	PVD thresholds	269
Figure 28.	AVD thresholds	270
Figure 29.	VBAT thresholds	271
Figure 30.	Temperature thresholds	272
Figure 31.	Switching V_{CORE} from VOS1 to VOS0	278
Figure 32.	V_{CORE} voltage scaling versus system power modes	279
Figure 33.	Power control modes detailed state diagram	281
Figure 34.	Dynamic voltage scaling in Run mode	284
Figure 35.	Dynamic voltage scaling behavior with D1, D2 and system in Stop mode	285
Figure 36.	Dynamic Voltage Scaling D1, D2, system Standby mode	286
Figure 37.	Dynamic voltage scaling behavior with D1 and D2 in DStandby mode and D3 in autonomous mode	288
Figure 38.	EXTI, RCC and PWR interconnections	312
Figure 39.	Timing diagram of SRAM4-to-LPUART1 transfer with BDMA and D3 domain in Autonomous mode	316
Figure 40.	BDMA and DMAMUX2 interconnection	318
Figure 41.	Timing diagram of LPUART1 transmission with D3 domain in Autonomous mode	321
Figure 42.	RCC Block diagram	324
Figure 43.	System reset circuit	327

Figure 44.	Boot sequences versus system states	333
Figure 45.	Top-level clock tree	335
Figure 46.	HSE/LSE clock source	336
Figure 47.	PLL block diagram	343
Figure 48.	PLLs Initialization Flowchart	346
Figure 49.	Core and bus clock generation	348
Figure 50.	Kernel clock distribution for SAls and DFSDM	355
Figure 51.	Kernel clock distribution for SPIs and SPI/I2S	356
Figure 52.	Kernel clock distribution for I2Cs	357
Figure 53.	Kernel clock distribution for UARTs, USARTs and LPUART1	357
Figure 54.	Kernel clock distribution for LTDC	358
Figure 55.	Kernel clock distribution for SDMMC, QUADSPI and FMC	358
Figure 56.	Kernel clock distribution for USB ⁽²⁾	359
Figure 57.	Kernel clock distribution for Ethernet	360
Figure 58.	Kernel clock distribution for ADCs, SWPMI, RNG and FDCAN ⁽²⁾	361
Figure 59.	Kernel clock distribution for LPTIMs and HDMI-CEC ⁽²⁾	362
Figure 60.	Peripheral allocation example	365
Figure 61.	Kernel Clock switching	368
Figure 62.	Peripheral kernel clock enable logic details	371
Figure 63.	Bus clock enable logic	376
Figure 64.	RCC mapping overview	378
Figure 65.	CRS block diagram	503
Figure 66.	CRS counter behavior	505
Figure 67.	HSEM block diagram	515
Figure 68.	Procedure state diagram	516
Figure 69.	Interrupt state diagram	519
Figure 70.	Basic structure of an I/O port bit	528
Figure 71.	Basic structure of a 5-Volt tolerant I/O port bit	528
Figure 72.	Input floating/pull up/pull down configurations	533
Figure 73.	Output configuration	534
Figure 74.	Alternate function configuration	535
Figure 75.	High impedance-analog configuration	535
Figure 76.	Analog inputs connected to ADC inputs	536
Figure 77.	MDMA block diagram	608
Figure 78.	DMA block diagram	637
Figure 79.	Peripheral-to-memory mode	641
Figure 80.	Memory-to-peripheral mode	642
Figure 81.	Memory-to-memory mode	643
Figure 82.	FIFO structure	648
Figure 83.	BDMA block diagram	671
Figure 84.	DMAMUX block diagram	699
Figure 85.	Synchronization mode of the DMAMUX request line multiplexer channel	702
Figure 86.	Event generation of the DMA request line multiplexer channel	702
Figure 87.	DMA2D block diagram	717
Figure 88.	Intel 8080 16-bit mode (RGB565)	724
Figure 89.	Intel 8080 18/24-bit mode (RGB888)	724
Figure 90.	EXTI block diagram	763
Figure 91.	Configurable event triggering logic CPU wakeup	765
Figure 92.	Configurable event triggering logic Any wakeup	767
Figure 93.	Direct event triggering logic CPU Wakeup	768
Figure 94.	Direct event triggering logic Any Wakeup	769
Figure 95.	D3 domain Pending request clear logic	770

Figure 96.	CRC calculation unit block diagram	795
Figure 97.	FMC block diagram	802
Figure 98.	FMC memory banks (default mapping)	805
Figure 99.	Mode 1 read access waveforms	816
Figure 100.	Mode 1 write access waveforms	817
Figure 101.	Mode A read access waveforms	819
Figure 102.	Mode A write access waveforms	820
Figure 103.	Mode 2 and mode B read access waveforms	822
Figure 104.	Mode 2 write access waveforms	822
Figure 105.	Mode B write access waveforms	823
Figure 106.	Mode C read access waveforms	825
Figure 107.	Mode C write access waveforms	825
Figure 108.	Mode D read access waveforms	828
Figure 109.	Mode D write access waveforms	828
Figure 110.	Muxed read access waveforms	831
Figure 111.	Muxed write access waveforms	831
Figure 112.	Asynchronous wait during a read access waveforms	834
Figure 113.	Asynchronous wait during a write access waveforms	834
Figure 114.	Wait configuration waveforms	837
Figure 115.	Synchronous multiplexed read mode waveforms - NOR, PSRAM (CRAM)	837
Figure 116.	Synchronous multiplexed write mode waveforms - PSRAM (CRAM)	839
Figure 117.	NAND Flash controller waveforms for common memory access	853
Figure 118.	Access to non 'CE don't care' NAND-Flash	854
Figure 119.	Burst write SDRAM access waveforms	864
Figure 120.	Burst read SDRAM access	865
Figure 121.	Logic diagram of Read access with RBURST bit set (CAS=2, RPIPE=0)	866
Figure 122.	Read access crossing row boundary	868
Figure 123.	Write access crossing row boundary	868
Figure 124.	Self-refresh mode	871
Figure 125.	Power-down mode	872
Figure 126.	QUADSPI block diagram when dual-flash mode is disabled	883
Figure 127.	QUADSPI block diagram when dual-flash mode is enabled	883
Figure 128.	An example of a read command in quad mode	885
Figure 129.	An example of a DDR command in quad mode	888
Figure 130.	nCS when CKMODE = 0 (T = CLK period)	896
Figure 131.	nCS when CKMODE = 1 in SDR mode (T = CLK period)	897
Figure 132.	nCS when CKMODE = 1 in DDR mode (T = CLK period)	897
Figure 133.	nCS when CKMODE = 1 with an abort (T = CLK period)	897
Figure 134.	DLYB block diagram	913
Figure 135.	ADC block diagram	921
Figure 136.	ADC Clock scheme	924
Figure 137.	ADC clock scheme	925
Figure 138.	ADC1 connectivity	927
Figure 139.	ADC2 connectivity	928
Figure 140.	ADC3 connectivity	929
Figure 141.	ADC calibration	933
Figure 142.	Updating the ADC offset calibration factor	933
Figure 143.	Mixing single-ended and differential channels	934
Figure 144.	Enabling / Disabling the ADC	937
Figure 145.	Analog to digital conversion time	943
Figure 146.	Stopping ongoing regular conversions	944
Figure 147.	Stopping ongoing regular and injected conversions	944

Figure 148. Triggers are shared between ADC master and ADC slave	946
Figure 149. Injected conversion latency	949
Figure 150. Example of JSQR queue of context (sequence change)	952
Figure 151. Example of JSQR queue of context (trigger change)	953
Figure 152. Example of JSQR queue of context with overflow before conversion	953
Figure 153. Example of JSQR queue of context with overflow during conversion	954
Figure 154. Example of JSQR queue of context with empty queue (case JQM=0)	954
Figure 155. Example of JSQR queue of context with empty queue (case JQM=1)	955
Figure 156. Flushing JSQR queue of context by setting JADSTP=1 (JQM=0). Case when JADSTP occurs during an ongoing conversion.	955
Figure 157. Flushing JSQR queue of context by setting JADSTP=1 (JQM=0). Case when JADSTP occurs during an ongoing conversion and a new trigger occurs.	956
Figure 158. Flushing JSQR queue of context by setting JADSTP=1 (JQM=0). Case when JADSTP occurs outside an ongoing conversion	956
Figure 159. Flushing JSQR queue of context by setting JADSTP=1 (JQM=1)	957
Figure 160. Flushing JSQR queue of context by setting ADDIS=1 (JQM=0)	957
Figure 161. Flushing JSQR queue of context by setting ADDIS=1 (JQM=1)	958
Figure 162. Single conversions of a sequence, software trigger	960
Figure 163. Continuous conversion of a sequence, software trigger	960
Figure 164. Single conversions of a sequence, hardware trigger	961
Figure 165. Continuous conversions of a sequence, hardware trigger	961
Figure 166. Right alignment (offset disabled, unsigned value)	963
Figure 167. Right alignment (offset enabled, signed value)	963
Figure 168. Left alignment (offset disabled, unsigned value)	964
Figure 169. Left alignment (offset enabled, signed value)	964
Figure 170. Example of overrun (OVRMOD = 0)	966
Figure 171. Example of overrun (OVRMOD = 1)	967
Figure 172. AUTDLY=1, regular conversion in continuous mode, software trigger	970
Figure 173. AUTDLY=1, regular HW conversions interrupted by injected conversions (DISCEN=0; JDISCEN=0)	971
Figure 174. AUTDLY=1, regular HW conversions interrupted by injected conversions. (DISCEN=1, JDISCEN=1)	972
Figure 175. AUTDLY=1, regular continuous conversions interrupted by injected conversions	973
Figure 176. AUTDLY=1 in auto- injected mode (JAUTO=1)	973
Figure 177. Analog watchdog guarded area	974
Figure 178. ADCy_AWDx_OUT signal generation (on all regular channels)	976
Figure 179. ADCy_AWDx_OUT signal generation (AWDx flag not cleared by SW)	976
Figure 180. ADCy_AWDx_OUT signal generation (on a single regular channel)	977
Figure 181. ADCy_AWDx_OUT signal generation (on all injected channels)	977
Figure 182. 16-bit result oversampling with 10-bits right shift and rounding	978
Figure 183. Triggered regular oversampling mode (TROVS bit = 1)	980
Figure 184. Regular oversampling modes (4x ratio)	981
Figure 185. Regular and injected oversampling modes used simultaneously	981
Figure 186. Triggered regular oversampling with injection	982
Figure 187. Oversampling in auto-injected mode	982
Figure 188. Dual ADC block diagram ⁽¹⁾	985
Figure 189. Injected simultaneous mode on 4 channels: dual ADC mode	986
Figure 190. Regular simultaneous mode on 16 channels: dual ADC mode	988
Figure 191. Interleaved mode on 1 channel in continuous conversion mode: dual ADC mode	989
Figure 192. Interleaved mode on 1 channel in single conversion mode: dual ADC mode	990
Figure 193. Interleaved conversion with injection	990

Figure 194.	Alternate trigger: injected group of each ADC	991
Figure 195.	Alternate trigger: 4 injected channels (each ADC) in discontinuous mode	992
Figure 196.	Alternate + regular simultaneous	993
Figure 197.	Case of trigger occurring during injected conversion	993
Figure 198.	Interleaved single channel CH0 with injected sequence CH11, CH12	994
Figure 199.	Two Interleaved channels (CH1, CH2) with injected sequence CH11, CH12 - case 1: Master interrupted first	994
Figure 200.	Two Interleaved channels (CH1, CH2) with injected sequence CH11, CH12 - case 2: Slave interrupted first	994
Figure 201.	DMA Requests in regular simultaneous mode when DAMDF=0b00	995
Figure 202.	DMA requests in regular simultaneous mode when DAMDF=0b10	996
Figure 203.	DMA requests in interleaved mode when DAMDF=0b10	996
Figure 204.	Temperature sensor channel block diagram	998
Figure 205.	VBAT channel block diagram	1000
Figure 206.	VREFINT channel block diagram	1000
Figure 207.	Dual-channel DAC block diagram	1053
Figure 208.	Data registers in single DAC channel mode	1056
Figure 209.	Data registers in dual DAC channel mode	1056
Figure 210.	Timing diagram for conversion with trigger disabled TEN = 0	1057
Figure 211.	DAC LFSR register calculation algorithm	1059
Figure 212.	DAC conversion (SW trigger enabled) with LFSR wave generation	1059
Figure 213.	DAC triangle wave generation	1060
Figure 214.	DAC conversion (SW trigger enabled) with triangle wave generation	1060
Figure 215.	DAC Sample and hold mode phase diagram	1063
Figure 216.	Comparator functional block diagram	1093
Figure 217.	Comparator hysteresis	1096
Figure 218.	Comparator output blanking	1096
Figure 219.	Output redirection	1098
Figure 220.	Scaler block diagram	1100
Figure 221.	Standalone mode: external gain setting mode	1110
Figure 222.	Follower configuration	1111
Figure 223.	PGA mode, internal gain setting (x2/x4/x8/x16), inverting input not used	1112
Figure 224.	PGA mode, internal gain setting (x2/x4/x8/x16), inverting input used for filtering	1113
Figure 225.	PGA mode, non-inverting gain setting (x2/x4/x8/x16) or inverting gain setting (x-1/x-3/x-7/x-15)	1114
Figure 226.	Example configuration	1114
Figure 227.	PGA mode, non-inverting gain setting (x2/x4/x8/x16) or inverting gain setting (x-1/x-3/x-7/x-15) with filtering	1115
Figure 228.	Example configuration	1115
Figure 229.	Single DFSDM block diagram	1129
Figure 230.	Input channel pins redirection	1133
Figure 231.	Channel transceiver timing diagrams	1136
Figure 232.	Clock absence timing diagram for SPI	1137
Figure 233.	Clock absence timing diagram for Manchester coding	1138
Figure 234.	First conversion for Manchester coding (Manchester synchronization)	1140
Figure 235.	DFSDM_CHyDATINR registers operation modes and assignment	1144
Figure 236.	Example: Sinc3 filter response	1146
Figure 237.	DCMI block diagram	1186
Figure 238.	DCMI signal waveforms	1187
Figure 239.	Timing diagram	1189
Figure 240.	Frame capture waveforms in snapshot mode	1191

Figure 241. Frame capture waveforms in continuous grab mode	1192
Figure 242. Coordinates and size of the window after cropping	1192
Figure 243. Data capture waveforms	1193
Figure 244. Pixel raster scan order	1194
Figure 245. LTDC block diagram	1210
Figure 246. LCD-TFT synchronous timings	1214
Figure 247. Layer window programmable parameters	1217
Figure 248. Blending two layers with background	1220
Figure 249. Interrupt events	1221
Figure 250. JPEG codec block diagram	1245
Figure 251. RNG block diagram	1265
Figure 252. Entropy source model	1266
Figure 253. RNG initialization overview	1269
Figure 254. CRYPT block diagram	1280
Figure 255. AES-ECB mode overview	1283
Figure 256. AES-CBC mode overview	1284
Figure 257. AES-CTR mode overview	1285
Figure 258. AES-GCM mode overview	1286
Figure 259. AES-GMAC mode overview	1286
Figure 260. AES-CCM mode overview	1287
Figure 261. Example of suspend mode management	1293
Figure 262. DES/TDES-ECB mode encryption	1294
Figure 263. DES/TDES-ECB mode decryption	1295
Figure 264. DES/TDES-CBC mode encryption	1296
Figure 265. DES/TDES-CBC mode decryption	1297
Figure 266. AES-ECB mode encryption	1299
Figure 267. AES-ECB mode decryption	1300
Figure 268. AES-CBC mode encryption	1301
Figure 269. AES-CBC mode decryption	1302
Figure 270. Message construction for the Counter mode	1304
Figure 271. AES-CTR mode encryption	1305
Figure 272. AES-CTR mode decryption	1306
Figure 273. Message construction for the Galois/counter mode	1308
Figure 274. Message construction for the Galois Message Authentication Code mode	1313
Figure 275. Message construction for the Counter with CBC-MAC mode	1314
Figure 276. 64-bit block construction according to the data type (IN FIFO)	1321
Figure 277. 128-bit block construction according to the data type	1323
Figure 278. HASH block diagram	1348
Figure 279. Message data swapping feature	1350
Figure 280. HASH suspend/resume mechanism	1356
Figure 281. High-resolution timer block diagram	1374
Figure 282. Timer A..E overview	1379
Figure 283. Continuous timer operation	1380
Figure 284. Single-shot timer operation	1381
Figure 285. Timer reset resynchronization (prescaling ratio above 32)	1382
Figure 286. Repetition rate vs HRTIM_REPxR content in continuous mode	1383
Figure 287. Repetition counter behavior in single-shot mode	1384
Figure 288. Compare events action on outputs: set on compare 1, reset on compare 2	1385
Figure 289. Timing unit capture circuitry	1387
Figure 290. Auto-delayed overview (Compare 2 only)	1388
Figure 291. Auto-delayed compare	1389
Figure 292. Push-pull mode block diagram	1391

Figure 293. Push-pull mode example	1392
Figure 294. Complementary outputs with deadtime insertion	1392
Figure 295. Deadtime insertion vs deadtime sign (1 indicates negative deadtime)	1393
Figure 296. Complementary outputs for low pulse width (SDTRx = SDTFx = 0)	1394
Figure 297. Complementary outputs for low pulse width (SDTRx = SDTFx = 1)	1394
Figure 298. Complementary outputs for low pulse width (SDTRx = 0, SDTFx = 1)	1394
Figure 299. Complementary outputs for low pulse width (SDTRx = 1, SDTFx=0)	1395
Figure 300. Master timer overview	1396
Figure 301. External event conditioning overview (1 channel represented)	1399
Figure 302. Latency to external events falling edge (counter reset and output set)	1402
Figure 303. Latency to external events (output reset on external event)	1402
Figure 304. Event blanking mode	1403
Figure 305. Event postpone mode	1403
Figure 306. External trigger blanking with edge-sensitive trigger	1405
Figure 307. External trigger blanking, level sensitive triggering	1405
Figure 308. Event windowing mode	1406
Figure 309. External trigger windowing with edge-sensitive trigger	1407
Figure 310. External trigger windowing, level sensitive triggering	1407
Figure 311. Delayed Idle mode entry	1409
Figure 312. Burst mode and delayed protection priorities (DIDL = 0)	1410
Figure 313. Burst mode and delayed protection priorities (DIDL = 1)	1411
Figure 314. Balanced Idle protection example	1412
Figure 315. Output management overview	1422
Figure 316. HRTIM output states and transitions	1422
Figure 317. Burst mode operation example	1424
Figure 318. Burst mode trigger on external event	1426
Figure 319. Delayed burst mode entry with deadtime enabled and IDLESx = 1	1428
Figure 320. Delayed Burst mode entry during deadtime	1429
Figure 321. Burst mode exit when the deadtime generator is enabled	1430
Figure 322. Burst mode emulation example	1432
Figure 323. Carrier frequency signal insertion	1432
Figure 324. HRTIM outputs with Chopper mode enabled	1433
Figure 325. Fault protection circuitry (FAULT1 fully represented, FAULT2..5 partially)	1434
Figure 326. Fault signal filtering (FLTxF[3:0]= 0010: f _{SAMPLING} = f _{HRTIM} , N = 4)	1435
Figure 327. Auxiliary outputs	1437
Figure 328. Auxiliary and main outputs during burst mode (DIDLx = 0)	1438
Figure 329. Deadtime distortion on auxiliary output when exiting burst mode	1438
Figure 330. Counter behavior in synchronized start mode	1442
Figure 331. ADC trigger selection overview	1443
Figure 332. Combining several updates on a single hrtim_dac_trgx output	1444
Figure 333. DMA burst overview	1448
Figure 334. Burst DMA operation flowchart	1449
Figure 335. Registers update following DMA burst transfer	1450
Figure 336. Buck converter topology	1452
Figure 337. Dual Buck converter management	1453
Figure 338. Synchronous rectification depending on output current	1453
Figure 339. Buck with synchronous rectification	1454
Figure 340. 3-phase interleaved buck converter	1455
Figure 341. 3-phase interleaved buck converter control	1456
Figure 342. Transition mode PFC	1456
Figure 343. Transition mode PFC waveforms	1457
Figure 344. Advanced-control timer block diagram	1547

Figure 345. Counter timing diagram with prescaler division change from 1 to 2	1549
Figure 346. Counter timing diagram with prescaler division change from 1 to 4	1549
Figure 347. Counter timing diagram, internal clock divided by 1	1551
Figure 348. Counter timing diagram, internal clock divided by 2	1551
Figure 349. Counter timing diagram, internal clock divided by 4	1552
Figure 350. Counter timing diagram, internal clock divided by N	1552
Figure 351. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)	1553
Figure 352. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)	1553
Figure 353. Counter timing diagram, internal clock divided by 1	1555
Figure 354. Counter timing diagram, internal clock divided by 2	1555
Figure 355. Counter timing diagram, internal clock divided by 4	1556
Figure 356. Counter timing diagram, internal clock divided by N	1556
Figure 357. Counter timing diagram, update event when repetition counter is not used	1557
Figure 358. Counter timing diagram, internal clock divided by 1, TIMx_ARR = 0x6	1558
Figure 359. Counter timing diagram, internal clock divided by 2	1559
Figure 360. Counter timing diagram, internal clock divided by 4, TIMx_ARR=0x36	1559
Figure 361. Counter timing diagram, internal clock divided by N	1560
Figure 362. Counter timing diagram, update event with ARPE=1 (counter underflow)	1560
Figure 363. Counter timing diagram, Update event with ARPE=1 (counter overflow)	1561
Figure 364. Update rate examples depending on mode and TIMx_RCR register settings	1562
Figure 365. External trigger input block	1563
Figure 366. TIM1/TIM8 ETR input circuitry	1563
Figure 367. Control circuit in normal mode, internal clock divided by 1	1564
Figure 368. TI2 external clock connection example	1565
Figure 369. Control circuit in external clock mode 1	1566
Figure 370. External trigger input block	1566
Figure 371. Control circuit in external clock mode 2	1567
Figure 372. Capture/compare channel (example: channel 1 input stage)	1568
Figure 373. Capture/compare channel 1 main circuit	1568
Figure 374. Output stage of capture/compare channel (channel 1, idem ch. 2 and 3)	1569
Figure 375. Output stage of capture/compare channel (channel 4)	1569
Figure 376. Output stage of capture/compare channel (channel 5, idem ch. 6)	1570
Figure 377. PWM input mode timing	1572
Figure 378. Output compare mode, toggle on OC1	1574
Figure 379. Edge-aligned PWM waveforms (ARR=8)	1575
Figure 380. Center-aligned PWM waveforms (ARR=8)	1576
Figure 381. Generation of 2 phase-shifted PWM signals with 50% duty cycle	1578
Figure 382. Combined PWM mode on channel 1 and 3	1579
Figure 383. 3-phase combined PWM signals with multiple trigger pulses per period	1580
Figure 384. Complementary output with dead-time insertion	1581
Figure 385. Dead-time waveforms with delay greater than the negative pulse	1581
Figure 386. Dead-time waveforms with delay greater than the positive pulse	1582
Figure 387. Break and Break2 circuitry overview	1584
Figure 388. Various output behavior in response to a break event on BRK (OSS1 = 1)	1586
Figure 389. PWM output state following BRK and BRK2 pins assertion (OSS1=1)	1587
Figure 390. PWM output state following BRK assertion (OSS1=0)	1588
Figure 391. Output redirection	1588
Figure 392. Clearing TIMx_OCxREF	1589
Figure 393. 6-step generation, COM example (OSSR=1)	1590
Figure 394. Example of one pulse mode	1591
Figure 395. Retriggerable one pulse mode	1593
Figure 396. Example of counter operation in encoder interface mode	1594

Figure 397. Example of encoder interface mode with TI1FP1 polarity inverted.	1595
Figure 398. Measuring time interval between edges on 3 signals.	1596
Figure 399. Example of Hall sensor interface	1598
Figure 400. Control circuit in reset mode	1599
Figure 401. Control circuit in Gated mode	1600
Figure 402. Control circuit in trigger mode.	1601
Figure 403. Control circuit in external clock mode 2 + trigger mode	1602
Figure 404. General-purpose timer block diagram	1651
Figure 405. Counter timing diagram with prescaler division change from 1 to 2	1653
Figure 406. Counter timing diagram with prescaler division change from 1 to 4	1653
Figure 407. Counter timing diagram, internal clock divided by 1	1654
Figure 408. Counter timing diagram, internal clock divided by 2	1655
Figure 409. Counter timing diagram, internal clock divided by 4	1655
Figure 410. Counter timing diagram, internal clock divided by N	1656
Figure 411. Counter timing diagram, Update event when ARPE=0 (TIMx_ARR not preloaded).	1656
Figure 412. Counter timing diagram, Update event when ARPE=1 (TIMx_ARR preloaded).	1657
Figure 413. Counter timing diagram, internal clock divided by 1	1658
Figure 414. Counter timing diagram, internal clock divided by 2	1658
Figure 415. Counter timing diagram, internal clock divided by 4	1659
Figure 416. Counter timing diagram, internal clock divided by N	1659
Figure 417. Counter timing diagram, Update event when repetition counter is not used	1660
Figure 418. Counter timing diagram, internal clock divided by 1, TIMx_ARR=0x6	1661
Figure 419. Counter timing diagram, internal clock divided by 2	1662
Figure 420. Counter timing diagram, internal clock divided by 4, TIMx_ARR=0x36	1662
Figure 421. Counter timing diagram, internal clock divided by N	1663
Figure 422. Counter timing diagram, Update event with ARPE=1 (counter underflow).	1663
Figure 423. Counter timing diagram, Update event with ARPE=1 (counter overflow).	1664
Figure 424. Control circuit in normal mode, internal clock divided by 1	1665
Figure 425. TI2 external clock connection example.	1665
Figure 426. Control circuit in external clock mode 1	1666
Figure 427. External trigger input block	1667
Figure 428. Control circuit in external clock mode 2	1668
Figure 429. Capture/Compare channel (example: channel 1 input stage)	1668
Figure 430. Capture/Compare channel 1 main circuit	1669
Figure 431. Output stage of Capture/Compare channel (channel 1).	1669
Figure 432. PWM input mode timing	1671
Figure 433. Output compare mode, toggle on OC1.	1673
Figure 434. Edge-aligned PWM waveforms (ARR=8).	1674
Figure 435. Center-aligned PWM waveforms (ARR=8).	1676
Figure 436. Generation of 2 phase-shifted PWM signals with 50% duty cycle	1677
Figure 437. Combined PWM mode on channels 1 and 3	1678
Figure 438. Clearing TIMx_OCxREF	1679
Figure 439. Example of one-pulse mode.	1680
Figure 440. Retriggerable one-pulse mode.	1682
Figure 441. Example of counter operation in encoder interface mode	1683
Figure 442. Example of encoder interface mode with TI1FP1 polarity inverted	1684
Figure 443. Control circuit in reset mode	1685
Figure 444. Control circuit in gated mode	1686
Figure 445. Control circuit in trigger mode.	1687
Figure 446. Control circuit in external clock mode 2 + trigger mode	1688
Figure 447. Master/Slave timer example	1688

Figure 448. Gating TIM2 with OC1REF of TIM3	1689
Figure 449. Gating TIM2 with Enable of TIM3	1690
Figure 450. Triggering TIM2 with update of TIM3	1691
Figure 451. Triggering TIM2 with Enable of TIM3	1691
Figure 452. Triggering TIM3 and TIM2 with TIM3 TI1 input.	1692
Figure 453. General-purpose timer block diagram (TIM12).	1727
Figure 454. General-purpose timer block diagram (TIM13/TIM14)	1728
Figure 455. Counter timing diagram with prescaler division change from 1 to 2	1730
Figure 456. Counter timing diagram with prescaler division change from 1 to 4	1730
Figure 457. Counter timing diagram, internal clock divided by 1	1731
Figure 458. Counter timing diagram, internal clock divided by 2	1732
Figure 459. Counter timing diagram, internal clock divided by 4	1732
Figure 460. Counter timing diagram, internal clock divided by N	1733
Figure 461. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded).	1733
Figure 462. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded).	1734
Figure 463. Control circuit in normal mode, internal clock divided by 1	1735
Figure 464. TI2 external clock connection example.	1735
Figure 465. Control circuit in external clock mode 1	1736
Figure 466. Capture/compare channel (example: channel 1 input stage)	1737
Figure 467. Capture/compare channel 1 main circuit	1737
Figure 468. Output stage of capture/compare channel (channel 1).	1738
Figure 469. PWM input mode timing	1740
Figure 470. Output compare mode, toggle on OC1.	1742
Figure 471. Edge-aligned PWM waveforms (ARR=8)	1743
Figure 472. Combined PWM mode on channel 1 and 2	1744
Figure 473. Example of one pulse mode.	1745
Figure 474. Retriggerable one pulse mode	1746
Figure 475. Measuring time interval between edges on 2 signals	1747
Figure 476. Control circuit in reset mode	1748
Figure 477. Control circuit in gated mode	1749
Figure 478. Control circuit in trigger mode	1750
Figure 479. TIM15 block diagram	1781
Figure 480. TIM16/TIM17 block diagram	1782
Figure 481. Counter timing diagram with prescaler division change from 1 to 2	1784
Figure 482. Counter timing diagram with prescaler division change from 1 to 4	1784
Figure 483. Counter timing diagram, internal clock divided by 1	1786
Figure 484. Counter timing diagram, internal clock divided by 2	1786
Figure 485. Counter timing diagram, internal clock divided by 4	1787
Figure 486. Counter timing diagram, internal clock divided by N	1787
Figure 487. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded).	1788
Figure 488. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded).	1788
Figure 489. Update rate examples depending on mode and TIMx_RCR register settings	1790
Figure 490. Control circuit in normal mode, internal clock divided by 1	1791
Figure 491. TI2 external clock connection example.	1791
Figure 492. Control circuit in external clock mode 1	1792
Figure 493. Capture/compare channel (example: channel 1 input stage)	1793
Figure 494. Capture/compare channel 1 main circuit	1793
Figure 495. Output stage of capture/compare channel (channel 1).	1794

Figure 496. Output stage of capture/compare channel (channel 2 for TIM15)	1794
Figure 497. PWM input mode timing	1796
Figure 498. Output compare mode, toggle on OC1	1798
Figure 499. Edge-aligned PWM waveforms (ARR=8)	1799
Figure 500. Combined PWM mode on channel 1 and 2	1800
Figure 501. Complementary output with dead-time insertion.	1801
Figure 502. Dead-time waveforms with delay greater than the negative pulse.	1801
Figure 503. Dead-time waveforms with delay greater than the positive pulse.	1802
Figure 504. Break circuitry overview	1804
Figure 505. Output behavior in response to a break	1806
Figure 506. Example of one pulse mode	1808
Figure 507. Retriggerable one pulse mode	1809
Figure 508. Measuring time interval between edges on 2 signals	1811
Figure 509. Control circuit in reset mode	1812
Figure 510. Control circuit in gated mode	1813
Figure 511. Control circuit in trigger mode	1814
Figure 512. Basic timer block diagram.	1865
Figure 513. Counter timing diagram with prescaler division change from 1 to 2	1867
Figure 514. Counter timing diagram with prescaler division change from 1 to 4	1867
Figure 515. Counter timing diagram, internal clock divided by 1	1868
Figure 516. Counter timing diagram, internal clock divided by 2	1869
Figure 517. Counter timing diagram, internal clock divided by 4	1869
Figure 518. Counter timing diagram, internal clock divided by N	1870
Figure 519. Counter timing diagram, update event when ARPE = 0 (TIMx_ARR not preloaded).	1870
Figure 520. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded).	1871
Figure 521. Control circuit in normal mode, internal clock divided by 1	1872
Figure 522. Low-power timer block diagram (LPTIM1 and LPTIM2)	1879
Figure 523. Low-power timer block diagram (LPTIM3)	1880
Figure 524. Low-power timer block diagram (LPTIM4 and LPTIM5)	1880
Figure 525. Glitch filter timing diagram	1885
Figure 526. LPTIM output waveform, single counting mode configuration	1887
Figure 527. LPTIM output waveform, Single counting mode configuration and Set-once mode activated (WAVE bit is set).	1887
Figure 528. LPTIM output waveform, Continuous counting mode configuration	1888
Figure 529. Waveform generation	1889
Figure 530. Encoder mode counting sequence	1893
Figure 531. Watchdog block diagram	1908
Figure 532. Window watchdog timing diagram	1909
Figure 533. Independent watchdog block diagram	1914
Figure 534. RTC block overview	1924
Figure 535. Detailed RTC block diagram	1925
Figure 536. Tamper detection	1926
Figure 537. I2C block diagram	1971
Figure 538. I2C bus protocol	1973
Figure 539. Setup and hold timings	1975
Figure 540. I2C initialization flowchart	1978
Figure 541. Data reception	1979
Figure 542. Data transmission	1980
Figure 543. Slave initialization flowchart	1983
Figure 544. Transfer sequence flowchart for I2C slave transmitter,	

	NOSTRETCH= 0	1985
Figure 545.	Transfer sequence flowchart for I2C slave transmitter, NOSTRETCH= 1	1986
Figure 546.	Transfer bus diagrams for I2C slave transmitter	1987
Figure 547.	Transfer sequence flowchart for slave receiver with NOSTRETCH=0	1988
Figure 548.	Transfer sequence flowchart for slave receiver with NOSTRETCH=1	1989
Figure 549.	Transfer bus diagrams for I2C slave receiver	1989
Figure 550.	Master clock generation	1991
Figure 551.	Master initialization flowchart	1993
Figure 552.	10-bit address read access with HEAD10R=0	1993
Figure 553.	10-bit address read access with HEAD10R=1	1994
Figure 554.	Transfer sequence flowchart for I2C master transmitter for $N \leq 255$ bytes	1995
Figure 555.	Transfer sequence flowchart for I2C master transmitter for $N > 255$ bytes	1996
Figure 556.	Transfer bus diagrams for I2C master transmitter	1997
Figure 557.	Transfer sequence flowchart for I2C master receiver for $N \leq 255$ bytes	1999
Figure 558.	Transfer sequence flowchart for I2C master receiver for $N > 255$ bytes	2000
Figure 559.	Transfer bus diagrams for I2C master receiver	2001
Figure 560.	Timeout intervals for $t_{\text{LOW:SEXT}}$, $t_{\text{LOW:MEXT}}$	2006
Figure 561.	Transfer sequence flowchart for SMBus slave transmitter N bytes + PEC	2010
Figure 562.	Transfer bus diagrams for SMBus slave transmitter (SBC=1)	2010
Figure 563.	Transfer sequence flowchart for SMBus slave receiver N Bytes + PEC	2012
Figure 564.	Bus transfer diagrams for SMBus slave receiver (SBC=1)	2013
Figure 565.	Bus transfer diagrams for SMBus master transmitter	2014
Figure 566.	Bus transfer diagrams for SMBus master receiver	2016
Figure 567.	I2C interrupt mapping diagram	2022
Figure 568.	USART block diagram	2042
Figure 569.	Word length programming	2045
Figure 570.	Configurable stop bits	2047
Figure 571.	TC/TXE behavior when transmitting	2050
Figure 572.	Start bit detection when oversampling by 16 or 8	2051
Figure 573.	usart_ker_ck clock divider block diagram	2054
Figure 574.	Data sampling when oversampling by 16	2055
Figure 575.	Data sampling when oversampling by 8	2056
Figure 576.	Mute mode using Idle line detection	2063
Figure 577.	Mute mode using address mark detection	2064
Figure 578.	Break detection in LIN mode (11-bit break length - LBDL bit is set)	2067
Figure 579.	Break detection in LIN mode vs. Framing error detection	2068
Figure 580.	USART example of synchronous master transmission	2069
Figure 581.	USART data clock timing diagram in synchronous master mode (M bits =00)	2069
Figure 582.	USART data clock timing diagram in synchronous master mode (M bits = 01)	2070
Figure 583.	USART data clock timing diagram in synchronous slave mode (M bits =00)	2071
Figure 584.	ISO 7816-3 asynchronous protocol	2073
Figure 585.	Parity error detection using the 1.5 stop bits	2075
Figure 586.	IrDA SIR ENDEC block diagram	2079
Figure 587.	IrDA data modulation (3/16) - Normal mode	2079
Figure 588.	Transmission using DMA	2081
Figure 589.	Reception using DMA	2082
Figure 590.	Hardware flow control between 2 USARTs	2082
Figure 591.	RS232 RTS flow control	2083

Figure 592. RS232 CTS flow control	2084
Figure 593. Wakeup event verified (wakeup event = address match, FIFO disabled)	2087
Figure 594. Wakeup event not verified (wakeup event = address match, FIFO disabled)	2087
Figure 595. LPUART block diagram	2130
Figure 596. LPUART word length programming	2132
Figure 597. Configurable stop bits	2134
Figure 598. TC/TXE behavior when transmitting	2136
Figure 599. lpuart_ker_ck clock divider block diagram	2139
Figure 600. Mute mode using Idle line detection	2143
Figure 601. Mute mode using address mark detection	2144
Figure 602. Transmission using DMA	2146
Figure 603. Reception using DMA	2147
Figure 604. Hardware flow control between 2 LPUARTs	2148
Figure 605. RS232 RTS flow control	2148
Figure 606. RS232 CTS flow control	2149
Figure 607. Wakeup event verified (wakeup event = address match, FIFO disabled)	2152
Figure 608. Wakeup event not verified (wakeup event = address match, FIFO disabled)	2152
Figure 609. SPI2S block diagram	2182
Figure 610. Full-duplex single master/ single slave application	2184
Figure 611. Half-duplex single master/ single slave application	2184
Figure 612. Simplex single master/single slave application (master in transmit-only/ slave in receive-only mode)	2186
Figure 613. Master and three independent slaves at star topology	2187
Figure 614. Master and three slaves at circular (daisy chain) topology	2188
Figure 615. Multi-master application	2189
Figure 616. Scheme of SS control logic	2191
Figure 617. Data flow timing control (SSOE=1, SSOM=0, SSM=0)	2191
Figure 618. SS interleaving pulses between data (SSOE=1, SSOM=1, SSM=0)	2192
Figure 619. Data clock timing diagram	2194
Figure 620. Data alignment when data size is not equal to 8-bit, 16-bit or 32-bit	2195
Figure 621. Packing data in FIFO for transmission and reception	2202
Figure 622. TI mode transfer	2204
Figure 623. Low-power mode application example	2209
Figure 624. Waveform examples	2215
Figure 625. Master I2S Philips protocol waveforms (16/32-bit full accuracy)	2216
Figure 626. I2S Philips standard waveforms	2216
Figure 627. Master MSB Justified 16-bit or 32-bit full-accuracy length	2217
Figure 628. Master MSB justified 16 or 24-bit data length	2217
Figure 629. Slave MSB justified	2218
Figure 630. LSB justified 16 or 24-bit data length	2218
Figure 631. Master PCM when the frame length is equal the data length	2219
Figure 632. Master PCM standard waveforms (16 or 24-bit data length)	2219
Figure 633. Slave PCM waveforms	2220
Figure 634. Start-up sequence, I2S Philips standard, master	2222
Figure 635. Start-up sequence, I2S Philips standard, slave	2223
Figure 636. Stop sequence, I2S Philips standard, master	2224
Figure 637. I ² S clock generator architecture	2224
Figure 638. Data Format	2227
Figure 639. Handling of underrun situation	2228

Figure 640. Handling of overrun situation	2229
Figure 641. Frame error detection, with FIXCH=0	2230
Figure 642. Frame error detection, with FIXCH=1	2230
Figure 643. SAI functional block diagram	2257
Figure 644. Audio frame	2261
Figure 645. FS role is start of frame + channel side identification (FSDEF = TRIS = 1)	2263
Figure 646. FS role is start of frame (FSDEF = 0)	2264
Figure 647. Slot size configuration with FBOFF = 0 in SAI_xSLOTR	2265
Figure 648. First bit offset	2265
Figure 649. Audio block clock generator overview	2266
Figure 650. PDM typical connection and timing	2270
Figure 651. Detailed PDM interface block diagram	2271
Figure 652. Start-up sequence	2272
Figure 653. SAI_ADR format in TDM, 32-bit slot width	2273
Figure 654. SAI_ADR format in TDM, 16-bit slot width	2274
Figure 655. SAI_ADR format in TDM, 8-bit slot width	2275
Figure 656. AC'97 audio frame	2278
Figure 657. Example of typical AC'97 configuration on devices featuring at least 2 embedded SAIs (three external AC'97 decoders)	2279
Figure 658. SPDIF format	2280
Figure 659. SAI_xDR register ordering	2281
Figure 660. Data companding hardware in an audio block in the SAI	2285
Figure 661. Tristate strategy on SD output line on an inactive slot	2286
Figure 662. Tristate on output data line in a protocol like I2S	2287
Figure 663. Overrun detection error	2288
Figure 664. FIFO underrun event	2288
Figure 665. SPDIFRX block diagram	2324
Figure 666. S/SPDIF Sub-frame format	2325
Figure 667. S/SPDIF block format	2325
Figure 668. S/SPDIF Preambles	2326
Figure 669. Channel coding example	2327
Figure 670. SPDIFRX decoder	2328
Figure 671. Noise filtering and edge detection	2328
Figure 672. Thresholds	2330
Figure 673. Synchronization flowchart	2332
Figure 674. Synchronization process scheduling	2333
Figure 675. SPDIFRX States	2334
Figure 676. SPDIFRX_FMTx_DR register format	2336
Figure 677. Channel/user data format	2337
Figure 678. S/SPDIF overrun error when RXSTEO = 0	2339
Figure 679. S/SPDIF overrun error when RXSTEO = 1	2340
Figure 680. SPDIFRX interface interrupt mapping diagram	2341
Figure 681. S1 signal coding	2358
Figure 682. S2 signal coding	2358
Figure 683. SWPMI block diagram	2360
Figure 684. SWP bus states	2363
Figure 685. SWP frame structure	2364
Figure 686. SWPMI No software buffer mode transmission	2365
Figure 687. SWPMI No software buffer mode transmission, consecutive frames	2366
Figure 688. SWPMI Multi software buffer mode transmission	2368
Figure 689. SWPMI No software buffer mode reception	2370
Figure 690. SWPMI single software buffer mode reception	2371

Figure 691. SWPMI Multi software buffer mode reception	2373
Figure 692. SWPMI single buffer mode reception with CRC error.	2374
Figure 693. MDIOS block diagram	2387
Figure 694. MDIO protocol write frame waveform	2388
Figure 695. MDIO protocol read frame waveform	2388
Figure 696. SDMMC “no response” and “no data” operations.	2401
Figure 697. SDMMC (multiple) block read operation.	2401
Figure 698. SDMMC (multiple) block write operation.	2402
Figure 699. SDMMC (sequential) stream read operation	2402
Figure 700. SDMMC (sequential) stream write operation	2402
Figure 701. SDMMC block diagram.	2404
Figure 702. SDMMC Command and data phase relation	2406
Figure 703. Control unit	2408
Figure 704. Command/Response path	2409
Figure 705. Command path state machine (CPSM)	2410
Figure 706. Data path	2416
Figure 707. DDR mode data packet clocking	2417
Figure 708. DDR mode CRC status / boot acknowledgment clocking.	2417
Figure 709. Data path state machine (DPSM).	2418
Figure 710. CLKMUX unit	2428
Figure 711. Hardware flow timing	2432
Figure 712. Asynchronous interrupt generation	2434
Figure 713. Synchronous interrupt period data read	2434
Figure 714. Synchronous interrupt period data write	2435
Figure 715. Asynchronous interrupt period data read	2436
Figure 716. Asynchronous interrupt period data write	2436
Figure 717. Clock stop with SDMMC_CK for DS, HS, SDR12, SDR25.	2439
Figure 718. Clock stop with SDMMC_CK for DDR50, SDR50, SDR104.	2439
Figure 719. ReadWait with SDMMC_CK < 50 MHz.	2440
Figure 720. ReadWait with SDMMC_CK > 50 MHz.	2441
Figure 721. CMD12 stream timing	2443
Figure 722. CMD5 Sleep Awake procedure	2445
Figure 723. Normal boot mode operation	2447
Figure 724. Alternative boot mode operation.	2448
Figure 725. Command response R1b busy signaling	2449
Figure 726. SDMMC state control	2450
Figure 727. Card cycle power / power up diagram	2451
Figure 728. CMD11 signal voltage switch sequence.	2452
Figure 729. Voltage switch transceiver typical application.	2454
Figure 730. CAN subsystem	2484
Figure 731. FDCAN block diagram	2486
Figure 732. Transceiver delay measurement	2491
Figure 733. Pin control in bus monitoring mode	2493
Figure 734. Pin control in loop back mode.	2495
Figure 735. Message RAM configuration.	2496
Figure 736. Standard message ID filter path	2499
Figure 737. Extended message ID filter path.	2500
Figure 738. Example of mixed configuration dedicated Tx buffers / Tx FIFO	2506
Figure 739. Example of mixed configuration dedicated Tx buffers / Tx queue	2506
Figure 740. Bit timing	2508
Figure 741. Bypass operation	2510
Figure 742. FSM calibration.	2511

Figure 743. Cycle time and global time synchronization	2526
Figure 744. TTCAN level 0 and level 2 drift compensation	2527
Figure 745. Level 0 schedule synchronization state machine	2534
Figure 746. Level 0 master to slave relation	2535
Figure 747. USB1 OTG_HS high-speed block diagram (OTG_HS1)	2618
Figure 748. USB2 OTG_HS high-speed block diagram (OTG_HS2)	2618
Figure 749. SOF connectivity (SOF trigger output to TIM and ITR1 connection)	2629
Figure 750. Updating OTG_HFIR dynamically (RLDCTRL = 1)	2632
Figure 751. Device-mode FIFO address mapping and AHB FIFO access mapping	2633
Figure 752. Host-mode FIFO address mapping and AHB FIFO access mapping	2634
Figure 753. Interrupt hierarchy	2638
Figure 754. Transmit FIFO write task	2743
Figure 755. Receive FIFO read task	2744
Figure 756. Normal bulk/control OUT/SETUP	2745
Figure 757. Bulk/control IN transactions	2749
Figure 758. Normal interrupt OUT	2752
Figure 759. Normal interrupt IN	2757
Figure 760. Isochronous OUT transactions	2759
Figure 761. Isochronous IN transactions	2762
Figure 762. Normal bulk/control OUT/SETUP transactions - DMA	2764
Figure 763. Normal bulk/control IN transaction - DMA	2766
Figure 764. Normal interrupt OUT transactions - DMA mode	2767
Figure 765. Normal interrupt IN transactions - DMA mode	2768
Figure 766. Normal isochronous OUT transaction - DMA mode	2769
Figure 767. Normal isochronous IN transactions - DMA mode	2770
Figure 768. Receive FIFO packet read	2776
Figure 769. Processing a SETUP packet	2778
Figure 770. Bulk OUT transaction	2785
Figure 771. TRDT max timing case	2794
Figure 772. A-device SRP	2795
Figure 773. B-device SRP	2796
Figure 774. A-device HNP	2797
Figure 775. B-device HNP	2799
Figure 776. Ethernet high-level block diagram	2807
Figure 777. DMA transmission flow (standard mode)	2810
Figure 778. DMA transmission flow (OSP mode)	2812
Figure 779. Receive DMA flow	2814
Figure 780. Overview of MAC transmission flow	2817
Figure 781. MAC reception flow	2819
Figure 782. Packet filtering sequence	2824
Figure 783. Networked time synchronization	2831
Figure 784. Propagation delay calculation in clocks supporting peer-to-peer path correction	2832
Figure 785. System time update using Fine method	2837
Figure 786. TCP segmentation offload overview	2844
Figure 787. Header and payload fields of segmented packets	2846
Figure 788. Supported PHY interfaces	2855
Figure 789. SMA Interface block	2855
Figure 790. MDIO packet structure	2856
Figure 791. Write data packet	2857
Figure 792. Read data packet	2858
Figure 793. Media independent interface (MII) signals	2858

Figure 794. RMI block diagram	2860
Figure 795. Transmission bit order	2861
Figure 796. Receive bit order	2862
Figure 797. Descriptor ring structure	2879
Figure 798. DMA descriptor ring	2880
Figure 799. Transmit descriptor (read format)	2881
Figure 800. Transmit descriptor write-back format	2885
Figure 801. Transmit context descriptor format	2890
Figure 802. Receive normal descriptor (read format)	2893
Figure 803. Receive normal descriptor (write-back format)	2896
Figure 804. Receive context descriptor	2903
Figure 805. Generation of ETH_DMAISR flags	2920
Figure 806. HDMI-CEC block diagram	3039
Figure 807. Message structure	3039
Figure 808. Blocks	3040
Figure 809. Bit timings	3040
Figure 810. Signal free time	3041
Figure 811. Arbitration phase	3041
Figure 812. SFT of three nominal bit periods	3041
Figure 813. Error bit timing	3042
Figure 814. Error handling	3044
Figure 815. TXERR detection	3045
Figure 816. Block diagram of debug infrastructure	3056
Figure 817. Power domains of debug infrastructure	3058
Figure 818. Clock domains of debug infrastructure	3059
Figure 819. SWD successful data transfer	3062
Figure 820. JTAG TAP state machine	3063
Figure 821. Debug and access port connections	3074
Figure 822. APB-D CoreSight component topology	3082
Figure 823. Global timestamp distribution	3090
Figure 824. Embedded cross trigger	3097
Figure 825. Mapping of trigger inputs to outputs	3099
Figure 826. ETF state transition diagram	3128
Figure 827. Cortex-M7 CoreSight topology	3197